具身智能世界模型综合综述

A Comprehensive Survey on World Models for Embodied AI

来源：https://arxiv.org/abs/2510.16732

原文 PDF：23 页。制作日期：2026-10-22。

译稿说明：这是 Google Translate 生成、经固定术语表约束的逐页机器翻译初稿，未经逐句人工学术审校。中文翻译覆盖全部可提取文本，包含正文、图表文字、附录与参考文献文本；图片内未能提取的文字、数学公式的二维布局和复杂表格请对照原文。排版提取可能造成少量顺序或符号错误，尤其是双栏图表、公式、脚注。不能用机器译稿替代原文进行公式推导或直接引用。常用英文术语和缩写予以保留。

PDF 采用原文页与中文译页交替编排，原始图表及公式完整保留在原文页。章节中的数字“原文第 X 页”始终指原始 PDF 物理页码，而非论文印刷页码或译稿页码。参考文献在原文页保留完整英文条目，检索时使用原文。学习文档中的例子和伪代码是导读，不属于作者原文。

核心摘要校订

根据原文摘要单独校订。正文仍为逐页机器翻译初稿；原始图表、公式和英文条目请查看紧邻原文页。

具身智能（Embodied AI）要求智能体能够感知、行动，并预判动作如何改变未来世界状态。世界模型（World Models）作为捕捉环境动态的内部模拟器，通过前向与反事实轨迹展开支持感知、预测和决策。本文提出具身世界模型的统一框架，形式化问题设定与学习目标，并建立三条分类轴：（1）功能：决策耦合（Decision-Coupled）与通用（General-Purpose）；（2）时间建模：序列仿真与推断（Sequential Simulation and Inference）与全局差异预测（Global Difference Prediction）；（3）空间表示：全局潜向量（Global Latent Vector）、词元特征序列（Token Feature Sequence）、空间潜网格（Spatial Latent Grid）与分解式渲染表示（Decomposed Rendering Representation）。本文系统梳理机器人、自动驾驶与一般视频任务中的数据和指标，涵盖像素预测质量、状态层理解与任务表现；进行代表模型的定量比较；并指出统一数据集缺乏、评价应更关注物理一致性而非只看像素保真度、模型表现与实时效率的权衡，以及长时程时间一致性和误差累积等挑战。

A Comprehensive Survey on World Models for Embodied AI

Xinqing Li, Xin He, Le Zhang, *Member, IEEE*, Min Wu, *Senior Member, IEEE*, Xiaoli Li, *Fellow, IEEE*, and Yun Liu

arXiv:2510.16732v3 [cs.CV] 25 Jun 2026

***Abstract*—Embodied AI requires agents that perceive, act, and anticipate how actions reshape future world states. World models serve as internal simulators that capture environment dynamics, enabling forward and counterfactual rollouts to support perception, prediction, and decision making. This survey presents a unified framework for world models in embodied AI. Specifically, we formalize the problem setting and learning objectives, and propose a three-axis taxonomy encompassing: (1) Functionality, *Decision-Coupled vs. General-Purpose*; (2) Temporal Modeling, *Sequential Simulation and Inference vs. Global Difference Prediction*; (3) Spatial Representation, *Global Latent Vector*, *Token Feature Sequence*, *Spatial Latent Grid*, and *Decomposed Rendering Representation*. We systematize data resources and metrics across robotics, autonomous driving, and general video settings, covering pixel prediction quality, state-level understanding, and task performance. Furthermore, we offer a quantitative comparison of state-of-the-art models and distill key open challenges, including the scarcity of unified datasets and the need for evaluation metrics that assess physical consistency over pixel fidelity, the trade-off between model performance and the computational efficiency required for real-time control, and the core modeling difficulty of achieving long-horizon temporal consistency while mitigating error accumulation. Finally, we maintain a curated bibliography at https://github.com/Li-Zn-H/AwesomeWorldModels.**

***Index Terms*—World Models, Embodied AI, Temporal Modeling, Spatial Representation.**

I. Introduction

EMBODIED AI aims to equip agents to perceive complex, multimodal environments, act within them, and anticipate how their actions will alter future world states [1], [2]. Central to this capability is the world model, an internal simulator that captures environment dynamics to support both forward and counterfactual rollouts for perception, prediction, and decision making [3], [4]. This survey focuses on world models that yield actionable predictions for embodied agents, distinguishing them from static scene descriptors or purely generative visual models that do not capture controllable dynamics.

Cognitive science suggests humans construct internal models of the world by integrating sensory inputs. These models do not merely predict and simulate future events but also shape perception and guide action [5]–[7]. Motivated by this view, early AI research on world models was rooted in model-based reinforcement learning (RL), where latent state-transition models were used to improve sample efficiency and planning performance [8]. The seminal work of Ha and Schmidhuber [9] crystallized the term world model and inspired the Dreamer series [10]–[12], highlighting how learned dynamics can drive imagination-based policy optimization. More recently, advances in large-scale generative modeling and multimodal learning have expanded world models beyond their initial focus on policy learning into general-purpose environment simulators capable of high-fidelity future prediction, exemplified by models like Sora [13] and V-JEPA 2 [14]. This expansion has diversified functional roles, temporal modeling strategies, and spatial representations, while introducing inconsistencies in terminology and taxonomy across sub-communities.

Faithfully capturing environment dynamics requires addressing both the temporal evolution of states and the spatial encoding of scenes [3]. Long-horizon rollouts are susceptible to error accumulation, which establishes coherence as a central challenge in video prediction and policy imagination [15], [16]. Similarly, coarse or 2D-centric layouts provide insufficient geometric detail for handling challenges such as occlusion, object permanence, and geometry-aware planning. In contrast, volumetric or 3D occupancy representations such as neural fields [17] and structured voxel grids [18] provide geometric structure that better supports forecasting and control. Taken together, these points establish temporal modeling and spatial representation as core design dimensions that fundamentally influence the predictive horizon, physical fidelity, and downstream performance of embodied agents.

Several recent surveys have organized the rapidly growing literature on world models. Overall, these surveys followed two main approaches. The first is a function-oriented perspective. For example, Ding *et al.* [4] categorized relevant works based on the two core functions of understanding and prediction, while Zhu *et al.* [19] presented a framework based on the core capabilities of world models. The second approach is application-driven, focusing on specific domains such as autonomous driving. Notably, Guan *et al.* [20] and Feng *et al.* [21] provided overviews of world-model techniques for

Xinqing Li and Yun Liu are with the College of Computer Science and the Academy for Advanced Interdisciplinary Studies, Nankai University, Tianjin 300350, China (e-mail: lixinqing@mail.nankai.edu.cn; liuyun@nankai.edu.cn).

Xin He is with the School of Computer Science and Engineering, Tianjin University of Technology, Tianjin 300384, China (e-mail: hexin@email.tjut.edu.cn).

Le Zhang is with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, Sichuan, China (e-mail: lezhang@uestc.edu.cn).

Min Wu is with the Institute for Infocomm Research (I2R), Agency for Science, Technology and Research (A*STAR), 138632, Singapore (e-mail: wumin@i2r.a-star.edu.sg).

Xiaoli Li is with the Information Systems Technology and Design (ISTD) Pillar, Singapore University of Technology and Design (SUTD), 487372, Singapore (e-mail: xiaoli_li@sutd.edu.sg).

Corresponding author: Yun Liu (e-mail: liuyun@nankai.edu.cn)

This work was supported in part by the National Natural Science Foundation of China (No. 62576176) and in part by the Fundamental Research Funds for the Central Universities (Nankai University, No. 070-63253235).

原文第 1 页 | 中文翻译初稿

1
具身智能世界模型综合综述

李新清、何新、张乐、会员，IEEE、吴敏、高级会员、IEEE、李晓丽、院士、IEEE 和刘云

术语保留：世界模型（World Models）；具身智能（Embodied AI）

摘要 -具身智能（Embodied AI）需要能够感知、行动并预测行动如何重塑未来世界状态的智能体。 世界模型（World Models）作为捕获环境动态的内部模拟器，使前向和 反事实（Counterfactual）展开轨迹（Rollouts）支持感知、预测、

和决策。本篇综述在具身智能（Embodied AI）中提出了世界模型（World Models）的统一框架

。具体来说，我们形式化了问题设置和学习目标，并提出了一个三轴分类法，包括：（1）功能，决策耦合（Decision-Coupled）与 通用（General-Purpose）；(2) 时间建模，序列仿真与推断（Sequential Simulation and Inference）与 全局差异预测（Global Difference Prediction）；(3) 空间表示，全局潜向量（Global Latent Vector）、词元特征序列（Token Feature Sequence）、空间潜网格（Spatial Latent Grid）和 分解式渲染表示（Decomposed Rendering Representation）

。我们将机器人、自动驾驶和一般视频设置的数据资源和指标系统化，涵盖像素预测质量、状态级理解和任务性能

。此外，我们对最先进的模型进行了定量比较，并提炼了关键的开放挑战，包括统一数据集的稀缺性和评估像素保真度物理一致性的评估指标的需要，模型性能和实时控制所需的计算效率之间的权衡，以及在减轻误差累积的同时实现 长时程（Long-Horizon）时间一致性的核心建模难度。最后，我们在 https://github.com/Li-Zn-H/AwesomeWorldModels 上维护了一份精选的参考书目。

索引术语 -世界模型（World Models）、具身智能（Embodied AI）、时间建模、空间表示。

一、简介

在 MBODIED 多模态环境中，人工智能旨在装备其中的智能体，感知并预测复杂的情况，他们的行为将如何改变未来的世界状态 [1], [2]。此功能的核心是 世界模型（World Model），它是一个内部模拟器，可捕获环境动态以支持前向和 反事实（Counterfactual）展开轨迹（Rollouts）进行感知、预测和决策 [3]、[4]。本篇综述重点关注 世界模型（World Models）

李新庆、刘云（南开大学计算机学院和高等交叉研究院，天津 300350）（e-mail：lixinqing@mail.nankai.edu.cn；liuyun@nankai.edu.cn）。

何鑫（天津工业大学计算机科学与工程学院，天津 300384）（e-mail：hexin@email.tjut.edu.cn）

张乐，电子科技大学信息与通信工程学院，四川省成都市 611731（e-mail：lezhang@uestc.edu.cn）。

Min Wu 就职于信息通信研究所 (I2R)、科学技术研究局 (A*STAR)，邮编：138632，新加坡（电子邮件：wumin@i2r.a-star.edu.sg）。

李晓丽 (Xiaoli Li) 就职于新加坡科技设计大学 (SUTD) 信息系统技术与设计 (ISTD) 支柱部门，邮编：487372，新加坡（电子邮件：xiaoli_li@sutd.edu.sg）。

通讯作者: 刘云(e-mail: liuyun@nankai.edu.cn)

该工作得到了国家自然科学基金（No. 62576176）和中央高校基本科研业务费专项资金（南开大学，No. 070-63253235）的部分支持。

为具体智能体产生可操作的预测，将它们与静态场景描述符或不捕获可控动态的纯粹生成视觉模型区分开来。

认知科学表明，人类通过整合感官输入来构建世界的内部模型。这些模型不仅可以预测和模拟未来事件，还可以塑造感知并指导行动[5]-[7]。受这种观点的推动，对 世界模型（World Models） 的早期人工智能研究植根于基于模型的 强化学习（Reinforcement Learning）(RL)，其中使用潜在状态转换模型来提高样本效率和规划性能 [8]。 Ha 和 Schmidhuber [9] 的开创性工作明确了 世界模型（World Model） 术语，并启发了 Dreamer 系列 [10]-[12]，强调了学习动态如何驱动基于想象力的策略优化。最近，大规模生成建模和多模态学习的进步已经将 世界模型（World Models） 从最初关注的策略学习扩展到能够进行高保真未来预测的 通用（General-Purpose） 环境模拟器，例如 Sora [13] 和 V-JEPA 2 [14] 等模型。这种扩展具有多样化的功能角色、时间建模策略和空间表示，同时在子社区之间引入了术语和分类的不一致。

忠实地捕捉环境动态需要解决状态的时间演化和场景的空间编码[3]。 长时程（Long-Horizon） 展开轨迹（Rollouts） 容易受到错误累积的影响，这使得一致性成为视频预测和策略想象中的核心挑战[15]，[16]。同样，粗略或以 2D 为中心的布局提供的几何细节不足以应对遮挡、对象持久性和几何感知规划等挑战。相比之下，体积或 3D 占用表示（例如神经场 [17] 和结构化体素网格 [18]）提供了更好支持预测和控制的几何结构。总而言之，这些点将时间建模和空间表示建立为核心设计维度，从根本上影响具身智能体的预测范围、物理保真度和下游性能。

最近的几项调查整理了有关 世界模型（World Models） 的快速增长的文献。总体而言，这些调查遵循两种主要方法。第一个是功能导向的视角。例如，丁等人[4]根据理解和预测两个核心功能对相关工作进行了分类，而朱等人[19]则根据世界模型（World Models）的核心功能提出了一个框架。第二种方法是应用驱动，专注于自动驾驶等特定领域。值得注意的是，Guan et al. [20] 和 Feng et al. [21] 提供了世界模型技术的概述

autonomous driving. Complementing these perspectives, we adopt a three-axis taxonomy around decision coupling, temporal modeling, and spatial representation, and apply it uniformly across robotics, autonomous driving, and general-purpose video world models.

To address the lack of a unified taxonomy in the context of embodied AI, this work introduces a framework centered on the three core axes of functionality, temporal modeling, and spatial representation. At the functional level, this framework distinguishes between decision-coupled and general-purpose models. At the temporal level, it differentiates between Sequential Simulation and Inference versus Global Difference Prediction. Finally, at the spatial level, it encompasses a range of representations from latent features to explicit geometry and neural fields. This framework offers a unified structure for organizing existing approaches and integrates standardized datasets and evaluation metrics. This structure facilitates quantitative comparisons and provides a panoramic, actionable knowledge map for future research.

Fig. 1 presents an overview of the structure and taxonomy of this paper. We begin in §II by outlining the core concepts and theoretical foundations of world models. §III introduces our three-axis taxonomy and maps representative methods onto this framework. §IV surveys the datasets and evaluation metrics used for training and assessment. §V offers a quantitative comparison of state-of-the-art models. §VI discusses open challenges and promising research directions, and §VII concludes the survey.

II. Background

A. Core Concepts

As discussed in §I, a world model can be viewed as an internal simulator that implements a **compression–prediction–control** pipeline: high-dimensional observations are first **compressed** into a latent state, the dynamics model **predicts** how this state will evolve under actions, and a policy or planner **controls** the agent by reasoning over imagined latent rollouts. Operationally, this functionality decomposes into three tightly coupled aspects:

- **Simulation & Planning**, which uses learned dynamics to generate plausible scenarios, allowing agents to assess potential actions through imagination without real-world interaction.
- **Temporal Evolution**, which learns how the encoded state evolves, enabling temporally consistent rollouts.
- **Spatial Representation**, which encodes scene geometry at an appropriate fidelity, using formats such as latent tokens or neural fields to provide context for control.

Together, these three pillars instantiate the compression–prediction–control view and jointly determine how a world model navigates the fundamental trade-off between computational efficiency, long-horizon coherence, and visual fidelity. They provide the conceptual foundation for the taxonomy introduced in §III and are formalized in the mathematical framework that follows.

B. Mathematical Formulation of World Models

We formalize the environment interaction as a POMDP [24]. For notational consistency, we define a null initial action a_0 at $t = 0$, which allows the dynamics to be written uniformly. At each step $t \geq 1$, the agent receives an observation o_t and takes an action a_t, while the true state s_t remains unobserved. To handle this partial observability, world models infer a learned latent state z_t using a one-step filtering posterior, where the previous latent state z_{t-1} is assumed to summarize the relevant history. Finally, z_t is used to reconstruct o_t:

$$\begin{array}{ll} \text{Dynamics Prior:} & p_\theta(z_t \mid z_{t-1}, a_{t-1}) \\ \text{Filtered Posterior:} & q_\phi(z_t \mid z_{t-1}, a_{t-1}, o_t) \\ \text{Reconstruction:} & p_\theta(o_t \mid z_t) \end{array} . \tag{1}$$

Consistent with the Markovian structure, the joint distribution over observations and latent states factorizes as:

$$p_\theta(o_{1:T}, z_{0:T} \,|\, a_{0:T-1}) = p_\theta(z_0) \prod_{t=1}^{T} p_\theta(z_t \,|\, z_{t-1}, a_{t-1}) p_\theta(o_t \,|\, z_t). \tag{2}$$

To infer the latent states, we must approximate the intractable true posterior $p_\theta(z_{0:T} \mid o_{1:T}, a_{0:T-1})$ with a time-factorized variational distribution:

$$q_\phi(z_{0:T} \,|\, o_{1:T}, a_{0:T-1}) = q_\phi(z_0 \,|\, o_1) \prod_{t=1}^{T} q_\phi(z_t \,|\, z_{t-1}, a_{t-1}, o_t), \tag{3}$$

which indeed reduces to the action-free case when ignoring a inputs. Directly maximizing the log-likelihood $\log p_\theta(o_{1:T} \mid a_{0:T-1})$ is intractable. Instead, we optimize an ELBO using the approximate posterior q_ϕ, which provides a tractable objective for learning the model parameters:

$$\begin{aligned} \log p_\theta(o_{1:T} \,|\, a_{0:T-1}) &= \log \int p_\theta(o_{1:T}, z_{0:T} \,|\, a_{0:T-1})\, dz_{0:T} \\ &\geq \mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(o_{1:T}, z_{0:T} \mid a_{0:T-1})}{q_\phi(z_{0:T} \mid o_{1:T}, a_{0:T-1})} \right] =: \mathcal{L}(\theta, \phi). \end{aligned} \tag{4}$$

Under the assumption of Markov factorization for both p_θ and q_ϕ, this ELBO decomposes into a reconstruction objective and a KL regularization term:

$$\begin{aligned} \mathcal{L}(\theta,\phi) = &\sum_{t=1}^{T} \mathbb{E}_{q_\phi(z_t)}\big[\log p_\theta(o_t \mid z_t) \big] \\ &- D_{\mathrm{KL}}\big(q_\phi(z_{0:T} \mid o_{1:T}, a_{0:T-1}) \,\|\, p_\theta(z_{0:T} \mid a_{0:T-1}) \big). \end{aligned} \tag{5}$$

Modern world models thus adopt a reconstruction-regularization training paradigm: the likelihood term $\log p_\theta(o_t \,|\, z_t)$ encourages faithful observation prediction, and KL regularization terms align the filtered posterior $q_\phi(z_t \mid z_{t-1}, a_{t-1}, o_t)$ with the dynamics prior $p_\theta(z_t \mid z_{t-1}, a_{t-1})$. Such world models can be instantiated with recurrent models [25]–[27], Transformer-based architectures [28]–[30], or diffusion-based decoders [31]–[35]. In all cases, the learned latent trajectory $z_{1:T}$ serves as a compact, predictive memory to support downstream policy optimization, model-predictive control, and counterfactual reasoning in embodied AI.

原文第 2 页 | 中文翻译初稿

2

自动驾驶。作为对这些观点的补充，我们采用了围绕决策耦合、时间建模和空间表示的三轴分类法，并将其统一应用于机器人、自动驾驶和 通用（General-Purpose） 视频 世界模型（World Models）。

为了解决 具身智能（Embodied AI） 背景下缺乏统一分类的问题，这项工作引入了一个以功能、时间建模和空间表示三个核心轴为中心的框架。在功能层面，该框架区分了 决策耦合（Decision-Coupled） 和 通用（General-Purpose） 模型。在时间级别上，它区分 序列仿真与推断（Sequential Simulation and Inference） 和 全局差异预测（Global Difference Prediction）。最后，在空间层面，它涵盖了从潜在特征到显式几何和神经场的一系列表示。该框架提供了一个统一的结构来组织现有方法并集成标准化数据集和评估指标。这种结构有利于定量比较，并为未来的研究提供全景的、可操作的知识图谱。

图 1 概述了本文的结构和分类。我们从第二节开始概述 世界模型（World Models） 的核心概念和理论基础。 §III 介绍了我们的三轴分类法并将代表性方法映射到该框架上。 §IV 调查用于训练和评估的数据集和评估指标。 §V 提供了最先进模型的定量比较。 §VI 讨论了开放的挑战和有前途的研究方向，§VII 总结了调查。

二.背景

A。核心概念

正如第 §I 中所讨论的，世界模型（World Model） 可以被视为实现压缩-预测-控制管道的内部模拟器：高维观察首先被压缩到潜在状态，动态模型预测该状态将如何在行动下演变，策略或规划器通过对想象的潜在 展开轨迹（Rollouts） 进行推理来控制智能体。在操作上，此功能分解为三个紧密耦合的方面：

- 模拟和规划，它使用学习的动态来生成合理的场景，允许智能体通过想象来评估潜在的行动，而无需现实世界的交互。

- 时间演化，它学习编码状态如何演化，从而实现时间一致的 展开轨迹（Rollouts）。

- 空间表示，以适当的保真度对场景几何进行编码，使用潜在标记或神经场等格式来提供控制上下文。

这三个支柱共同实例化了压缩-预测-控制视图，并共同确定 世界模型（World Model） 如何在计算效率、长时程（Long-Horizon） 一致性和视觉保真度之间进行基本权衡。它们为第 III 节中引入的分类法提供了概念基础，并在随后的数学框架中形式化。

_B。 世界模型（World Models）_的数学公式

我们将环境交互形式化为 POMDP [24]。为了符号上的一致性，我们在 t = 0 处定义了一个空初始动作 a 0 ，这使得动态可以统一写入。在每一步 t ≥ 1 处，智能体都会收到观察结果 ot_ 并采取行动 at ，而真实状态 st 仍然未被观察到

。为了处理这种部分可观察性，世界模型（World Models） 使用一步过滤后验推断学习的潜在状态_zt_，其中假设先前的潜在状态_zt−_ 1 总结了相关历史。最后，zt 用于重建 ot ：

动力学先验：pθ (zt | zt− 1 , at− 1)
滤波后验：q phi (zt | zt− 1 , at− 1 , ot) 。 (1)
重构：pθ (ot | zt)

与马尔可夫结构一致，观测值和潜在状态的联合分布因式分解为：

T

pθ (o 1: T , z 0: T | a 0: T − 1)= pθ (z 0) pθ (zt | zt− 1 , at− 1) pθ (ot | zt) .
t =1

(2)

为了推断潜在状态，我们必须用时间分解的变分分布来近似难以处理的真实后验 pθ (z 0: T | o 1: T , a 0: T − 1)：

T
qΦ (z 0: T | o 1: T , a 0: T − 1) = qΦ (z 0 | o 1) qΦ (zt | zt− 1 , at− 1 , ot) ,
t =1

(3)

当忽略 a 输入时，这确实减少了无动作的情况。直接最大化对数似然 log pθ (o 1: T | a 0: T − 1) 是很棘手的。相反，我们使用近似后验 qphi 来优化 ELBO，它为学习模型参数提供了一个易于处理的目标：

log pθ (o 1: T | a 0: T − 1) =log pθ (o 1: T , z 0: T | a 0: T − 1) dz 0: T

≥ E qφ log pθ (o 1: T , z 0: T | a 0: T − 1) =: L (θ, φ) .
qφ (z 0: T | o 1: T , a 0: T − 1)
(4)

在 pθ 和 qphi 均进行马尔可夫分解的假设下，该 ELBO 分解为重建目标和 KL 正则化项：

T
L (θ,φ) = E qφ (zt) log pθ (ot | zt)
t =1
− D KL qφ (z 0: T | o 1: T , a 0: T − 1) ‖ pθ (z 0: T | a 0: T − 1) .
(5)

现代 世界模型（World Models） 因此采用重建正则化训练范例：似然项 log pθ (ot | zt) 鼓励忠实的观察预测，而 KL 正则化项将滤波后验 qphi (zt | zt− 1 , at− 1 , ot) 与动态先验 pθ (zt | zt− 1 ,在− 1)。这样的世界模型（World Models） 可以用循环模型 [25]-[27]、基于 Transformer 的架构 [28]-[30] 或基于扩散的解码器 [31]-[35] 进行实例化。在所有情况下，学习到的潜在轨迹 z 1: T 充当紧凑的预测存储器，以支持具身智能（Embodied AI） 中的下游策略优化、模型预测控制和 反事实（Counterfactual） 推理。

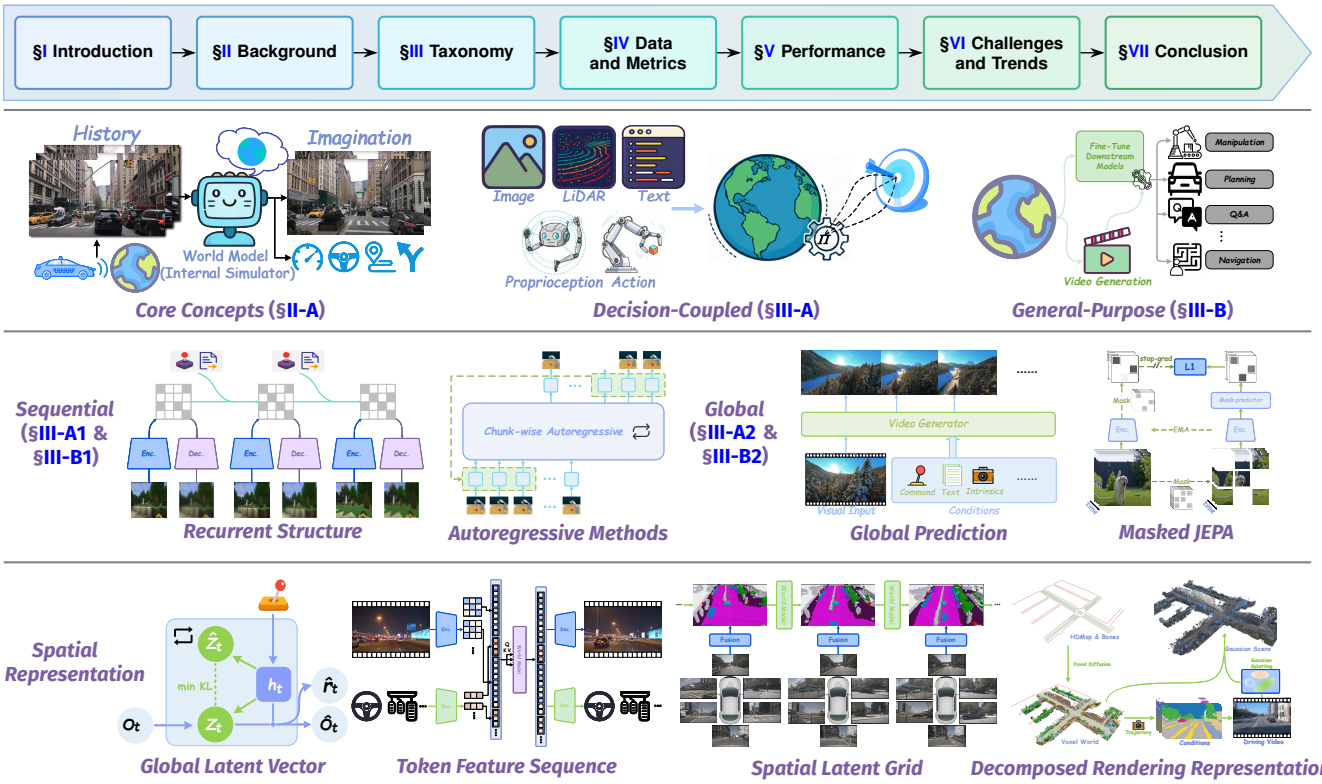


Fig. 1. Structure of this survey. The figure classifies world models along three axes and illustrates representative methods for each, providing a unified view of the field. Figure design inspired in part by [12], [14], [22], [23].

III. Taxonomy

We categorize world models along three core dimensions, which provide the foundation for the subsequent analysis in this survey.

The first dimension, decision coupling, distinguishes between **Decision-Coupled** and **General-Purpose** world models. Decision-Coupled models are task-specific, learning dynamics tightly aligned with a particular control objective, which improves sample efficiency and closed-loop performance but weakens generalization beyond the training distribution. In contrast, General-Purpose models are task-agnostic simulators that prioritize broad coverage of visual and physical phenomena, easing transfer across tasks and domains, yet their pretraining objective often misaligns with downstream control, leading to suboptimal or brittle behavior unless explicitly mitigated.

The second dimension, temporal reasoning, delineates two distinct paradigms of prediction. **Sequential Simulation and Inference** models dynamics autoregressively, unfolding future states one step at a time. This offers fine-grained control and naturally supports closed-loop planning, but suffers from error accumulation over long horizons and a computational cost that scales linearly with the rollout length. In contrast, **Global Difference Prediction** directly estimates future states in parallel, amortizing computation across time and reducing wall-clock latency for long-horizon imagination, but its global updates weaken closed-loop interactivity and tend to smooth over fine-grained local dynamics, making it less suited to tasks that require frequent feedback and precise step-wise control.

The third dimension, spatial representation, comprises four primary strategies used in current research to model spatial states. They lie at different points along the coherence-fidelity-efficiency trade-off:

1) **Global Latent Vector** encodes world states into a compact vector, enabling efficient real-time rollouts but sacrificing fine-grained spatial detail.
2) **Token Feature Sequence** discretizes world states into token sequences to model token-token dependencies via attention, enabling fine-grained multimodal modeling but typically requiring large data, large models, and high inference cost.
3) **Spatial Latent Grid** injects spatial priors by encoding scenes into BEV or voxel grids, preserving local topology and supporting multi-view fusion and map-like planning, but incurring large memory cost, limited resolution, and weaker adaptability to unstructured environments.
4) **Decomposed Rendering Representation** decomposes scenes into renderable primitives (*e.g.*, 3DGS [36], NeRF [37]) and reconstructs observations via differentiable rendering, achieving geometry-consistent, high-fidelity and object-level controllable views, but with expensive training and limited ability to handle rapid dynamics or topology changes.

The following tables apply this taxonomy to classify representative works. Tab. I reviews approaches in robotics, while Tab. II focuses on autonomous driving. Together, they provide a roadmap for the detailed analyses in the subsequent sections.

原文第 3 页 | 中文翻译初稿

3

§IV 数据 §VI 挑战
§I 简介 §II 背景 §III 分类 §V 性能 §VII 结论
以及指标和趋势
历史想象力微调操作
下游
模型
规划
图像 LiDAR 文本
问答
Π
世界模型（World Model） 导航
（内部模拟器）本体感觉动作视频生成
核心概念（§II-A）决策耦合（Decision-Coupled）（§III-A）通用（General-Purpose）（§III-B）
……stop-grad // L1
...
顺序（§III-A1和块式自回归（§全局III-A2和视频生成器掩模Enc.EMA掩模） PredictorEnc.
§III-B1) Enc. Dec. Enc. Dec. Enc. §III-B2)
… …… Mask
命令文本内在函数
… 视觉输入条件
循环结构自回归方法全局预测 Masked JEPA
… ...
空间 Z^t Enc. K VQ Dec. Fusion Fusion Fusion HDMap & Boxes体素扩散高斯场景SplattingGaussian
表示 min KL ht ^rt
... Enc。十二月...
O t Zt O ^t 轨迹
体素世界条件驾驶视频
全局潜向量（Global Latent Vector） 词元特征序列（Token Feature Sequence） 空间潜网格（Spatial
Latent Grid） 分解式渲染表示（Decomposed Rendering Representation）
世界模型（World Model） 世界模型（World Model）
…
…
…
世界模型（World Model）
…
…

图 1. 本篇综述的结构。该图沿三个轴对 世界模型（World Models） 进行分类，并说明了每个轴的代表性

方法，提供了该领域的统一视图。人物设计的部分灵感来自于[12]、[14]、[22]、[23]。
三．分类学

我们按照三个核心维度对世界模型（World Models）进行分类，为本篇综述的后续分析提供基础。

第一个维度，决策耦合，区分 决策耦合（Decision-Coupled） 和 通用（General-Purpose） 世界模型（World Models）。 决策耦合（Decision-Coupled） 模型是特定于任务的学习动态，与特定控制目标紧密结合，这提高了样本效率和闭环性能，但削弱了训练分布之外的泛化能力。相比之下，通用（General-Purpose） 模型是与任务无关的模拟器，优先考虑视觉和物理现象的广泛覆盖，简化跨任务和领域的传输，但其预训练目标通常与下游控制不一致，导致次优或脆弱的行为，除非明确缓解。

第二个维度，时间推理，描绘了两种不同的预测范式。 序列仿真与推断（Sequential Simulation and Inference） 对动力学进行自回归建模，一次一步地展现未来状态。这提供了细粒度的控制，自然支持闭环规划，但会遭受长期误差积累以及与 展开轨迹（Rollout） 长度线性缩放的计算成本的影响。相比之下，全局差异预测（Global Difference Prediction） 直接并行估计未来状态，跨时间摊销计算并减少 长时程（Long-Horizon） 想象力的挂钟延迟，但其全局更新削弱了闭环交互性，并且倾向于平滑细粒度的局部动态，使其不太适合需要频繁反馈和精确逐步控制的任务。

第三个维度，空间表示，包括当前研究中用于建模空间状态的四种主要策略。它们位于一致性与保真度效率权衡的不同点：

- 1) 全局潜向量（Global Latent Vector） 将世界状态编码为紧凑向量，实现高效的实时 展开轨迹（Rollouts），但牺牲了细粒度的空间细节。

- 2) 词元特征序列（Token Feature Sequence） 将世界状态离散化为令牌序列，通过注意力对令牌-令牌依赖关系进行建模，从而实现细粒度的多模态建模，但通常需要大数据、大模型和高推理成本。

- 3）空间潜网格（Spatial Latent Grid）通过将场景编码到BEV或体素网格中来注入空间先验，保留局部拓扑并支持多视图融合和类地图规划，但会产生较大的内存成本、有限的分辨率以及对非结构化环境的适应性较弱。

- 4) 分解式渲染表示（Decomposed Rendering Representation） 将场景分解为可渲染基元（例如，3DGS [36]、NeRF [37]），并通过可微分渲染重建观察结果，实现几何一致、高保真和对象级可控视图，但训练成本高昂，处理快速动态或拓扑变化的能力有限。

下表应用此分类法对代表性作品进行分类。标签。我回顾了机器人技术的方法，而 Tab. II重点关注自动驾驶。它们共同为后续部分的详细分析提供了路线图。

TABLE I
A SUMMARY OF REPRESENTATIVE WORLD MODELS IN ROBOTICS AND GENERAL-PURPOSE DOMAINS.

Paper	Publication	Taxonomy[1]	Characteristics[2]	Datasets Platform											Modality						Reality[4]
				DMC	Atari	RLBench	SSv2	OXE	Meta-world	Franka	RT-1	LIBERO	Other(s)	Total[3]	RGB	Action	Proprio.	Depth	Language	Other(s)	
PlaNet [38]	ICML'19	Dec/Seq/GLV	RSSM	✓										1	✓	✓					
Dreamer [10]	ICLR'20	Dec/Seq/GLV	RSSM	✓	✓								✓	3	✓	✓					
GLAMOR [39]	ICLR'21	Dec/Seq/GLV	IDM	✓	✓									2	✓	✓					
DreamerV2 [11]	ICLR'21	Dec/Seq/GLV	RSSM	✓	✓									2	✓	✓					
TransDreamer [28]	arXiv'22	Dec/Seq/GLV	TSSM	✓	✓								✓	4	✓	✓					
Iso-Dream [40]	NeurIPS'22	Dec/Seq/GLV	IDM	✓									✓	4	✓	✓					
MWM [41]	CoRL'22	Dec/Seq/TFS	RSSM	✓		✓			✓					3	✓	✓					
Inner Monologue [42]	CoRL'22	Dec/Seq/TFS	CoT										✓	3	✓	✓			✓		✓
DayDreamer [43]	CoRL'22	Dec/Seq/GLV	RSSM										✓	4	✓	✓	✓	✓			✓
TWM [29]	ICLR'23	Dec/Seq/TFS	Transformer		✓									1	✓	✓					
IRIS [44]	ICLR'23	Dec/Seq/TFS	Transformer		✓									1	✓	✓					
WorldDreamer [45]	arXiv'24	Gen/Glo/TFS	Transformer										✓	4	✓	✓			✓		
Statler [46]	ICRA'24	Dec/Seq/TFS	LLM										✓	2	✓	✓		✓	✓		✓
Pandora [47]	arXiv'24	Gen/Seq/TFS	Video Diffusion				✓						✓	2	✓				✓		
DWL [48]	RSS'24	Dec/Seq/GLV	MLP										✓	4		✓	✓			✓	✓
RoboDreamer [49]	ICML'24	Dec/Glo/TFS	IDM			✓					✓			2	✓	✓			✓	✓	
Genie [50]	ICML'24	Gen/Seq/TFS	Transformer								✓		✓	3	✓	✓					
V-JEPA [51]	TMLR'24	Gen/Glo/TFS	JEPA				✓						✓	6	✓						
PreLAR [52]	ECCV'24	Dec/Seq/GLV	RSSM			✓	✓		✓					3	✓	✓					
ManiGaussian [53]	ECCV'24	Dec/Seq/DRR	3DGS			✓								1	✓	✓	✓	✓	✓		
ECoT [54]	CoRL'24	Dec/Glo/TFS	CoT										✓	3	✓	✓			✓		✓
VidMan [55]	NeurIPS'24	Dec/Glo/TFS	IDM			✓		✓					✓	4	✓	✓	✓		✓		
iVideoGPT [56]	NeurIPS'24	Gen/Seq/TFS	Transformer				✓	✓	✓				✓	6	✓	✓					
EnerVerse [34]	arXiv'25	Dec/Seq/SLG	Video Diffusion								✓	✓	✓	4	✓	✓			✓		✓
GLAM [57]	AAAI'25	Dec/Seq/GLV	Mamba		✓									1	✓	✓					
NavCoT [58]	TPAMI'25	Dec/Seq/TFS	CoT										✓	4	✓	✓			✓		
DreamerV3 [12]	Nature'25	Dec/Seq/GLV	RSSM	✓	✓								✓	8	✓	✓	✓				
MineWorld [59]	arXiv'25	Dec/Seq/TFS	Transformer										✓	1	✓	✓					
DreMa [60]	ICLR'25	Dec/Seq/DRR	3DGS			✓				✓				2	✓	✓		✓	✓	✓	✓
S2-SSM [61]	arXiv'25	Gen/Seq/TFS	Mamba										✓	1	✓					✓	
RLVR-World [62]	arXiv'25	Gen/Seq/TFS	RLVR							✓			✓	3	✓	✓			✓		
StateSpaceDiffuser [63]	arXiv'25	Gen/Seq/TFS	Mamba										✓	2	✓	✓					
DeepVerse [64]	arXiv'25	Gen/Seq/TFS	DiT										✓	1	✓			✓	✓	✓	
ORV [65]	arXiv'25	Gen/Glo/SLG	DiT								✓		✓	4	✓	✓		✓	✓	✓	
V-JEPA 2 [14]	arXiv'25	Gen/Glo/TFS	JEPA				✓			✓			✓	15	✓	✓	✓		✓		✓
NWM [66]	CVPR'25	Dec/Seq/TFS	DiT										✓	6	✓	✓					
WorldVLA [67]	arXiv'25	Dec/Seq/TFS	Transformer									✓		1	✓	✓			✓		
World4Omni [68]	arXiv'25	Gen/Seq/TFS	VLM			✓							✓	2	✓	✓			✓	✓	✓
Dyn-O [69]	arXiv'25	Dec/Seq/TFS	Mamba										✓	1	✓	✓					
DINO-WM [70]	ICML'25	Dec/Seq/SLG	Transformer	✓									✓	3	✓	✓	✓				
EVA [71]	ICML'25	Gen/Seq/TFS	RoG					✓			✓		✓	4	✓	✓			✓		
AdaWorld [72]	ICML'25	Gen/Seq/TFS	Video Diffusion				✓	✓				✓	✓	6	✓	✓					
MindJourney [73]	arXiv'25	Gen/Seq/SLG	VLM										✓	2	✓	✓			✓		
GAF [74]	arXiv'25	Dec/Seq/DRR	4DGS			✓								1	✓	✓	✓			✓	
Yume [75]	arXiv'25	Gen/Seq/TFS	DiT										✓	1	✓	✓			✓		
villa-X [76]	arXiv'25	Dec/Glo/TFS	IDM				✓	✓			✓	✓	✓	5	✓	✓	✓		✓		✓
AETHER [77]	ICCV'25	Gen/Glo/SLG	DiT										✓	6	✓	✓		✓		✓	
TesserAct [78]	ICCV'25	Dec/Glo/SLG	IDM			✓	✓				✓		✓	4	✓			✓	✓	✓	
MineDreamer [79]	IROS'25	Dec/Seq/TFS	CoI										✓	3	✓	✓			✓		
ManiGaussian++ [80]	IROS'25	Dec/Seq/DRR	3DGS			✓							✓	2	✓	✓	✓	✓	✓		✓
LeWorldModel [81]	arXiv'26	Dec/Seq/GLV	JEPA	✓									✓	4	✓	✓					
DreamZero [82]	arXiv'26	Dec/Seq/TFS	Video Diffusion							✓			✓	4	✓	✓	✓		✓		✓
PointWorld [83]	CVPR'26	Dec/Seq/SLG	Transformer							✓			✓	2	✓	✓	✓	✓		✓	✓
Cosmos Policy [84]	arXiv'26	Dec/Glo/TFS	Video Diffusion							✓		✓	✓	3+	✓	✓	✓		✓	✓	✓
V-JEPA 2.1 [85]	arXiv'26	Gen/Glo/TFS	JEPA				✓			✓			✓	15+	✓	✓	✓	✓	✓	✓	✓
Cosmos 3 [86]	arXiv'26	Gen/Glo/TFS	Transformer							✓	✓	✓	✓	12+	✓	✓	✓	✓	✓	✓	✓

[1] **Taxonomy**: Abbreviations for the taxonomy categories defined in §III.
[2] **Characteristics**: Representative backbone or core technical approach.
[3] **Total**: Number of data platforms used. Underlined entries denote newly proposed or aggregated datasets.
[4] **Reality**: The check mark (✓) indicates validation on a physical robot.

A. Decision-Coupled World Models

1) Sequential Simulation and Inference: **Global Latent Vector.** Early decision-coupled world models combined sequential inference with global latent states. These approaches primarily use Recurrent Neural Networks (RNNs) for efficient real-time and long-horizon prediction.

Ha and Schmidhuber [9] introduced an early world model that encodes observations into a latent space and employs an RNN to model dynamics for policy optimization. Building on this, PlaNet [38] introduced the Recurrent State-Space Model (RSSM), which fuses deterministic memory with stochastic components to enable robust long-horizon imagination. The successor models Dreamer, DreamerV2, and DreamerV3 [10]–[12] further advanced this formulation, inspiring a broad line of subsequent research.

Building on RSSM, several variants modify or eliminate the decoder to better capture dynamics. For example, Dreaming [117] uses contrastive learning and linear methods to mitigate state shifts, whereas DreamerPro [118] replaces the decoder with prototypes to suppress visual distractions. To fur-

原文第 4 页 | 中文翻译初稿

4

表一

机器人领域和 通用（General-Purpose） 领域代表性 世界模型（World Models） 的概述。

|||||||数据集 P
d|平台|||模态|||
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|论文|出版物|分类1|特征2|DMC
Atari
RLBench|SSv2|OXE
Meta-worl|F ranka
RT-1|LIBERO
其他|总计3
RGB|动作
固有。
深度|语言|现实4
其他|
|PlaNet [38]|ICML’19|Dec/Seq/GLV|RSSM|✓|||||1
✓|✓|||
|梦想家 [10]|ICLR’20|Dec/Seq/GLV|RSSM|✓
✓||||✓|3
✓|✓|||
|GLAMOR [39]
|ICLR’21
|Dec/Seq/GLV
|IDM
|✓
✓

|||||2
✓

|✓
|||
|DreamerV2 [11]|ICLR’21|Dec/Seq/GLV|RSSM|✓
✓|||||2
✓|✓|||
|TransDreamer [28]|arXiv'22|Dec/Seq/GLV|TSSM|✓
✓||||✓|4
✓|✓|||
|Iso-Dream [40]|NeurIPS’22|Dec/Seq/GLV|IDM|✓||||✓|4

✓|✓|||
|MWM [41]|CoRL’22|Dec/Seq/TFS|RSSM|✓
✓||✓|||3
✓|✓|||
|内心独白[42]|CoRL’22|Dec/Seq/TFS|CoT|||||✓|3
✓|✓|✓|✓|
|DayDreamer [43]|CoRL’22|Dec/Seq/GLV|RSSM|||||✓|4
✓|✓
✓
✓||✓|
|TWM [29]|ICLR’23|Dec/Seq/TFS|变压器|✓|||||1
✓|✓|||
|IRIS [44]
|ICLR’23
|Dec/Seq/TFS
|变压器
|✓||||1
✓

|✓
|||
|WorldDreamer [45]|arXiv'24|Gen/Glo/TFS|变压器|||||✓|4
✓|✓|✓||
|Statler [46]|ICRA'24|Dec/Seq/TFS|LLM|||||✓|2
✓|✓
✓|✓|✓|
|Pandora [47]|arXiv’24|Gen/Seq/TFS|视频扩散||✓|||✓|2
✓||✓||
|DWL [48]|RSS’24|Dec/Seq/GLV|MLP|||||✓|4|✓
✓||✓
✓|
|RoboDreamer [49]|ICML’24|Dec/Glo/TFS|IDM|✓|||✓||2
✓|✓|✓|✓|
|Genie [50]|ICML’24|Gen/Seq/TFS|变压器||||✓|✓|3
✓|✓|||
|V-JEPA [51]|TMLR’24|Gen/Glo/TFS|JEPA||✓|||✓|6
✓||||
|PreLAR [52]|ECCV’24|Dec/Seq/GLV|RSSM|✓|✓|✓|||3
✓|✓|||
|ManiGaussian [53]|ECCV’24|Dec/Seq/DRR|3DGS|✓|||||1
✓|✓
✓
✓|✓||

|ECOT [54]|CoRL'24|Dec/Glo/TFS|CoT|||||✓|3
✓|✓|✓|✓|
|VidMan [55]|NeurIPS’24|Dec/Glo/TFS|IDM|✓||✓||✓|4
✓|✓
✓|✓||
|iVideoGPT [56]|NeurIPS’24|Gen/Seq/TFS|变压器||✓|✓
✓||✓|6
✓|✓|||
|EnerVerse [34]|arXiv’25|Dec/Seq/SLG|视频传播||||✓|✓
✓|4
✓|✓|✓|✓|
|GLAM [57]|AAAI'25|Dec/Seq/GLV|Mamba|✓|||||1
✓|✓|||
|NavCoT [58]|TPAMI’25|Dec/Seq/TFS|CoT|||||✓|4
✓|✓|✓||
|DreamerV3 [12]|Nature’25|Dec/Seq/GLV|RSSM|✓
✓||||✓|8
✓|✓
✓|||
|MineWorld [59]|arXiv’25|Dec/Seq/TFS|变压器|||||✓|1
✓|✓|||
|DreMa [60]
|ICLR'25
'|Dec/Seq/DRR
|3DGS
|✓|||✓||2
✓

|✓
✓|✓|✓
✓
|
|S2-SSM [61]|arXiv25|Gen/Seq/TFS|Mamba|||||✓|1
✓|||✓|
|RLVR-World [62]|arXiv'25|Gen/Seq/TFS|RLVR||||✓|✓|3
✓|✓|✓||
|StateSpaceDiffuser [63]|arXiv’25|Gen/Seq/TFS|Mamba|||||✓|2
✓|✓|||
|DeepVerse [64]|arXiv’25|Gen/Seq/TFS|DiT|||||✓|1
✓|✓|✓|✓|
|ORV [65]|arXiv’25|Gen/Glo/SLG|DiT||||✓|✓|4
✓|✓
✓|✓|✓|

|V-JEPA 2 [14]|arXiv'25|Gen/Glo/TFS|JEPA||✓||✓|✓|15
✓|✓
✓|✓|✓|
|NWM [66]|CVPR’25|Dec/Seq/TFS|DiT|||||✓|6
✓|✓|||
|WorldVLA [67]|arXiv'25|Dec/Seq/TFS|变压器|||||✓|1
✓|✓|✓||
|World4Omni [68]|arXiv'25|Gen/Seq/TFS|VLM|✓||||✓|2
✓|✓|✓|✓
✓|
|Dyn-O [69]|arXiv'25|Dec/Seq/TFS|Mamba|||||✓|1
✓|✓|||
|DINO-WM [70]|ICML’25|Dec/Seq/SLG|变压器|✓||||✓|3
✓|✓
✓|||
|EVA [71]|ICML'25|Gen/Seq/TFS|RoG|||✓|✓|✓|4
✓|✓|✓||
|AdaWorld [72]|ICML’25|Gen/Seq/TFS|视频扩散||✓|✓||✓
✓|6
✓|✓|||
|MindJourney [73]|arXiv'25|Gen/Seq/SLG|VLM|||||✓|2
✓|✓|✓||
|GAF

[74]|arXiv'25|Dec/Seq/DRR|4DGS|✓|||||1
✓|✓
✓||✓|
|Yume [75]|arXiv'25|Gen/Seq/TFS|DiT|||||✓|1
✓|✓|✓||
|villa-X [76]|arXiv’25|Dec/Glo/TFS|IDM||✓|✓|✓|✓
✓|5
✓|✓
✓|✓|✓|
|AETHER [77]|ICCV’25|Gen/Glo/SLG|DiT|||||✓|6
✓|✓
✓||✓|
|TesserAct [78]|ICCV’25|Dec/Glo/SLG|IDM|✓|✓||✓|✓|4
✓|✓|✓|✓|
|MineDreamer [79]|IROS’25|Dec/Seq/TFS|CoI|||||✓|3
✓|✓|✓||
|ManiGaussian++ [80]|IROS’25|Dec/Seq/DRR|3DGS|✓||||✓|2
✓|✓
✓

✓|✓|✓|
|LeWorldModel [81]|arXiv'26|Dec/Seq/GLV|JEPA|✓||||✓|4
✓|✓|||
|DreamZero [82]|arXiv'26|Dec/Seq/TFS|视频扩散||||✓|✓|4
✓|✓
✓|✓|✓|
|PointWorld [83]|CVPR’26|Dec/Seq/SLG|变压器||||✓|✓|2
✓|✓
✓
✓||✓
✓|
|Cosmos 策略 [84]
VJEPA21[85]|arXiv’26
rXi’26|Dec/Glo/TFS
Gn/Gl/TFS|视频扩散
JEPA||✓||✓
✓|✓
✓
✓|3+
✓
15+
✓|✓
✓
✓
✓
✓|✓
✓|✓
✓
✓
✓|
|- 。
Cosmos 3 [86]|av
arXiv’26|eo
Gen/Glo/TFS|变压器||||✓
✓|✓
✓|12+
✓|✓
✓
✓|✓|✓
✓|

1 分类法：第 III 节中定义的分类法类别的缩写。 2 特征：具有代表性的骨干或核心技术方法。 3 总计：使用的数据平台数量。带下划线的条目表示新提议或聚合的数据集。 4 现实：复选标记 (✓) 表示在物理机

器人上进行验证。
一个。 决策耦合（Decision-Coupled） 世界模型（World Models）

1) 序列仿真与推断（Sequential Simulation and Inference）: 全局潜向量（Global Latent Vector）。早期的 决策耦合（Decision-Coupled） 世界模型（World Models） 将顺序推理与全局潜在状态相结合。这些方法主要使用循环神经网络 (RNN) 进行高效的实时和 长时程（Long-Horizon） 预测。

Ha 和 Schmidhuber [9] 引入了早期的 世界模型（World Model），它将观察结果编码到潜在空间中，并使用 RNN 来建模动态以进行策略优化。在此基础上，PlaNet [38] 引入了循环状态空间模型

(RSSM)，它将确定性内存与随机组件融合在一起，以实现强大的 长时程（Long-Horizon） 想象力。后续模型 Dreamer、DreamerV2 和 DreamerV3 [10]-[12] 进一步推进了这一公式，激发了广泛的后续研究。

在 RSSM 的基础上，多个变体修改或消除了解码器，以更好地捕捉动态。例如，Dreaming [117] 使用对比学习和线性方法来减轻状态转移，而 DreamerPro [118] 用原型替换解码器以抑制视觉干扰。为了毛皮-

TABLE II
A SUMMARY OF REPRESENTATIVE WORLD MODELS FOR THE AUTONOMOUS DRIVING DOMAIN.

Paper	Publication	Taxonomy[1]	Characteristics[2]	Datasets Platform								Input Modality							
				CARLA	nuScenes	nuPlan	Waymo	Occ3D	OpenDV	Other(s)	Total[3]	RGB	Motion	Map	LiDAR	Bound box	Language	Occpancy	Other(s)
MILE [87]	NeurIPS'22	Dec/Seq/GLV	RSSM	✓							1	✓	✓	✓					
Copilot4D [88]	ICLR'24	Gen/Seq/SLG	Video Diffusion		✓					✓	3		✓		✓				
SEM2 [89]	TITS'24	Dec/Seq/GLV	RSSM	✓							1	✓	✓	✓	✓				
MagicDrive3D [90]	arXiv'24	Gen/Glo/DRR	3DGS		✓						1	✓	✓	✓		✓	✓		✓
OccSora [91]	arXiv'24	Gen/Glo/SLG	Diffusion		✓			✓			2		✓					✓	
Delphi [92]	arXiv'24	Gen/Seq/SLG	Video Diffusion		✓						1	✓		✓		✓	✓		
DriveWorld [93]	CVPR'24	Dec/Seq/SLG	RSSM		✓					✓	2	✓	✓				✓	✓	
Drive-WM [94]	CVPR'24	Dec/Glo/SLG	Video Diffusion		✓						1	✓	✓	✓		✓	✓		✓
ViDAR [95]	CVPR'24	Gen/Seq/SLG	Transformer		✓						1	✓	✓		✓				
GenAD [96]	CVPR'24	Gen/Seq/TFS	Video Diffusion		✓	✓	✓		✓	✓	4	✓	✓				✓		
OccLLaMA [18]	arXiv'24	Dec/Seq/SLG	Transformer		✓			✓		✓	3	✓	✓				✓	✓	
DriveDreamer [97]	ECCV'24	Dec/Seq/SLG	GRU		✓						1	✓	✓	✓		✓	✓		
GenAD [98]	ECCV'24	Dec/Seq/SLG	GRU		✓						1	✓							
OccWorld [99]	ECCV'24	Dec/Seq/SLG	Transformer		✓			✓			2	✓	✓		✓			✓	
DOME [100]	arXiv'24	Gen/Seq/SLG	DiT		✓			✓			2	✓	✓					✓	
TOKEN [101]	CoRL'24	Dec/Glo/TFS	Transformer		✓					✓	2	✓	✓	✓			✓		
Vista [102]	NeurIPS'24	Gen/Seq/SLG	Video Diffusion		✓		✓		✓	✓	4	✓	✓				✓		
DriveDreamer-2 [103]	AAAI'25	Gen/Glo/SLG	Video Diffusion		✓						1	✓	✓	✓		✓	✓		
DTT [104]	arXiv'25	Dec/Seq/DRR	Transformer		✓			✓			2	✓	✓				✓	✓	
DynamicCity [105]	ICLR'25	Gen/Glo/SLG	DiT	✓	✓		✓	✓			4	✓	✓				✓	✓	
LidarDM [106]	ICRA'25	Gen/Seq/SLG	Diffusion				✓			✓	3		✓	✓					
FutureSightDrive [107]	arXiv'25	Dec/Seq/TFS	CoT(VLM)		✓					✓	3	✓	✓				✓		
GEM [108]	CVPR'25	Gen/Seq/SLG	Video Diffusion		✓				✓		1	✓	✓						✓
GaussianWorld [109]	CVPR'25	Gen/Seq/DRR	Transformer		✓					✓	2	✓	✓						
MaskGWM [110]	CVPR'25	Gen/Glo/TFS	DiT		✓		✓		✓		3	✓	✓				✓		
DriveDreamer4D [111]	CVPR'25	Gen/Glo/DRR	4DGS		✓	✓			✓	3	✓		✓	✓	✓	✓	✓		✓
ReconDreamer [112]	CVPR'25	Gen/Glo/DRR	3DGS		✓		✓			✓	3	✓	✓	✓	✓	✓			
WoTE [113]	ICCV'25	Dec/Seq/SLG	Transformer	✓		✓					2	✓	✓		✓				
HERMES [114]	ICCV'25	Gen/Glo/SLG	LLM		✓					✓	4	✓	✓				✓		
InfiniCube [22]	ICCV'25	Gen/Seq/DRR	3DGS				✓				1	✓	✓	✓		✓	✓		✓
DriVerse [115]	ACMMM'25	Gen/Seq/TFS	DiT		✓		✓				2	✓	✓				✓		✓
DINO-Foresight [116]	NeurIPS'25	Gen/Glo/TFS	Transformer		✓					✓	2	✓							✓

[1] **Taxonomy**: Abbreviations for the taxonomy categories defined in §III.
[2] **Characteristics**: Representative backbone or core technical approach.
[3] **Total**: Number of data platforms used. Underlined entries denote newly proposed or aggregated datasets.

ther enhance robustness, HRSSM [25] was proposed, featuring a dual-branch architecture that aligns latent observations and shares information without reconstruction. Beyond architectural refinements, DisWM [119] disentangles semantic knowledge from video content, distilling it into a world model that enables cross-domain generalization. LeWorldModel [81] extends this compact-latent line beyond RSSM-style objectives by jointly training a pixel encoder and action-conditioned Transformer predictor with a next-embedding loss and SIGReg anti-collapse regularization for reward-free latent planning.

A unifying theme across recent RSSM extensions is transferability, which reflects generalization across modalities, tasks, and embodiments for robust real-world robotics. At the representation level, PreLAR [52] learns implicit action abstractions to bridge video-pretrained representations and control fine-tuning. Similarly, Wang *et al.* [120] used optical flow as an embodiment-agnostic action representation to refine behavior-cloned policies, facilitating transfer across diverse embodiments. SENSEI [121] distilled a Vision-Language Model (VLM) to derive semantic rewards and employed an RSSM that learns to predict and propagate these rewards internally. Under limited supervision, SR-AIF [122] exploits prior preference learning and self-revision to enable adaptive learning in sparse-reward, continuous-control settings. To mitigate the Sim-to-Real (S2R) gap, ReDRAW [123] is pretrained in simulation and adapted to the real environment using a limited amount of reward-free data, applying residual corrections to the latent dynamics. To handle mismatches, AdaWM [124] identifies discrepancies between the learned dynamics and the planner and selectively fine-tunes critical components. Other methods like WMP [125] address S2R transfer for challenging tasks, and DayDreamer [43] demonstrated sample-efficient deployment on physical robots. To broaden transfer, FOUNDER [126] grounds representations from foundation models in the world-model state space, using temporal-distance prediction to handle flexible goals, and LUMOS [127] introduced a language-conditioned imitation framework that operates on-policy in latent space with intrinsic rewards, enabling zero-shot transfer to real-world robotics.

RSSM-based models have also been developed for autonomous driving. MILE [87] leverages offline expert data to enable imagined future states for planning. SEM2 [89] integrates semantic filtering with multi-source sampling to extract driving-relevant features and balance data distribution. Popov *et al.* [128] addressed covariate shift through a latent generative world model that realigns policies with expert states. For safety, VL-SAFE [129] supervises world models using safety scores derived from a VLM to generate safe trajectories. Finally, CALL [130] extended the RSSM framework to Multi-Agent RL by introducing ego-centric information sharing to enhance planning capabilities.

原文第 5 页 | 中文翻译初稿

5

表二

自动驾驶领域代表性 世界模型（World Models） 的摘要。

					数据集	平台			输入	pu	t模态	
论文	出版物	分类1	特征2	CARLA
nuScenes	nuPlan
Waymo	Occ3D
OpenDV
其他	Total3	RGB	Motion
M	ap	LiDAR
绑定框
语言	入住人数
其他
MILE [87]	NeurIPS’22	Dec/Seq/GLV	RSSM	✓			1	✓	✓
✓		
Copilot4D [88]	ICLR’24	Gen/Seq/SLG	视频扩散	✓		✓	3		✓		✓	
SEM2 [89]	TITS’24	Dec/Seq/GLV	RSSM	✓			1	✓	✓
✓	✓	
MagicDrive3D [90]	arXiv'24	Gen/Glo/DRR	3DGS	✓			1	✓	✓
✓	✓
✓	✓
OccSora [91]	arXiv'24	Gen/Glo/SLG	扩散	✓		✓	2		✓			✓
Delphi [92]	arXiv'24	Gen/Seq/SLG	视频扩散	✓			1	✓		✓	✓
✓	
DriveWorld [93]	CVPR’24	Dec/Seq/SLG	RSSM	✓		✓	2	✓	✓		✓	✓
Drive-WM [94]	CVPR’24	Dec/Glo/SLG	视频扩散	✓			1	✓	✓
✓	✓
✓	✓
ViDAR [95]	CVPR’24	Gen/Seq/SLG	变压器	✓			1	✓	✓		✓	
GenAD [96]	CVPR’24	Gen/Seq/TFS	视频扩散	✓	✓
✓	✓
✓	4	✓	✓		✓	
OccLLaMA [18]	arXiv’24	Dec/Seq/SLG	变压器	✓		✓
✓	3	✓	✓		✓	✓
DriveDreamer [97]	ECCV’24	Dec/Seq/SLG	GRU	✓			1	✓	✓
✓	✓
✓	
GenAD [98]	ECCV’24	Dec/Seq/SLG	GRU	✓			1	✓				

|OccWorld [99]|ECCV’24|Dec/Seq/SLG|变压器|✓||✓|2|✓|✓||✓|✓|
|DOME [100]|arXiv’24|Gen/Seq/SLG|DiT|✓||✓|2|✓|✓|||✓|
|代币 [101]|CoRL’24|Dec/Glo/TFS|变压器|✓||✓|2|✓|✓
|✓|✓||
|Vista [102]|NeurIPS’24|Gen/Seq/SLG|视频扩散|✓|✓|✓
✓|4|✓|✓||✓||
|DriveDreamer-2 [103]|AAAI’25|Gen/Glo/SLG|视频扩散|✓|||1|✓|✓
|✓|✓
✓||
|DTT [104]|arXiv’25|Dec/Seq/DRR|变压器|✓||✓|2|✓|✓||✓|✓|
|DynamicCity [105]|ICLR’25|Gen/Glo/SLG|DiT|✓
✓|✓|✓|4|✓|✓||✓|✓|
|LidarDM [106]|ICRA’25|Gen/Seq/SLG|扩散||✓|✓|3||✓
✓		
FutureSightDrive [107]	arXiv’25	Dec/Seq/TFS	CoT(VLM)	✓		✓	3	✓	✓		✓	
GEM [108]	CVPR’25	Gen/Seq/SLG	视频扩散	✓		✓	1	✓	✓			✓
GaussianWorld [109]	CVPR’25	Gen/Seq/DRR	Transformer	✓		✓	2	✓	✓			
MaskGWM [110]	CVPR’25	Gen/Glo/TFS	DiT	✓	✓	✓	3	✓	✓		✓	
DriveDreamer4D [111]	CVPR’25	Gen/Glo/DRR	4DGS	✓	✓	✓
3	✓		✓
✓	✓
✓
✓	✓
ReconDreamer [112]	CVPR’25	Gen/Glo/DRR	3DGS	✓	✓	✓	3	✓	✓
✓	✓
✓	
WoTE [113]	ICCV’25	Dec/Seq/SLG	变压器	✓	✓		2	✓	✓		✓	
HERMES [114]	ICCV’25	Gen/Glo/SLG	LLM	✓		✓	4	✓	✓		✓	
InfiniCube [22]	ICCV’25	Gen/Seq/DRR	3DGS		✓		1	✓	✓
✓	✓
✓	✓
DriVerse [115]	ACMMM’25	Gen/Seq/TFS	DiT	✓	✓		2	✓	✓		✓	✓
DINO-Foresight [116]	NeurIPS’25	Gen/Glo/TFS	变压器	✓		✓	2	✓				✓

1 分类法：第 III 节中定义的分类法类别的缩写。 2 特征：具有代表性的骨干或核心技术方法。 3 总计：使用的数据平台数量。带下划线的条目表示新提议或聚合的数据集。

为了增强鲁棒性，提出了 HRSSM [25]，其特点是双分支架构，可以对齐潜在观察并共享信息而无需重建。除了架构改进之外，DisWM [119] 将语义知识从视频内容中分离出来，将其提炼成 世界模型（World Model），从而实现跨域泛化

。 LeWorldModel [81] 通过联合训练像素编码器和动作条件 Transformer 预测器，使用下一个嵌入损失和 SIGReg 抗崩溃正则化来实现无奖励潜在规划，从而将这种紧凑潜在线扩展到 RSSM 式目标之外。

最近 RSSM 扩展的一个统一主题是可转移性，它反映了稳健的现实世界机器人技术的模式、任务和实施例的泛化。在表示层面，PreLAR [52] 学习隐式动作抽象来桥接视频预训练表示并控制微调。类似地，Wang_et al._[120]使用光流作为与实施例无关的动作表示来细化行为克隆策略，促进跨不同实施例的转移。SENSEI [121]提炼了视觉语言模型（VLM）来导出语义奖励，并采用 RSSM 来学习在内部预测和传播这些奖励。在有限的监督下，SR-AIF [122] 利用先验偏好学习和自我修正来实现稀疏奖励、连续控制设置中的自适应学习。为了缩小 仿真到现实（Sim-to-Real）(S2R) 差距，ReDRAW [123] 在模拟中进行了预训练并适应

使用有限数量的无奖励数据来模拟真实环境，对潜在动态应用剩余修正。为了处理不匹配，AdaWM [124] 识别学习动态和规划器之间的差异，并有选择地微调关键组件。其他方法如 WMP [125] 解决了具有挑战性的任务的 S2R 传输，而 DayDreamer [43] 则展示了在物理机器人上的样本高效部署。为了扩大迁移，FOUNDER [126] 将来自 基础模型（Foundation Models） 的表示建立在世界模型状态空间中，使用时间距离预测来处理灵活的目标，而 LUMOS [127] 引入了一种语言条件模仿框架，该框架在具有内在奖励的潜在空间中进行策略操作，从而实现了向现实世界机器人的零样本迁移。

基于 RSSM 的模型也被开发用于自动驾驶。 MILE [87] 利用离线专家数据来实现想象的未来状态进行规划。SEM2 [89]将语义过滤与多源采样相结合，以提取驾驶相关特征并平衡数据分布。 Popov et al. [128] 通过潜在的生成 世界模型（World Model） 解决了协变量转变，该生成 世界模型（World Model） 将策略与专家国家重新调整。为了安全起见，VL-SAFE [129] 使用从 VLM 导出的安全分数来监督 世界模型（World Models），以生成安全轨迹。最后，CALL [130]通过引入以自我为中心的信息共享来增强规划能力，将 RSSM 框架扩展到 MultiAgent RL。

In contrast to RSSM, TransDreamer [28] introduced a Transformer State-Space Model (TSSM) that replaces the recurrent core in Dreamer, thereby substantially enhancing its capacity to capture long-horizon dependencies. The complementary OSVI-WM [131] employs a causal Transformer for one-shot Imitation Learning (IL), autoregressively predicting the future latent trajectory and decoding it into physical waypoints for robot control.

Some approaches continue to employ RNNs to capture temporal dependencies. On the modeling side, RWM [132] introduced a dual-autoregressive, domain-agnostic neural simulator for long-horizon prediction. X-MOBILITY [133], in contrast, disentangles modeling from policy learning, using multi-head decoders for large-scale pretraining followed by supervised fine-tuning to derive strong policies. For humanoid locomotion, DWL [48] and WMR [134] adopted end-to-end (E2E) frameworks. These frameworks reconstruct states from partial observations using either denoising or a gradient-blocked state estimator, which enables zero-shot transfer across complex real-world terrains.

Recently, State Space Models (SSMs), exemplified by Mamba, have emerged as alternatives to RNNs and Transformers, combining linear-time complexity with long-horizon modeling capacity. Building on this, GLAM [57] improves both fidelity and efficiency via a Mamba-based parallel framework that integrates global and local modules to capture contextual and fine-grained dynamics.

Beyond forward temporal modeling, Inverse Dynamics Modeling (IDM) is a key paradigm in world model construction. An IDM infers the actions required to transition between an initial and a target state. Agrawal *et al.* [135] integrated a forward model with an IDM for multi-step prediction, establishing the basis for subsequent research. More recent work includes GLAMOR [39], which trains an object-conditioned IDM to predict the actions necessary to reach a specified target. In Dreamer-style agents, Iso-Dream [40] leverages IDM to decompose the world model into controllable and uncontrollable components, using the rollout of uncontrollable states to guide policy learning.

Token Feature Sequence. The Token Feature Sequence paradigm centers on modeling dependencies among discrete tokens. This representation supports causal inference, multimodal integration, and the reuse of Large Language Model (LLM).

Recent RSSM-centric studies have begun to exploit token-level dependencies to strengthen representation learning and temporal reasoning. For example, MWM [41] decouples visual tokens from RSSM-based dynamics via a masked autoencoder, improving both performance and data efficiency. NavMorph [136] introduced a self-evolving RSSM with a contextual evolution memory for online adaptation. For temporal abstraction, WISTER [137] employed action-conditioned contrastive predictive coding to train a TSSM that captures high-level temporal features. Similarly, TWM [29] used a Transformer to align multimodal tokens with historical states during training, while relying on a lightweight policy at inference. To handle long-horizon tasks, some approaches integrate LLMs with RSSMs to decompose objectives into subtasks. EvoAgent [138], for example, uses an LLM to guide low-level actions and regularizes RSSM updates. RoboHorizon [139], in contrast, enhances task recognition with dense rewards and leverages key task segments via a masked autoencoder.

In autonomous driving, token-based sequence representations are increasingly adopted to model cross-modal interactions and spatiotemporal structures. DrivingWorld [140] pairs next-state prediction for temporal dynamics with next-token prediction for spatial structure. For multimodal control, Doe-1 [141] formulates closed-loop driving as autoregressive prediction over perception-description-action tokens, which unifies perception, prediction, and planning, and DrivingGPT [23] interleaves vision and action tokens and casts world modeling and trajectory planning as next-token prediction. To enhance diversity and safety, LatentDriver [142] models future actions as a mixture distribution and actuates the world model with planner-sampled intermediate actions. At the same time, Vasudevan *et al.* [143] proposed an adaptive model that predicts surrounding agents for safe navigation.

The token-based paradigm also extends to broader robotics. Within RL, IRIS [44] and TWM [144] leverage discrete tokens to enable data-efficient policy learning via imagined or hybrid rollouts. DyWA [145] improves action learning by conditioning on trajectory dynamics and jointly predicting future states with single-view point-cloud and proprioceptive modalities. EgoAgent [146] interleaves state-action sequence modeling within a Transformer, enabling unified perception, prediction, and action inference. Tokenized representations unify multimodal inputs, including vision, language, and action (VLA), enabling generalist agents with cross-domain adaptability, as shown in WorldVLA [67]. DreamZero [82] further instantiates this paradigm as a World Action Model built on a pretrained video diffusion backbone, autoregressively predicting future video and actions from language, visual history, and proprioception to support real-time closed-loop robot control. Recent studies encode environmental states as discrete symbolic tokens and condition next-token prediction on action, as demonstrated by DCWM [147] and TrajWorld [148].

Recent studies have strengthened the link between tokenized representations and planning, particularly through object-centric approaches. These models, such as CarFormer [149], the work of Jeong *et al.* [150], and Dyn-O [69], represent scenes as a collection of slots. CarFormer autoregressively models the relationships between these slots in BEV. Jeong *et al.* added language-guided manipulation, while Dyn-O used a Mamba with dropout scheduling for temporal modeling and to disentangle static from dynamic elements. Δ-IRIS [151] introduced a hybrid Transformer that integrates tokens with stochastic Δ-tokens to capture dynamics. D^2PO [152] employed preference learning to jointly optimize state prediction and action selection, enhancing the model's understanding of underlying dynamics. For efficiency, MineWorld [59] accelerated token generation by predicting sequences in parallel and introduced an IDM as a controllability metric. Meanwhile, PIVOT-R [153] and ReOI [154] incorporated VLMs into control. PIVOT-R parses instructions to produce waypoint-based plans that an action module decodes into low-level controls, whereas ReOI detects

原文第 6 页 | 中文翻译初稿

6

与 RSSM 相比，TransDreamer [28] 引入了 Transformer 状态空间模型（TSSM），它取代了 Dreamer 中的循环核心，从而大大增强了其捕获 长时程（Long-Horizon） 依赖项的能力。互补的 OSVI-WM [131] 采用因果 Transformer 进行一次性 模仿学习（Imitation Learning）(IL)，自回归预测未来的潜在轨迹并将其解码为机器人控制的物理路径点。

一些方法继续使用 RNN 来捕获时间依赖性。在建模方面，RWM [132] 引入了用于 长时程（Long-Horizon）预测的双自回归、与领域无关的神经模拟器。相比之下，X-MOBILITY [133]将建模与策略学习分开，使用多头解码器进行大规模预训练，然后进行监督微调以得出强大的策略。对于人形运动，DWL [48]和 WMR [134]采用了端到端（E2E）框架。这些框架使用去噪或梯度阻塞状态估计器从部分观测中重建状态，从而实现跨复杂现实世界地形的零样本传输。

最近，以 Mamba 为代表的状态空间模型 (SSM) 已成为 RNN 和 Transformer 的替代品，将线性时间复杂度与 长时程（Long-Horizon） 建模能力相结合。在此基础上，GLAM [57] 通过基于 Mamba 的并行框架提高了保真度和效率，该框架集成了全局和本地模块以捕获上下文和细粒度的动态。

除了正向时间建模之外，逆动态建模 (IDM) 是 世界模型（World Model） 构造的关键范例。 IDM 推断在初始状态和目标状态之间转换所需的操作。 Agrawal等人[135]将前向模型与IDM集成以进行多步预测，为后续研究奠定了基础。最近的工作包括 GLAMOR [39]，它训练对象条件 IDM 来预测达到指定目标所需的操作。在Dreamer风格的智能体中，Iso-Dream [40]利用IDM将世界模型（World Model）分解为可控和不可控组件，使用不可控状态的展开轨迹（Rollout）来指导策略学习。

词元特征序列（Token Feature Sequence）。 词元特征序列（Token Feature Sequence） 范式以离散令牌之间的依赖关系建模为中心。这种表示支持因果推理、多模式集成和大型语言模型 (LLM) 的重用。

最近以 RSSM 为中心的研究已经开始利用标记级依赖关系来加强表示学习和时间推理。例如，MWM [41]通过屏蔽自动编码器将视觉标记与基于 RSSM 的动态解耦，从而提高性能和数据效率。 NavMorph [136]引入了一种自我进化的 RSSM，具有用于在线适应的上下文进化记忆。对于时间抽象，WISTER [137] 采用动作条件对比预测编码来训练捕获高级时间特征的 TSSM。类似地，TWM [29] 在训练期间使用 Transformer 将多模态标记与历史状态对齐，同时在推理时依赖轻量级策略。为了处理 长时程（Long-Horizon） 任务，一些方法集成了

法学硕士与 RSSM 将目标分解为子任务

。例如，EvoAgent [138] 使用 LLM 来指导低级操作并规范 RSSM 更新。相比之下，RoboHorizon [139]通过密集奖励增强任务识别，并通过屏蔽自动编码器利用关键任务片段。

在自动驾驶中，越来越多地采用基于令牌的序列表示来建模跨模式交互和时空结构。 DrivingWorld [140]将时间动态的下一状态预测与空间结构的下一标记预测配对。对于多模态控制，Doe-1 [141] 将闭环驾驶制定为感知-描述-动作标记的自回归预测，从而统一了感知、预测和规划，而 DrivingGPT [23] 交错了视觉和动作标记，并将世界建模和轨迹规划作为下一个标记预测。为了增强多样性和安全性，LatentDriver [142] 将未来动作建模为混合分布，并使用规划器采样的中间动作来驱动 世界模型（World Model）。与此同时，Vasudevan 等人[143]提出了一种自适应模型，可以预测周围的智能体以实现安全导航。

基于代币的范例还扩展到更广泛的机器人技术。在 RL 中，IRIS [44] 和 TWM [144] 利用离散代币通过想象或混合 展开轨迹（Rollouts） 实现数据高效的策略学习。 DyWA [145] 通过调节轨迹动态并使用单视点云和本体感受模式联合预测未来状态来改进动作学习。 EgoAgent [146] 在 Transformer 中交错状态-动作序列建模，从而实现统一的感知、预测和动作推理。标记化表示统一了多模态输入，包括视觉、语言和动作（VLA），使通才智能体具有跨域适应性，如 WorldVLA [67] 所示。 DreamZero [82] 进一步将这种范例实例化为建立在预训练视频扩散主干上的世界动作模型，从语言、视觉历史和本体感觉中自回归预测未来的视频和动作，以支持实时闭环机器人控制。最近的研究将环境状态编码为离散的符号标记，并条件下一个标记的动作预测，如 DCWM [147] 和 TrajWorld [148] 所证明的那样。

最近的研究加强了标记化表示和规划之间的联系，特别是通过以对象为中心的方法。这些模型，例如 CarFormer [149]、Jeong et al. [150] 和 Dyn-O [69] 的工作，将场景表示为槽的集合。 CarFormer 对 BEV 中这些插槽之间的关系进行自回归建模。 Jeong 等人添加了语言引导的操作，而 Dyn-O 使用带有 dropout 调度的 Mamba 进行时间建模，并将静态元素与动态元素分开。 Δ-IRIS [151] 引入了一种混合 Transformer，它将令牌与随机 Δ- 令牌集成以捕获动态。 D2 PO [152] 采用偏好学习来联合优化状态预测和动作选择，增强模型对潜在动态的理解。为了提高效率，MineWorld [59] 通过并行预测序列来加速令牌生成，并引入 IDM 作为可控性度量。同时，PIVOT-R [153] 和 ReOI [154] 将 VLM 纳入控制中

。 PIVOT-R 解析指令以生成基于航路点的计划，操作模块将其解码为低级控制，而 ReOI 则检测

implausible prediction elements, reimagines the distractors, and reintegrates the corrected content.

Based on tokenization, some studies employ autoregressive diffusion to achieve stable generation and long-horizon planning. Epona [155] decouples spatiotemporal modeling, handled by a Transformer, from long-horizon multimodal generation, realized through trajectory and vision Diffusion Transformers (DiTs). Goff *et al.* [156] used a DiT to instantiate the state transition, which enables on-policy training and multi-second closed-loop rollouts. SceneDiffuser++ [157] pushes this further to city-scale traffic simulation, applying multi-tensor diffusion over agents and traffic lights to produce stable closed-loop rollouts. For navigation, NWM [66] introduced an efficient conditional DiT to simulate visual trajectories for zero-shot planning.

Another emerging direction is to inject explicit reasoning into the world model using LLMs and Chain-of-Thought (CoT). NavCoT [58] decomposes navigation into imagination, filtering, and prediction, enabling parameter-efficient in-domain training, and ECoT [54] leverages a pipeline of foundation models to generate reasoning labels for training a VLA policy. Variants like MineDreamer [79] introduced Chain-of-Imagination (CoI), where a multimodal LLM imagines future observations to steer diffusion and guide actions, and FSDrive [107] generates physics-constrained future scenes and treats them as CoT supervision, enabling VLMs to function as IDMs for planning.

Other approaches directly couple LLMs with world models to operationalize planning and data generation. Dyna-Think [158] fuses reasoning and acting via a distilled LLM, and RIG [159] unifies reasoning and imagination end-to-end generalist policy. In terms of explicit dynamics and long-horizon, Gkountouras *et al.* [160] trained a causal world-model simulator that grounds an LLM nvironment causal reasoning and planning skills, Statler [46] enables LLMs to keep a structured world-state, using a reader for planning and a writer for updating, and Inner Monologue [42] incorporates a closed-loop feedback into LLMs, enabling agents to reason and deliberate more akin to human thinking. Finally, WoMAP [161] synthesized 3DGS scenes and trained a world model that refines VLM instructions for precise execution.

Spatial Latent Grid. By encoding features on geometry-aligned grids or incorporating explicit spatial priors, this paradigm preserves locality, enables efficient convolutional or attention-based updates and streaming rollouts.

In autonomous driving, many studies couple RNN-based dynamics with spatial grids to guide planning. For instance, DriveDreamer [97] and GenAD [98] adopt GRU-based dynamics over grid or instance-centric tokens to predict motion and decode trajectories. In contrast, DriveWorld [93] and Raw2Drive [162] instantiate RSSM dynamics on BEV tokens. DriveWorld conditions on tokens and actions for joint prediction, while Raw2Drive adopts a dual-stream design for spatiotemporal learning.

Numerous studies focus on autoregressively predicting future 3D occupancy representations to enable motion planning for autonomous driving. One strand discretizes scenes into occupancy tokens for sequential prediction, exemplified by OccWorld [99] and RenderWorld [163]. Another strand directly forecasts volumetric features or embeddings, as in Drive-OccWorld [164] and PreWorld [165]. Self-supervised variants predict future representations from current cues. For example, LAW [166] conditions on current representations and trajectories, SSR [167] compresses scenes into sparse BEV tokens for future BEV features, and NeMo [168] voxelizes multi-frame images and predicts occupancy to support imitation-based planning. Building on these representations, FASTopoWM [169] employs a unified decoder to align fast and slow systems from vehicle poses, enabling lane-topology reasoning, and WoTE [113] simulates candidate trajectories in BEV and selects among them with a reward model. Extending the paradigm, OccLLaMA [18] unifies occupancy, actions, and text within a single token vocabulary and employs a LLaMA for next token forecasting, planning, and question answering.

Beyond autonomous driving, similar formulations have been extended to broader domains of robotics. WMNav [170] leverages a VLM to maintain a curiosity-driven value map and adopts phased decision making to enable zero-shot, object-driven navigation. RoboOccWorld [171] targets indoor robotics by predicting fine-grained 3D occupancy with a pose-conditioned autoregressive Transformer, thereby supporting exploration and decision making. To achieve high-fidelity dynamics, EnerVerse [34] applies chunk-wise autoregressive video diffusion with a sparse memory mechanism to produce 4D latent dynamics and integrates 4DGS to mitigate the S2R gap in robotic execution. For manipulation, ParticleFormer [172] forecasts future point clouds with a Transformer-based particletized dynamics model, enabling robust handling of multi-object and multi-material interactions. Scaling this point-flow view, PointWorld [83] unifies scene states and robot actions as 3D point flows, using a point-cloud Transformer backbone to predict action-conditioned full-scene displacements for MPC-based real-world manipulation from RGB-D observations. At the representation level, DINO-WM [70] learns dynamics in the DINOv2 feature space and predicts future states to support zero-shot planning.

Decomposed Rendering Representation. This paradigm represents scenes with explicit renderable primitives such as NeRFs and 3DGS, updating them to simulate dynamics and render future observations. It provides view-consistent forecasts, object-level compositionality, and seamless integration with physics priors and digital twins, thereby supporting long-horizon rollouts.

Building on 3DGS, GAF [74] augments splats with learnable motion attributes to forecast future states and refines initial actions with diffusion. ManiGaussian [53] predicts per-point variations to generate future Gaussian scenes for manipulation under current states and actions, and ManiGaussian++ [80] adds a hierarchical leader-follower design with task-oriented splats to model primitive deformations for multi-body and bimanual skills. Within simulation and digital twin coupling, DreMa [60] integrates GS with a physics simulator to build twins for data synthesis in imitation learning, Abou-Chakra *et al.* [173] introduced a dual Gaussian-Particle representation where Gaussian points are attached to particles driven by visual loss

原文第 7 页 | 中文翻译初稿

7

难以置信的预测元素，重新想象干扰因素，并重新整合纠正的内容。

基于标记化，一些研究采用自回归扩散来实现稳定生成和长时程（Long-Horizon）规划。 Epona [155] 将由 Transformer 处理的时空建模与 长时程（Long-Horizon） 多模态生成解耦，通过轨迹和视觉扩散变压器 (DiTs) 实现。 Goff 等人[156]使用 DiT 来实例化状态转换，从而实现同策略训练和多秒闭环 展开轨迹（Rollouts）。 SceneDiffuser++ [157] 将其进一步推向城市规模的交通模拟，在智能体和交通灯上应用多张量扩散以产生稳定的闭环 展开轨迹（Rollouts）。对于导航，NWM [66]引入了一种有效的条件 DiT 来模拟零样本规划的视觉轨迹。

另一个新兴方向是使用 LLM 和思想链 (CoT) 将显式推理注入 世界模型（World Model）。 NavCoT [58] 将导航分解为想象、过滤和预测，从而实现参数高效的域内训练，ECoT [54] 利用 基础模型（Foundation Models） 的管道来生成用于训练 VLA 策略的推理标签。像 MineDreamer [79] 这样的变体引入了想象链 (CoI)，其中多模式 LLM 想象未来的观察结果来引导扩散和指导行动，而 FSDrive [107] 生成物理约束的未来场景并将其视为 CoT 监督，使 VLM 能够充当 IDM 进行规划。

其他方法直接将法学硕士与 世界模型（World Models） 结合起来以实施规划和数据生成。 Dyna-Think [158] 通过精炼的法学硕士融合推理和行动，RIG [159] 统一推理和想象力端到端的通才策略。在显式动力学和 长时程（Long-Horizon） 方面，Gkountouras et al. [160] 训练了一个因果世界模型模拟器，该模拟器为 LLM 环境因果推理和规划技能奠定了基础，Statler [46] 使 LLM 能够保持结构化的世界状态，使用读者进行规划，使用作者进行更新，而 Inner Monologue [42] 将闭环反馈纳入 LLM 中，使智能体能够更类似于人类思维地进行推理和深思熟虑。最后，WoMAP [161] 合成了 3DGS 场景并训练了 世界模型（World Model），该 世界模型（World Model） 可以细化 VLM 指令以实现精确执行。

空间潜网格（Spatial Latent Grid）。通过对几何对齐网格上的特征进行编码或合并显式空间先验，该范例保留了局部性，实现了高效的卷积或基于注意力的更新和流式传输 展开轨迹（Rollouts）。

在自动驾驶中，许多研究将基于 RNN 的动力学与空间网格结合起来来指导规划。例如，DriveDreamer [97] 和 GenAD [98] 采用基于 GRU 的网格动力学或以实例为中心的令牌来预测运动和解码轨迹。相比之下，DriveWorld [93] 和 Raw2Drive [162] 在 BEV 代币上实例化 RSSM 动态。 DriveWorld 以 token 和 actions 为条件进行联合预测，而 Raw2Drive 采用双流设计进行时空学习。

许多研究重点关注自回归预测未来 3D 占用表示，以实现自动驾驶的运动规划。一股将场景离散化为占用标记以进行顺序预测，例如

OccWorld [99] 和 RenderWorld [163]。另一条链直接预测体积特征或嵌入，如 Drive-OccWorld [164] 和 PreWorld [165] 中所示。自我监督的变体根据当前线索预测未来的表征。例如，LAW [166] 以当前表示和轨迹为条件，SSR [167] 将场景压缩为稀疏 BEV 标记以用于未来的 BEV 特征，NeMo [168] 对多帧图像进行体素化并预测占用率以支持基于模仿的规划。基于这些表示，FASTopoWM [169] 采用统一的解码器根据车辆姿态来对齐快速和慢速系统，从而实现车道拓扑推理，而 WoTE [113] 模拟 BEV 中的候选轨迹并使用奖励模型在其中进行选择。 OccLLaMA [18] 扩展了该范式，将占用、操作和文本统一在一个令牌词汇表中，并采用 LLaMA 进行下一个令牌预测、规划和问题回答。

除了自动驾驶之外，类似的公式已扩展到更广泛的机器人领域。 WMNav [170] 利用 VLM 来维护好奇心驱动的价值图，并采用分阶段决策来实现零样本、对象驱动的导航。 RoboOccWorld [171] 通过使用姿势条

件自回归 Transformer 预测细粒度的 3D 占用情况来瞄准室内机器人，从而支持探索和决策。为了实现高保真动态，EnerVerse [34] 应用具有稀疏内存机制的块式自回归视频扩散来产生 4D 潜在动态，并集成 4DGS 以减轻机器人执行中的 S2R 差距。对于操纵，ParticleFormer [172] 使用基于 Transformer 的粒子化动力学模型预测未来的点云，从而能够稳健地处理多对象和多材料相互作用。通过缩放这个点流视图，PointWorld [83] 将场景状态和机器人动作统一为 3D 点流，使用点云 Transformer 主干来预测动作条件下的全场景位移，以便根据 RGB-D 观测进行基于 MPC 的现实世界操作。在表示级别，DINO-WM [70] 学习 DINOv2 特征空间中的动态并预测未来状态以支持零样本规划。

分解式渲染表示（Decomposed Rendering Representation）。该范例表示具有显式可渲染图元（例如 NeRF 和 3DGS）的场景，更新它们以模拟动态并渲染未来的观察结果。它提供视图一致的预测、对象级组合性以及与物理先验和数字孪生的无缝集成，从而支持长期 展开轨迹（Rollouts）。

GAF [74] 在 3DGS 的基础上，通过可学习的运动属性增强了 splats，以预测未来状态并通过扩散细化初始动作。 ManiGaussian [53] 预测每点变化，以生成未来的高斯场景，以便在当前状态和动作下进行操作，ManiGaussian++ [80] 添加了分层领导者-跟随者设计和面向任务的 splats，以对多体和双手技能的原始变形进行建模

。在模拟和数字孪生耦合中，DreMa [60] 将 GS 与物理模拟器集成，以构建孪生用于 模仿学习（Imitation Learning） 中的数据合成，Abou-Chakra et al. [173] 引入了双高斯粒子表示，其中高斯点附加到由视觉损失驱动的粒子

forces, DexSim2Real^2 [174] builds twins of articulated objects with generative models and uses sampling based planning for precise manipulation, PIN-WM [175] combines 3DGS with differentiable physics to estimate physical parameters from limited observations and generates digital cousins for zero-shot S2R policy learning, and PWTF [176] constructs an interactive twin that simulates candidate action outcomes and uses VLMs for evaluation and selection. At the representation level, DTT [104] adopts a triplane representation with multiscale Transformers to autoregressively capture incremental changes, forming a 4D world model for prediction and planning.

2) Global Difference Prediction: **Token Feature Sequence.** The compact Global Latent Vector representation discards fine-grained spatiotemporal detail and is therefore rarely used for global prediction. In contrast, Token Feature Sequences predict the future sequence in parallel, reducing error accumulation while enabling multimodal diversity.

On the representation side, TOKEN [101] tokenizes scenes into object-level tokens, aligning world representations with reasoning and leveraging LLMs to predict full future trajectories for long-tail scenarios. GeoDrive [177] extracts 3D representations, renders trajectory-conditioned views, and edits the position of vehicles to guide DiT in producing editable generations. For control, FLARE [178] aligns diffusion policies with latent future representations, avoiding pixel-space video generation and learning effectively from action-free videos. In a related vein, LaDi-WM [179] predicts future states through interactive diffusion in a latent space aligned with visual foundation models, integrating geometric and semantic features while iteratively refining the diffusion policy to improve performance and generalization. Cosmos Policy [84] adapts Cosmos-Predict2 into a unified visuomotor policy by injecting proprioception, action chunks, future observations, and values as latent frames, allowing one video diffusion model to serve as policy, world model, and value function for model-based planning. villa-X [76] and VidMan [55] both couple diffusion-based models with IDM for control. villa-X infers latent actions, aligns them with ego-centric forward dynamics, and maps them via joint diffusion, while VidMan adapts a pretrained video-diffusion model into an IDM using a self-attention adapter for accurate action prediction.

Spatial Latent Grid. Spatial grid models parallel-forecast BEV or voxel maps from ego-stabilized views, preserving locality and uncertainty while producing planner-ready maps for fast control.

Diffusion-based world models are commonly used for parallel generation. EmbodiedDreamer [180] couples differentiable physics with video diffusion to render photorealistic and physically consistent futures. TesserAct [78] reconstructs 4D spatiotemporal consistent scenes by jointly generating RGB, depth, and normal videos for IDM-based action learning. DFIT-OccWorld [181] reformulates prediction as decoupled voxel warping and adopts image-assisted single-stage training for reliable and efficient dynamic scene modeling. For instruction-conditioned control, RoboDreamer [49] decomposes instructions into low-level primitives that steer video diffusion, synthesizing novel compositional scenes beyond the training distribution while grounding execution via an IDM, and ManipDreamer [182] extends this design with an action-tree prior together with depth and semantic guidance to improve instruction following and temporal consistency.

On the planning side, 3DFlowAction [183] employs a pretrained 3D optical flow world model to treat future motion as a unified action cue, enabling label-free and cross-robot manipulation through closed-loop optimization. Imagine-2-Drive [184] integrates video diffusion with a multimodal diffusion policy to accelerate policy learning. Drive-WM [94] uses multi-view diffusion with image-based rewards to select safer trajectories, while World4Drive [185] leverages vision-based priors to construct intent-aware world models that support self-supervised multi-intent imagination. COMBO [186] composes multi-agent actions with diffusion, leverages VLMs to infer purposes, and integrates tree search for online cooperative planning.

B. General-Purpose World Models

1) Sequential Simulation and Inference: **Token Feature Sequence.** General-purpose models pretrain task-agnostic dynamics to capture environmental physics and generate future scenes, prioritizing transferability over specific tasks.

Some general world models increasingly pretrain on unlabeled video and use tokenized latent space for robust forecasting and generation. iVideoGPT [56] was pretrained on large-scale interaction videos for action-free forecasting and later adapted to downstream control. Genie [50] learned discrete latent actions and spatiotemporal tokens, enabling user-controllable interactive environments via autoregressive dynamics. RoboScape [187] jointly learned video generation with temporal depth and keypoint dynamics to improve physical realism. PACT [188] tokenized multimodal perception and action and trained a causal Transformer to obtain a unified representation for diverse tasks, and DINO-world [189] learns generalizable dynamics by predicting the temporal evolution of DINOv2 features from large-scale unlabeled video corpora. Building on language priors, EVA [71] introduced a Reflection-of-Generation (RoG) policy that used a VLM for iterative self-correction, strengthening long-horizon anticipation. In the same vein, Owl-1 [190] employs a VLM to forecast world dynamics conditioned on current states and generated fragments, explicitly guiding the subsequent fragments and enabling coherent long-horizon video synthesis, while World4Omni [68] employs a reflective world model, where a VLM refines subgoal images from an image generator, and integrates them with pretrained modules for zero-shot robotic manipulation.

Recent work adapts video diffusion models into controllable world models that autoregressively imagine future scenes. AdaWorld [72] introduced an action-aware pretraining scheme that extracted self-supervised latent actions between adjacent frames to condition diffusion, enabling efficient transfer with minimal interaction. Vid2World [191] adapts pretrained video diffusion models into autoregressive interactive world models via causalization and a causal action-guidance mechanism. GenAD [96] employs a two-stage strategy to adapt diffusion into a general video-prediction model conditioned on text and

原文第 8 页 | 中文翻译初稿

8

DexSim2Real2 [174] 使用生成模型构建铰接对象的双胞胎，并使用基于采样的规划进行精确操作，PIN-WM [175] 将 3DGS 与可微分物理相结合，从有限的观察中估计物理参数，并生成用于零样本 S2R 策略学习的数字表兄弟，PWTF [176] 构建一个交互式双胞胎，模拟候选动作结果并使用 VLM 进行评估和评估选择。在表示层面，DTT[104]采用具有多尺度Transformers的三平面表示来自回归捕获增量变化，形成 4D 世界模型（World Model）用于预测和规划。

2) 全局差异预测（Global Difference Prediction）: 词元特征序列（Token Feature Sequence）。紧凑的全局潜向量（Global Latent Vector） 表示丢弃细粒度的时空细节，因此很少用于全局预测。相比之下，令牌特征序列并行预测未来序列，减少错误积累，同时实现多模态多样性。

在表示方面，TOKEN [101] 将场景标记为对象级标记，将世界表示与推理保持一致，并利用 LLM 来预测长尾场景的完整未来轨迹。 GeoDrive [177] 提取 3D 表示，渲染轨迹条件视图，并编辑车辆的位置以指导 DiT 生成可编辑的生成。为了进行控制，FLARE [178]将扩散策略与潜在的未来表示相结合，避免像素空间视频生成并从无动作视频中有效学习。与此相关，LaDi-WM [179] 通过与视觉 基础模型（Foundation Models） 对齐的潜在空间中的交互扩散来预测未来状态，集成几何和语义特征，同时迭代地细化 扩散策略（Diffusion Policy） 以提高性能和泛化能力。 Cosmos Policy [84] 通过注入本体感觉、动作块、未来观察和值作为潜在框架，将 Cosmos-Predict2 调整为统一的视觉运动策略，允许一个视频扩散模型充当策略、世界模型（World Model） 和基于模型的规划的价值函数。 Villa-X [76] 和 VidMan [55] 都将基于扩散的模型与 IDM 结合起来进行控制。 Villa-X 推断潜在动作，将它们与以自我为中心的前向动态对齐，并通过联合扩散映射它们，而 VidMan 使用自注意力适配器将预训练的视频扩散模型调整到 IDM 中，以实现准确的动作预测。

空间潜网格（Spatial Latent Grid）。空间网格根据自我稳定视图对并行预测 BEV 或体素图进行建模，保留局部性和不确定性，同时生成可供规划者使用的地图以进行快速控制。

基于扩散的 世界模型（World Models） 通常用于并行生成。 EmbodiedDreamer [180] 将可微分物理与视频扩散结合起来，以渲染逼真且物理一致的未来。 TesserAct [78] 通过联合生成 RGB、深度和法线视频来重建 4D 时空一致场景，用于基于 IDM 的动作学习。 DFIT-OccWorld [181] 将预测重新表述为解耦体素扭曲，并采用图像辅助单阶段训练来实现可靠且高效的动态场景建模

。对于指令条件控制，RoboDreamer [49] 将指令分解为引导视频扩散的低级原语，合成超出训练的新颖的合成场景

分布，而 落地关联（Grounding） 通过 IDM 执行，ManipDreamer [182] 使用动作树先验以及深度和语义指导来扩展此设计，以提高指令跟踪和时间一致性。

在规划方面，3DFlowAction [183] 采用预训练的 3D 光流 世界模型（World Model） 将未来运动视为统一的动作线索，通过闭环优化实现无标签和跨机器人操作。 Imagine-2Drive [184] 将视频扩散与多模态 扩散策略（Diffusion Policy） 集成以加速策略学习。 Drive-WM [94] 使用多视图扩散和基于图像的奖励来选择更安全的轨迹，而 World4Drive [185] 利用基于视觉的先验来构建支持自我监督的多意图想象力的意图感知 世界模型（World Models）。 COMBO [186] 通过扩散组合多智能体动作，利用 VLM 来推断目的，并集成树搜索以进行在线合作规划。

B。 通用（General-Purpose） 世界模型（World Models）

1) 序列仿真与推断（Sequential Simulation and Inference）: 词元特征序列（Token Feature Sequence）。通用（General-Purpose） 对与任务无关的预训练动力学进行建模，以捕获环境物理并生成未来场景，将可转移性优先于特定任务。

一些通用的 世界模型（World Models） 越来越多地对未标记的视频进行预训练，并使用标记化的潜在空间进行稳健的预测和生成。 iVideoGPT [56] 在大规模交互视频上进行了预训练，以进行无动作预测，然后适应下游控制。 Genie [50] 学习了离散的潜在动作和时空标记，通过自回归动力学实现用户可控的交互环境。 RoboScape [187] 联合学习具有时间深度和关键点动态的视频生成，以提高物理真实感。 PACT [188] 标记化多模态感知和动作，并训练因果 Transformer 以获得不同任务的统一表示，而 DINO-world [189] 通过从大规模未标记视频语料库中预测 DINOv2 特征的时间演化来学习可概括的动态。基于语言先验，EVA [71] 引入了生成反射 (RoG) 策略，该策略使用 VLM 进行迭代自我修正，增强了 长时程（Long-Horizon） 的预期。同样，Owl-1 [190] 采用 VLM 来预测以当前状态和生成片段为条件的世界动态，明确指导后续片段并实现连贯的 长时程（Long-Horizon） 视频合成，而 World4Omni [68] 采用反射型 世界模型（World Model），其中 VLM 从图像生成器中细化子目标图像，并将其与预训练模块集成以进行零镜头机器人操作。

最近的工作将视频扩散模型改编成可控制的 世界模型（World Models），它可以自回归地想象未来的场景。AdaWorld [72] 引入了一种动作感知预训练方案，该方案提取相邻帧之间的自监督潜在动作以进行条件扩散，从而以最少的交互实现高效传输

。 Vid2World [191] 通过因果关系和因果动作指导机制将预训练的视频扩散模型改编为自回归交互式 世界模型（World Models）。 GenAD [96]采用两阶段策略将扩散适应于以文本和文本为条件的通用视频预测模型

actions, enabling large-scale driving simulation and planning. Pandora [47] uses an instruction-tuned LLM to autoregressively steer a separate video-diffusion generator for explicit, goal-directed control, while Yume [75] quantized camera motions into text tokens to guide a masked Video DiT, enabling autoregressive synthesis of dynamic 3D exploratory worlds.

To maintain geometric fidelity and long-horizon stability, recent methods couple explicit 3D priors with temporal-consistency modules in diffusion-based world models. At the geometric level, Geometry Forcing [192] aligns latent features with a geometric foundation model to inject explicit 3D priors, improving geometric consistency, while DeepVerse [64] integrates visual and geometric prediction targets and introduces a geometry-aware memory to sustain consistent long-horizon generation. For temporal stability, VRAG [193] proposed a Video Retrieval-Augmented Generation (RAG) framework that retrieves historical frames conditioned on a global state to stabilize autoregressive rollouts, StateSpaceDiffuser [63] combines Mamba with diffusion to alleviate long-term memory loss and content drift under short context windows, and Infinity-Drive [194] injects memory and adopts an adaptive loss within DiT, producing minute-scale driving videos with high fidelity, temporal consistency, and diverse content. Complementing these designs, LongDWM [195] mitigates error accumulation in long-horizon video generation through distillation—where a fine-grained DiT learns continuous motion to guide a coarse model, whereas MiLA [196] adopts a coarse-to-fine strategy that predicts sparse anchor frames and refines them during interpolation to improve temporal consistency and long-term fidelity. Finally, for dynamics and conditioning, Orbis [197] employs a continuous-space flow-matching formulation that demonstrates greater robustness than discrete-token schemes for long-horizon rollouts, and DriVerse [115] leverages multimodal trajectory prompts with latent motion alignment to synthesize long-horizon driving videos from a single image and navigation trajectories.

Sequential world models increasingly act as learned simulators, providing action-conditioned rollouts for policy evaluation and training. WorldGym [198] and WorldEval [199] generate action-conditioned rollouts and use VLM-based critics for evaluation, while WorldEval further leverages latent action representations to drive a DiT-based synthesizer. RLVR-World [62] fine-tunes world models with RL with Verifiable Rewards (RLVR), using explicit metrics to close the pretraining-task objective gap. For safety risk prediction, Guan *et al.* [200] presented a framework for autonomous driving accident prediction that augments data with a domain-informed world model and enhances spatiotemporal reasoning using graph and temporal convolutions.

Beyond diffusion, sequence models broadened the capacity for long-range consistency. Po *et al.* [201] integrated block-wise state space models for long-term memory with local attention for short-term coherence, enabling video generation with sustained memory and consistent dynamics. S2-SSM [61] employs a Mamba layer to model the independent evolution of object slots and a sparsity-regularized cross-attention mechanism to capture their causal interactions, enabling causal reasoning over the environment.

Spatial Latent Grid. Pretraining geometry-aligned spatial maps with self-supervised spatiotemporal objectives, the spatial latent grid paradigm preserves locality and enables efficient rollouts, multimodal fusion, and transferable planner-ready maps.

Building on this paradigm, structured-grid and physics-informed methods encode geometry and dynamics for controllable rollouts. PhyDNet [202] disentangles physical priors expressed as partial differential equations from visual factors, improving prediction. ViDAR [95] unifies semantics, geometry, and dynamics through a pre-training task of point cloud forecasting and a latent rendering operator, enabling a scalable foundation for downstream autonomous driving tasks. FOLIAGE [203] models dynamics with an Accretive Graph Network and a Transformer-based predictor, executing rollouts on simulated data. Complementing these grid and physics lines, MindJourney [73] couples VLMs with a controllable world model to render egocentric rollouts along planned camera trajectories, enabling multi-view reasoning.

Building on grid-based representations, diffusion-based forecasting has become dominant for stable long-horizon generation. Within grid-centric predictors, DOME [100] encodes observations into a continuous latent space and applies a spatiotemporal DiT for scene forecasting, Copilot4D [88] tokenizes point clouds and couples a spatiotemporal Transformer with discrete diffusion to improve fidelity and coherence, and LidarDM [106] generates layout-conditioned static scenes, composes them with dynamic objects, integrating LiDAR simulation to produce controllable videos. For long-video generation, Vista [102] adopts a two-stage large-scale training regime to produce controllable, high-fidelity, driving videos, and Delphi [92] enforces long-horizon multiview consistency through shared noise and feature alignment and a failure-driven framework to synthesize targeted data for improved planning. To strengthen long-horizon stability, GEM [108] achieves controllable ego-vision generation through large-scale training and fine-grained control over motion, dynamics, and posture, Zhou *et al.* [204] maintains a persistent RGB-D 3D memory map to guide subsequent frames, and STAGE [205] introduced hierarchical temporal feature transfer with multi-stage training.

Decomposed Rendering Representation. Scenes are decomposed into explicit primitives to synthesize view-consistent, simulatable trajectories over long horizons. Within this paradigm, GaussianWorld [109] models scene evolution as ego-motion, object dynamics, and newly observed regions, iteratively updating 3D Gaussian primitives to enable accurate and efficient dynamic perception. InfiniCube [22] introduced a hybrid pipeline that combines voxel-based generation, video synthesis, and dynamic Gaussian reconstruction, enabling large-scale dynamic 3D driving scenes conditioned on HDmaps, bounding boxes, and text. Complementarily, Wu *et al.* [206] augmented a video world model with long-term spatial memory grounded in reconstructed geometry and an episodic memory, which together condition sequential generation for long-range consistency.

2) Global Difference Prediction: **Token Feature Sequence.** For general-purpose world models, tokenized feature sequences

原文第 9 页 | 中文翻译初稿

9

行动，实现大规模驾驶模拟和规划。 Pandora [47] 使用指令调整的 LLM 来自回归引导单独的视频扩散生成器，以实现明确的目标定向控制，而 Yume [75] 将相机运动量化为文本标记以引导屏蔽视频 DiT，从而实现动态 3D 探索世界的自回归合成。

为了保持几何保真度和 长时程（Long-Horizon） 稳定性，最近的方法将显式 3D 先验与基于扩散的 世界模型（World Models） 中的时间一致性模块结合起来。在几何层面，Geometry Forcing [192] 将潜在特征与几何 基础模型（Foundation Model） 对齐，以注入显式 3D 先验，提高几何一致性，而 DeepVerse [64] 集成视觉和几何预测目标，并引入几何感知内存来维持一致的 长时程（Long-Horizon） 生成。为了实现时间稳定性，VRAG [193] 提出了一种视频检索增强生成（RAG）框架，该框架检索以全局状态为条件的历史帧以稳定自回归 展开轨迹（Rollouts），StateSpaceDiffuser [63] 将 Mamba 与扩散相结合，以减轻短上下文窗口下的长期记忆丢失和内容漂移，InfinityDrive [194] 注入内存并在 DiT 中采用自适应损失，生成分钟级驾驶视频高保真度、时间一致性和内容多样化。作为对这些设计的补充，LongDWM [195] 通过蒸馏减轻了 长时程（Long-Horizon） 视频生成中的错误累积，其中细粒度 DiT 学习连续运动来指导粗略模型，而 MiLA [196] 采用从粗到细的策略来预测稀疏锚帧并在插值期间对其进行细化，以提高时间一致性和长期保真度。最后，对于动力学和调节，Orbis [197] 采用连续空间流匹配公式，该公式比 长时程（Long-Horizon） 展开轨迹（Rollouts） 的离散令牌方案表现出更强的鲁棒性，而 DriVerse [115] 利用多模态轨迹提示和潜在运动对齐来从单个图像和导航轨迹合成 长时程（Long-Horizon） 驾驶视频。

顺序 世界模型（World Models） 越来越多地充当学习模拟器，为策略评估和训练提供动作条件 展开轨迹（Rollouts）。 WorldGym [198] 和 WorldEval [199] 生成动作条件 展开轨迹（Rollouts） 并使用基于 VLM 的批评者进行评估，而 WorldEval 进一步利用潜在动作表示来驱动基于 DiT 的合成器。 RLVR-World [62] 使用可验证奖励的 RL（RLVR）对 世界模型（World Models） 进行微调，使用显式指标来缩小预训练任务目标差距。对于安全风险预测，Guan 等人[200]提出了一种自动驾驶事故预测框架，该框架使用领域信息 世界模型（World Model） 来增强数据，并使用图和时间卷积增强时空推理。

除了扩散之外，序列模型还拓宽了远程一致性的能力。 Po et al. [201]将长期记忆的分块状态空间模型与短期一致性的局部注意力集成起来，从而实现具有持续记忆和一致动态的视频生成

。 S2-SSM [61] 采用 Mamba 层对对象槽的独立演化进行建模，并采用稀疏正则化交叉注意机制来捕获它们的因果交互，从而实现对环境的因果推理。

空间潜网格（Spatial Latent Grid）。 空间潜网格（Spatial Latent Grid） 范例使用自监督时空目标预训练几何对齐的空间地图，保留局部性并实现高效的 展开轨迹（Rollouts）、多模态融合和可转移的规划器就绪地图。

在此范例的基础上，结构化网格和物理信息方法对可控 展开轨迹（Rollouts） 的几何和动力学进行编码。 PhyDNet [202] 将表示为偏微分方程的物理先验与视觉因素分开，从而改进预测。 ViDAR [95]通过点云预测和潜在渲染算子的预训练任务统一了语义、几何和动力学，为下游自动驾驶任务提供了可扩展的基础。 FOLIAGE [203] 使用增值图网络和基于 Transformer 的预测器对动态进行建模，对模拟数据执行 展开轨迹（Rollouts）。作为对这些网格和物理线的补充，MindJourney [73] 将 VLM 与可控 世界模型（World Model） 结合起来，沿着计划的相机轨迹渲染以自我为中心的 展开轨迹（Rollouts），从而实现多视图推理。

以基于网格的表示为基础，基于扩散的预测已成为稳定 长时程（Long-Horizon） 生成的主导。在以网格为中心的预测器中，DOME [100] 将观测值编码到连续的潜在空间中，并应用时空 DiT 进行场景预测，Copilot4D [88] 对点云进行标记，并将时空 Transformer 与离散扩散相结合，以提高保真度和连贯性，LidarDM [106] 生成布局条件的静态场景，将它们与动态对象组合，集成 LiDAR 模拟以生成可控视频。对于长视频生成，Vista [102] 采用两阶段大规模训练机制来生成可控、高保真驾驶视频，而 Delphi [92] 通过共享噪声和特征对齐以及故障驱动框架来增强 长时程（Long-Horizon） 多视图一致性，以合成目标数据以改进规划。为了增强 长时程（Long-Horizon） 的稳定性，GEM [108] 通过大规模训练和对运动、动力学和姿势的细粒度控制来实现可控的自我视觉生成，Zhou et al. [204] 维护持久的 RGB-D 3D 内存映射来指导后续帧，STAGE [205] 引入了多阶段训练的分层时间特征传输。

分解式渲染表示（Decomposed Rendering Representation）。场景被分解为显式基元，以在长视野范围内合成视图一致、可模拟的轨迹。在这个范式中，GaussianWorld [109] 将场景演化建模为自我运动、物体动力学和新观察到的区域，迭代更新 3D 高斯基元以实现准确高效的动态感知。 InfiniCube [22] 引入了一种混合管道，结合了基于体素的生成、视频合成和动态高斯重建，实现了以高清地图、边界框和文本为条件的大规模动态 3D 驾驶场景

。作为补充，Wu等人[206]用基于重建几何的长期空间记忆和情景记忆增强了视频世界模型（World Model），它们共同条件连续生成以实现长期一致性。

2) 全局差异预测（Global Difference Prediction）: 词元特征序列（Token Feature Sequence）。对于 通用（General-Purpose） 世界模型（World Models），标记化特征序列

support global prediction via masked and generative modeling, enabling parallel long-horizon rollouts with global constraints and multimodal conditioning.

Within Joint-Embedding Predictive Architecture (JEPA) [207], V-JEPA [51] extends this architecture to video by predicting latent features of occluded spatiotemporal regions, learning generalizable representations for both appearance and motion without pixel reconstruction or contrastive learning. Building on this, V-JEPA 2 [14] scales pretraining to large-scale Internet videos with larger models and incorporates limited robot interaction data for post-training, transferring to robotic planning. V-JEPA 2.1 [85] further modifies this recipe with dense prediction over both visible and masked tokens, deep self-supervision, and image-video tokenizers, improving local spatiotemporal features while preserving global prediction and embodied planning capability. DINO-Foresight [116] similarly treats pretrained DINO/VFM features as semantic latent tokens, using a masked feature Transformer to forecast future driving-scene representations that can be decoded by segmentation, depth, and surface-normal heads. AD-L-JEPA [208] adapts JEPA to BEV LiDAR, predicting masked embeddings in a self-supervised manner. Beyond JEPA-style prediction, WorldDreamer [45] frames world modeling as masked visual sequence prediction to learn physics and motion for diverse video generation and editing, while MaskGWM [110] combines diffusion with masked feature reconstruction and a dual-branch masking strategy to improve long-horizon consistency and generalization.

In parallel, diffusion-based methods have become central to global-difference modeling. Sora [13] represents video as unified spacetime patches and uses a DiT to generate long, coherent sequences at scale. Extending this line toward Physical AI, Cosmos 3 [86] unifies language, image, video, audio, and action sequences within a mixture-of-transformers architecture, positioning omnimodal world models as general-purpose backbones for understanding, generation, simulation, and action prediction. ForeDiff [209] decouples conditioning from denoising by adding a deterministic predictive stream and employing a pretrained predictor to guide generation, improving accuracy and consistency. For domain-specific synthesis, AirScape [210] introduces an aerial video-intention dataset, applies supervised fine-tuning for controllability, and leverages VLMs to impose spatiotemporal constraints; Mars-Gen [211] builds a multimodal Mars dataset from NASA's sparse rover stereo imagery using geometric foundation models, then trains a controllable generator to produce visually realistic, geometry-consistent Martian videos. In clinical guidance, EchoWorld [212] proposes a motion-aware world model for echocardiography probe control, pretraining on region- and motion-outcome prediction and fine-tuning attention to fuse visual and motion cues for precise guidance.

Spatial Latent Grid. Spatial-grid models forecast voxel grids in parallel and fuse multi-view visual features into a unified map, learning a general-purpose world model.

Recent work converges on unified scene understanding and future prediction. UniFuture [213] couples Dual Latent Sharing with multi-scale latent interaction to jointly model appearance and depth in future driving scenes, and HERMES [114] integrates multiview BEV features into an LLM with world queries that link scene understanding to future prediction within a single framework. BEVWorld [214] maps images and LiDAR into a compact BEV latent space through a unified tokenizer and applies a latent BEV diffusion model for synchronized multimodal forecasting. Progress in grid and occupancy forecasting includes differentiable raycasting with a proxy reformulation for sensor agnostic motion learning by Khurana *et al.* [215], [216] and LiDAR to range images 3D spatiotemporal convolutions by Mersch *et al.* [217]. Cam4DOcc [218] established the first vision-only benchmark with an E2E 3D CNN baseline, and Liu *et al.* [219] enhanced cross-task transfer through high-ratio compression and latent flow matching.

On the generative front, tokenized 4D representations enable controllable scene synthesis. OccSora [91] uses a 4D tokenizer to derive compact representations for trajectory conditioned diffusion, and DynamicCity [105] encodes 4D occupancy as HexPlanes representations with a VAE and employs a conditional DiT for high fidelity controllable dynamics. Fidelity and consistency improve through decoupling ego-motion with scene evolution in COME [220], physics-informed constraints in DrivePhysica [221], cross-view point map alignment in Liu *et al.* [222], and photometric warping-based supervision in PosePilot [223]. For controllable conditioning, DriveDreamer 2 [103] translates prompts into agent trajectories and HDMaps for customizable video generation, EOT-WM [224] encodes ego and surrounding trajectories as trajectory videos for trajectory consistent synthesis, and ORV [65] uses 4D semantic occupancy sequences to guide action conditioned video with S2R transfer. AETHER [77] unifies dynamic 4D reconstruction, action-conditioned video prediction, and vision-based planning under training on synthetic 4D data and achieves zero-shot generalization to real-world scenarios.

Decomposed Rendering Representation. This paradigm performs global prediction by combining explicit 3D structure with video generative priors.

A trend is combining video generation with Gaussian Splatting. DriveDreamer4D [111] exploits complex driving trajectories such as lane changes to guide video synthesis and optimize a 4DGS model, which enhances reconstruction fidelity and spatiotemporal coherence from novel viewpoints. ReconDreamer [112] introduces an online restoration module together with progressive data reuse to correct artifacts in Gaussian-rendered views and enables reliable reconstruction of large-scale trajectories. MagicDrive3D [90] generates multi-view street scenes conditioned on BEV maps, 3D boxes, and text, and further converts the outputs into full 3D environments through fault-tolerant GS. In contrast, implicit-field methods replace GS with continuous neural representations. UnO [17] leverages future point clouds to learn a NeRF-style 4D occupancy field, which allows annotation-free prediction and achieves strong transfer performance beyond supervised baselines in point-cloud forecasting.

In summary, decision coupling specifies how strongly a world model is tied to a particular control objective, temporal modeling governs how predictions unfold over time, and

原文第 10 页 | 中文翻译初稿

10

通过屏蔽和生成建模支持全局预测，从而实现具有全局约束和多模态调节的并行 长时程（Long-Horizon） 展开轨迹（Rollouts）。

在联合嵌入预测架构（JEPA）[207]中，V-JEPA[51]通过预测被遮挡时空区域的潜在特征，学习外观和运动的通用表示而无需像素重建或对比学习，将该架构扩展到视频。在此基础上，V-JEPA 2 [14] 将预训练扩展到具有更大模型的大规模互联网视频，并纳入有限的机器人交互数据进行后训练，转移到机器人规划。V-JEPA 2.1 [85]通过对可见标记和屏蔽标记的密集预测、深度自监督和图像视频标记器进一步修改了这个配方，改进了局部时空特征，同时保留了全局预测和具体规划能力。 DINO-Foresight [116] 同样将预训练的 DINO/VFM 特征视为语义潜在标记，使用屏蔽特征 Transformer 来预测未来的驾驶场景表示，这些表示可以通过分割、深度和表面法线头进行解码。 AD-L-JEPA [208] 将 JEPA 应用于 BEV LiDAR，以自我监督的方式预测屏蔽嵌入。除了 JEPA 风格的预测之外，WorldDreamer [45] 将世界建模构建为屏蔽视觉序列预测，以学习物理和运动以进行不同的视频生成和编辑，而 MaskGWM [110] 将扩散与屏蔽特征重建和双分支屏蔽策略相结合，以提高 长时程（Long-Horizon） 的一致性和泛化性。

与此同时，基于扩散的方法已成为全局差异建模的核心。 Sora [13] 将视频表示为统一的时空补丁，并使用 DiT 大规模生成长的、连贯的序列。将这条路线扩展到物理 AI，Cosmos 3 [86] 在混合变形金刚架构中统一了语言、图像、视频、音频和动作序列，将全模态 世界模型（World Models） 定位为理解、生成、模拟和动作预测的通用骨干。 ForeDiff [209] 通过添加确定性预测流并使用预训练的预测器来指导生成，从而提高准确性和一致性，从而将调节与去噪分离。对于特定领域的合成，AirScape [210] 引入了航空视频意图数据集，应用有监督的微调来实现可控性，并利用 VLM 施加时空约束； MarsGen [211] 使用几何基础模型（Foundation Models） 从 NASA 的稀疏漫游车立体图像构建多模态火星数据集，然后训练可控生成器来生成视觉逼真、几何一致的火星视频。在临床指导中，EchoWorld [212] 提出了一种运动感知 世界模型（World Model），用于超声心动图探头控制、区域和运动结果预测的预训练以及微调注意力以融合视觉和运动线索以实现精确指导。

空间潜网格（Spatial Latent Grid）。空间网格模型并行预测体素网格，并将多视图视觉特征融合到统一的地图中，学习 通用（General-Purpose） 世界模型（World Model）。

最近的工作集中在统一场景理解和未来预测上

。 UniFuture [213]将双潜在共享与多尺度潜在交互结合起来共同建模外观

和未来驾驶场景的深度，HERMES [114] 将多视图 BEV 功能集成到 LLM 中，并通过世界查询将场景理解与单个框架内的未来预测联系起来。 BEVWorld [214]通过统一的标记器将图像和LiDAR映射到紧凑的BEV潜在空间中，并应用潜在的BEV扩散模型进行同步多模态预测。网格和占用预测方面的进展包括 Khurana et al. [215]、[216] 的可微光线投射和传感器无关运动学习的智能体重构，以及 Mersch et al. [217] 的 LiDAR 测距图像 3D 时空卷积。 Cam4DOcc [218] 使用 E2E 3D CNN 基线建立了第一个仅视觉基准，Liu 等人 [219] 通过高比率压缩和潜在的 流匹配（Flow Matching） 增强了跨任务传输。

在生成方面，标记化的 4D 表示可以实现可控的场景合成。 OccSora [91] 使用 4D 标记器来导出轨迹条件扩散的紧凑表示，DynamicCity [105] 使用 VAE 将 4D 占用编码为 HexPlanes 表示，并采用条件 DiT 来实现高保真可控动态。通过将自我运动与 COME [220] 中的场景演化解耦、DrivePhysica [221] 中的物理信息约束、Liu et al. [222] 中的跨视点地图对齐以及 PosePilot [223] 中基于光度扭曲的监督来提高保真度和

一致性。为了实现可控调节，DriveDreamer 2 [103] 将提示转换为智能体轨迹和 HDMap，以生成可定制的视频，EOT-WM [224] 将自我和周围轨迹编码为轨迹视频，以实现轨迹一致合成，ORV [65] 使用 4D 语义占用序列通过 S2R 传输来指导动作调节视频。 AETHER [77] 在合成 4D 数据的训练下统一了动态 4D 重建、动作条件视频预测和基于视觉的规划，并实现了对现实世界场景的零样本泛化。

分解式渲染表示（Decomposed Rendering Representation）。该范例通过将显式 3D 结构与视频生成先验相结合来执行全局预测。

一种趋势是将视频生成与高斯泼溅相结合。 DriveDreamer4D [111] 利用复杂的驾驶轨迹（例如变道）来指导视频合成并优化 4DGS 模型，从而从新颖的角度增强重建保真度和时空连贯性。 ReconDreamer [112] 引入了在线恢复模块以及渐进式数据重用，以纠正高斯渲染视图中的伪影，并实现大规模轨迹的可靠重建。 MagicDrive3D [90] 生成以 BEV 地图、3D 框和文本为条件的多视图街道场景，并通过容错 GS 进一步将输出转换为完整的 3D 环境。相比之下，隐式场方法用连续神经表示代替 GS。 UnO [17] 利用未来的点云来学习 NeRF 式的 4D 占用场，它允许无注释预测，并在点云预测中实现超越监督基线的强大传输性能。

总之，决策耦合指定了 世界模型（World Model） 与特定控制目标的关联程度，时间建模控制预测如何随时间展开，以及

spatial representation determines the level of geometric detail and locality. Different combinations of these axes induce characteristic trade-offs: decision-coupled RSSM-style models with global latent vectors favor sample efficiency and fast rollouts at the cost of visual detail; token-based simulators, especially with global prediction, improve fidelity and semantic richness but incur higher computational and data demands; grid-based and decomposed-rendering models provide strong geometric and multi-view coherence, yet remain challenging to deploy under strict latency and memory budgets.

IV. Data Resources & Metrics

World models in embodied AI are required to address diverse tasks spanning manipulation, navigation, and autonomous driving, requiring heterogeneous resources and rigorous evaluation. Accordingly, we present Data Resources in §IV-A and Metrics in §IV-B, focusing on the most widely adopted platforms and evaluation measures as a unified foundation for cross-domain assessment.

A. Data Resources

To meet the diverse demands of embodied AI, we categorize data resources into four categories: Simulation Platforms, Interactive Benchmarks, Offline Datasets, and Real-world Robot Platforms, as detailed in the following subsections. Tab. III provides a comprehensive overview of these resources.

1) Simulation Platforms: Simulation platforms provide controllable and scalable virtual environments for training and evaluating world models.

- **MuJoCo** [225] is a customizable physics engine widely adopted for its efficient robotic simulation of articulated systems and contact dynamics in robotics and control research.
- **NVIDIA Isaac** is an E2E, GPU-accelerated simulation stack that comprises Isaac Sim, Isaac Gym [228], and Isaac Lab [229]. It offers photorealistic rendering and large-scale RL capabilities.
- **CARLA** [226] is an open-source simulator based on Unreal Engine for urban autonomous driving, providing realistic rendering, diverse sensors, and closed-loop evaluation protocols.
- **Habitat** [227] is a high-performance simulator for embodied AI, specializing in photorealistic 3D indoor navigation.

2) Interactive Benchmarks: Interactive benchmarks offer standardized task suites and protocols for reproducible closed-loop evaluation of world models.

- **DeepMind Control (DMC)** [231] is a standard MuJoCo-based suite for control tasks, offering a consistent basis for comparing agents that learn from state or pixel-based observations.
- **Atari** [230] is a suite of pixel-based, discrete-action games for evaluating agent performance. The Atari100k [246] specifically assesses sample efficiency by limiting interaction to 100k steps.
- **Meta-World** [232] is a benchmark for multi-task and meta-RL, featuring 50 diverse robotic manipulation tasks with a Sawyer arm in MuJoCo under standardized evaluation protocols.
- **RLBench** [233] offers 100 simulated tabletop manipulation tasks with sparse rewards and rich, multi-modal observations, designed to test complex skills and rapid adaptation.
- **LIBERO** [235] is a benchmark for lifelong robotic manipulation, providing 130 procedurally generated tasks and human demonstrations to evaluate sample-efficient and continual learning.
- **nuPlan** [234] is a planning benchmark for autonomous driving, using a lightweight closed-loop simulator and over 1 500 h of real-world driving logs to evaluate long-horizon performance.

3) Offline Datasets: Offline datasets are large-scale, pre-collected trajectories that eliminate interactive rollouts and provide a foundation for reproducible evaluation and data-efficient pretraining of world models.

- **RT-1** [240] is a real-world dataset for robot learning, collected over 17 months with 13 Everyday Robots mobile manipulators. It contains 130 000 demonstrations spanning more than 700 tasks, pairing language instructions and image observations with discretized 11-DoF actions for the arm and mobile base.
- **Open X-Embodiment (OXE)** [242] is a corpus aggregating 60 sources from 21 institutions, spanning 22 robot embodiments, 527 skills, and over one million trajectories in a unified format for cross-embodiment training. Models trained on OXE demonstrate strong transfer beyond single-robot baselines, underscoring the effectiveness of cross-platform data sharing.
- **Habitat-Matterport 3D (HM3D)** [239] is a large-scale dataset of 1 000 indoor reconstructions with 112 500 m^2 navigable area, substantially expanding the scope and diversity of embodied AI simulation. Released for the Habitat platform, it provides the necessary metadata and resources for seamless use.
- **nuScenes** [237] is a large-scale multimodal driving dataset with a 360-degree sensor suite comprising six cameras, five radars, one LiDAR, and GPS/IMU. It contains 1 000 twenty-second scenes collected in Boston and Singapore with dense 3D annotations for 23 classes and HDMaps, providing a core benchmark for multimodal fusion and long-horizon prediction.
- **Waymo** [238] is a multimodal autonomous driving benchmark with 1 150 twenty-second scenes at 10 Hz from San Francisco, Phoenix, and Mountain View. It includes five LiDARs and five cameras, with about 12 million 3D and 2D annotations, making it a large-scale resource for modeling traffic dynamics.
- **Occ3D** [241] defines 3D occupancy prediction from surround-view images, providing voxel labels that distinguish free, occupied, and unobserved states. Occ3D-nuScenes contains about 40 000 frames at 0.4 m resolution, while Occ3D-Waymo offers about 200 000 frames at 0.05 m. This voxel-level supervision enables holistic scene understanding beyond bounding boxes.
- **Something-Something v2 (SSv2)** [236] is a video dataset

原文第 11 页 | 中文翻译初稿

11

空间表示决定了几何细节和局部性的水平。这些轴的不同组合会导致特征权衡：具有全局潜在向量的 决策耦合（Decision-Coupled） RSSM 型模型有利于样本效率和快速 展开轨迹（Rollouts），但以视觉细节为代价；基于令牌的模拟器，尤其是全局预测，可以提高保真度和语义丰富度，但会产生更高的计算和数据需求；基于网格和分解的渲染模型提供了强大的几何和多视图一致性，但在严格的延迟和内存预算下部署仍然具有挑战性。
四．数据资源和指标

具身智能（Embodied AI） 中的 世界模型（World Models） 需要解决涵盖操纵、导航和自动驾驶的各种任务，需要异构资源和严格的评估。因此，我们在§IV-A中介绍数据资源，在§IV-B中介绍指标，重点关注最广泛采用的平台和评估措施，作为跨领域评估的统一基础。
一个。数据资源

为了满足具身智能（Embodied AI）的多样化需求，我们将数据资源分为四类：仿真平台、交互式基准、离线数据集和现实世界机器人平台，具体内容将在以下小节中详细介绍。标签。 III 全面概述了这些资源。

1) 仿真平台： 仿真平台为训练和评估世界模型（World Models）提供可控且可扩展的虚拟环境。

- MuJoCo [225] 是一种可定制的物理引擎，因其对机器人和控制研究中的铰接系统和接触动力学的高效机器人模拟而被广泛采用。

- NVIDIA Isaac 是一个 E2E、GPU 加速的模拟堆栈，由 Isaac Sim、Isaac Gym [228] 和 Isaac Lab [229] 组成。它提供逼真的渲染和大规模 RL 功能。

- CARLA [226] 是一个基于 Unreal Engine 的开源模拟器，用于城市自动驾驶，提供逼真的渲染、多样化的传感器和闭环评估协议。

- Habitat [227] 是 具身智能（Embodied AI） 的高性能模拟器，专门从事逼真的 3D 室内导航。 2) 交互式基准测试： 交互式基准测试提供

- 用于 世界模型（World Models） 的可重复闭环评估的标准化任务套件和协议。

- DeepMind Control (DMC) [231] 是一个用于控制任务的标准 MuJoCo 套件，为比较从基于状态或基于像素的观察中学习的智能体提供了一致的基础。

- Atari [230] 是一套基于像素的离散动作游戏，用于评估智能体性能。 Atari100k [246] 通过将交互限制为 100k 步骤来专门评估样本效率。

- Meta-World [232] 是多任务和 metaRL 的基准，具有 50 个不同的机器人操作任务

- MuJoCo 中的 Sawyer 臂符合标准化评估协议。

- RLBench [233] 提供了 100 个模拟桌面操作任务，具有稀疏的奖励和丰富的多模式观察，旨在测试复杂的技能和快速适应。

- LIBERO [235] 是终身机器人操作的基准，提供 130 个程序生成的任务和人类演示，以评估样本效率和持续学习（Continual Learning）。

- nuPlan [234] 是自动驾驶的规划基准，使用轻量级闭环模拟器和超过 1 500 小时的真实驾驶日志来评估长时程（Long-Horizon） 性能。

3) 离线数据集： 离线数据集是大规模的、预先收集的轨迹，消除了交互式 展开轨迹（Rollouts），并为世界模型（World Models） 的可重复评估和数据高效预训练提供了基础。

- RT-1 [240] 是一个用于机器人学习的真实数据集，是在 17 个月内使用 13 个 Everyday Robots 移动机械手收集的。它包含 130,000 个演示，涵盖 700 多项任务，将语言指令和图像观察与手臂和移动底座的离散 11 自由度动作配对。

- Open X-Embodiment (OXE) [242] 是一个语料库，汇集了来自 21 个机构的 60 个来源，涵盖 22 个机器人实施例、527 种技能以及超过 100 万个统一格式的轨迹，用于跨实施例训练。在 OXE 上训练的模型表现出超越单机器人基线的强大传输能力，强调了跨平台数据共享的有效性。

- Habitat-Matterport 3D (HM3D) [239] 是一个包含 1 000 个室内重建的大型数据集，具有 112 500 m2 可导航区域，大大扩展了 具身智能（Embodied AI） 模拟的范围和多样性。它针对 Habitat 平台发布，提供无缝使用所需的元数据和资源。

- nuScenes [237] 是一个大规模多模式驾驶数据集，具有 360 度传感器套件，包括六个摄像头、五个雷达、一个 LiDAR 和 GPS/IMU。它包含在波士顿和新加坡收集的 1000 个二十秒场景，以及 23 个类别和 HDMap 的密集 3D 注释，为多模态融合和 长时程（Long-Horizon） 预测提供了核心基准。

- Waymo [238] 是一个多模式自动驾驶基准，包含来自旧金山、凤凰城和山景城的 1150 个 10 Hz 的二十秒场景。它包括五个 LiDAR 和五个摄像头，具有约 1200 万个 3D 和 2D 注释，使其成为交通动态建模的大规模资源。

- Occ3D [241] 定义了环绕视图图像的 3D 占用预测，提供区分空闲、占用和未观察状态的体素标签。Occ3D-nuScenes 包含约 40 000 个分辨率为 0.4 m 的帧，而 Occ3DWaymo 提供约 200 000 个分辨率为 0.05 m 的帧。这种体素级监督可以实现超越边界框的整体场景理解。

- Something-Something v2 (SSv2) [236] 是一个视频数据集

TABLE III
AN OVERVIEW OF DATA RESOURCES FOR TRAINING AND EVALUATING EMBODIED WORLD MODELS.

Category	Name	Year	Task	Input	Domain	Scale	Protocol[1]
Platform	MuJoCo [225]	2012	Continuous control	Proprio.	Sim	-	-
	CARLA [226]	2017	Driving simulation	RGB-D/Seg/LiDAR/Radar/GPS/IMU	Sim	-	✓
	Habitat [227]	2019	Embodied navigation	RGB-D/Seg/GPS/Compass	Sim	-	✓
	Isaac Gym [228]	2021	continuous control	Proprio.	Sim	-	-
	Isaac Lab [229]	2023	Robot learning suites	RGB-D/Seg/LiDAR/Proprio.	Sim	-	-
Benchmark	Atari [230]	2013	Discrete-action game	RGB/State	Sim	55+ Games	✓
	DMC [231]	2018	Continuous control	RGB/Proprio.	Sim	30+ Tasks	✓
	Meta-World [232]	2019	Multi-task manipulation	RGB/Proprio.	Sim	50 tasks	✓
	RLBench [233]	2020	Robotic manipulation	RGB-D/Seg/Proprio.	Sim	100 tasks	✓
	nuPlan [234]	2021	Driving planning	RGB/LiDAR/Map/Proprio.	Real	1.5k hours	✓
	LIBERO [235]	2023	Lifelong manipulation	RGB/Text/Proprio.	Sim	130 tasks	✓
Dataset	SSv2 [236]	2018	Video-action understanding	RGB/Text	Real	220k videos	169k/24k/27k
	nuScenes [237]	2020	Driving perception	RGB/LiDAR/Radar/GPS/IMU	Real	1k scenes	700/150/150
	Waymo [238]	2020	Driving perception	RGB/LiDAR	Real	1.15k scenes	798/202/150
	HM3D [239]	2021	Indoor navigation	RGB-D	Real	1k scenes	800/100/100
	RT-1 [240]	2022	Real-robot manipulation	RGB/Text	Real	130k+ trajectories	-
	Occ3D [241]	2023	3D occupancy	RGB/LiDAR	Real	1.9k scenes	600/150/150; 798/202/-
	OXE [242]	2024	Cross-embodiment pretraining	RGB-D/LiDAR/Text	Real	1M+ trajectories	-
	OpenDV [96]	2024	Driving video pretraining	RGB/Text	Real	2k+ hours	-
	VideoMix22M [14]	2025	Video pretraining	RGB	Real	22M+ samples	-
	VisionMix163M [85]	2026	Image-video pretraining	RGB	Real	163M+ samples	-
Robot	Franka Emika [243]	2022	Manipulation	Proprio.	Real	-	-
	Unitree Go1 [244]	2021	Quadruped locomotion	RGB-D/LiDAR/Proprio.	Real	-	-
	Unitree G1 [245]	2024	Humanoid manipulation	RGB-D/LiDAR/Proprio./Audio	Real	-	-

[1] **Protocol**: For interactive benchmarks, a check mark (✓) indicates available evaluation protocols. For datasets, it indicates official data splits are provided.

for fine-grained action understanding. It contains 220 847 clips across 174 categories, collected from crowd workers following textual prompts (*e.g.*, Putting something into something) with splits of 168 913 train, 24 777 val, and 27 157 test videos.

- **OpenDV** [96] is the largest large-scale video-text dataset for autonomous driving, proposed by GenAD, which supports video prediction and world-model pretraining. It contains 2 059 hours and 65.1 million frames from YouTube and seven public datasets, covering over 40 countries and 244 cities. The dataset provides command and context annotations to enable language- and action-conditioned prediction and planning.
- **VideoMix22M** [14] is a large-scale dataset introduced with V-JEPA 2 for self-supervised pretraining. It scales from 2 million to 22 million samples, drawn from YT-Temporal-1B [247], HowTo100M [248], Kinetics [249], SSv2, and ImageNet [250]. The largest source, YT-Temporal-1B, is curated with retrieval-based filtering to suppress noise, while ImageNet images are converted into static video clips for consistency.
- **VisionMix163M** [85] extends VideoMix-style self-supervised pretraining by combining SSv2, Kinetics, HowTo100M, YT-Temporal-1B, and LVD-142M image data. It replaces the ImageNet subset used in V-JEPA 2 with a much larger curated image source and rebalances video sampling toward motion-rich content to support dense and temporally consistent visual features.

4) Real-world Robot Platforms: Real-world robot platforms provide physical embodiments for interaction, enabling closed-loop evaluation, high-fidelity data collection, and S2R validation under real-world constraints.

- **Franka Emika** [243] is a 7-DoF collaborative robot arm with joint torque sensors for precise force control. Through the control interface, it supports 1 kHz torque control for contact-rich tasks, while its ROS integration makes it a versatile platform.
- **Unitree Go1** [244] is a cost-effective and widely adopted quadrupedal robot equipped with a panoramic depth-sensing suite, onboard computing of 1.5 TFLOPS, and a maximum speed of 4.7 m/s, establishing it as a standard platform for locomotion and embodied-AI research.
- **Unitree G1** [245] is a compact humanoid robot for research, offering up to 43-DoF and knee torques of 120 N·m, with integrated 3D LiDAR and depth cameras. With multimodal sensing, onboard compute, ROS support, and swappable batteries, this low-cost platform provides a practical real-robot testbed for training and evaluating embodied world models.

B. Metrics

Metrics evaluate the capability of world models to capture dynamics, generalize to unseen scenarios, and scale with additional resources. We organize them into three abstraction levels: §IV-B1 **Pixel Prediction Quality**, §IV-B2 **State-level Understanding**, and §IV-B3 **Task Performance**, representing a progression from low-level signal fidelity to high-level goal attainment.

1) Pixel Generation Quality: At the most fundamental level, world models are evaluated by their ability to reconstruct sensory inputs and generate realistic sequences. Metrics assess image fidelity, temporal consistency, and perceptual similarity, providing quantitative measures of the extent to which models capture raw environmental dynamics.

原文第 12 页 | 中文翻译初稿

12

表三

世界模型（World Models）训练和评估数据资源概述。

类别	名称	年份	任务	输入	域	规模	协议1
	MuJoCo [225]	2012	连续控制	Proprio。	Sim	-	-
rm	CARLA [226]	2017	驾驶模拟	RGB-D/Seg/LiDAR/雷达/GPS/IMU	Sim	-	✓
atfo	栖息地 [227]	2019	嵌入式导航	RGB-D/Seg/GPS/指南针	Sim	-	✓
PL	艾萨克健身房 [228]	2021	连续控制	Proprio。	Sim	-	-
	Isaac Lab [229]	2023	机器人学习套件	RGB-D/Seg/LiDAR/Proprio。	Sim	-	-
	Atari [230]	2013	离散动作游戏	RGB/状态	模拟	55+ 游戏	✓
ark	DMC [231]	2018	连续控制	RGB/Proprio。	Sim	30+ 任务	✓
m	元世界 [232]	2019	多任务操作	RGB/Proprio。	模拟	50 个任务	✓
nch	RLBench [233]	2020	机器人操纵	RGB-D/Seg/Proprio。	Sim	100 个任务	✓
Be	nuPlan [234]	2021	驾驶规划	RGB/LiDAR/地图/Proprio。	真实	1500小时	✓
	LIBERO [235]	2023	终生操作	RGB/文本/Proprio。	模拟	130 项任务	✓
	SSv2 [236]	2018	视频动作理解	RGB/文本	真实	220k 视频	169k/24k/27k
	nuScenes [237]	2020	驾驶感知	RGB/LiDAR/雷达/GPS/IMU	真实	1k场景	700/150/150
	Waymo [238]	2020	驾驶感知	RGB/LiDAR	真实	115k场景	798/202/150
t	HM3D [239]	2021	室内导航	RGB-D	真实	1k场景	800/100/100
ase	RT-1 [240]	2022	真实机器人操作	RGB/文本	真实	130k+ 轨迹	-
Dat	Occ3D [241]	2023	3D 占用	RGB/LiDAR	真实	1.9k 场景	600/150/150; 798/202/-
	OXE [242]	2024	跨实施例预训练	RGB-D/LiDAR/文本	真实	1M+ 轨迹	-
	OpenDV [96]	2024	驾驶视频预训练	RGB/文本	真实	2k+ 小时	-
	VideoMix22M [14]	2025	视频预训练	RGB	Real	22M+ 样本	-
	VisionMix163M [85]	2026	图像视频预训练	RGB	Real	163M+ 样本	-
ot	弗兰卡·艾米卡 [243]	2022	操纵	Proprio。	真实	-	-
ob	Unitree Go1 [244]	2021	四足运动	RGB-D/LiDAR/Proprio。	真实	-	-
R	Unitree G1 [245]	2024	人形操控	RGB-D/LiDAR/Proprio./Audio	Real	-	-

1 协议：对于交互式基准测试，复选标记 (✓) 表示可用的评估协议。对于数据集，它表示提供了官方数据分割。

- 用于细粒度的动作理解。它包含 174 个类别的 220 847 个剪辑，这些剪辑是从人群工作人员按照文本提示（例如，将某物放入某物）收集的，其中包含 168 913 个训练视频、24 777 个验证视频和 27 157 个测试视频。

- OpenDV[96]是GenAD提出的最大的自动驾驶大规模视频文本数据集，支持视频预测和世界模型预训练。它包含来自 YouTube 的 2059 小时和 6510 万帧以及七个公共数据集，覆盖 40 多个国家和 244 个城市。该数据集提供命令和上下文注释，以实现以语言和动作为条件的预测和规划。

- VideoMix22M [14] 是 V-JEPA 2 引入的用于自监督预训练的大规模数据集。它的样本规模从 200 万到 2200 万个，取自 YT-Temporal1B [247]、HowTo100M [248]、Kinetics [249]、SSv2 和 ImageNet [250]。最大的源 YT-Temporal-1B 采用基于检索的过滤来抑制噪声，同时将 ImageNet 图像转换为静态视频剪辑以保持一致性。

- VisionMix163M [85] 通过结合 SSv2、Kinetics、HowTo100M、YT-Temporal-1B 和 LVD-142M 图像数据扩展了 VideoMix 式的自监督预训练。它用更大的精选图像源取代了 V-JEPA 2 中使用的 ImageNet 子集，并将视频采样重新平衡为运动丰富的内容，以支持密集且时间一致的视觉特征。

- 4) 现实世界的机器人平台： 现实世界的机器人平台

- 提供交互的物理实施例，在现实世界的约束下实现闭环评估、高保真数据收集和 S2R 验证。

- Franka Emika [243] 是一款 7-DoF 协作机器人手臂，配有关节扭矩传感器，可实现精确的力控制。通过控制接口，它支持 1 kHz 扭矩控制，用于接触丰富的任务，而其 ROS 集成使其成为一个多功能平台。

- Unitree Go1 [244] 是一款经济高效且广泛采用的四足机器人，配备全景深度传感套件、板载计算能力为 1.5 TFLOPS、最大速度为 4.7 m/s，使其成为运动和嵌入式人工智能研究的标准平台。

- Unitree G1 [245] 是一款用于研究的紧凑型人形机器人，提供高达 43 个自由度和 120 N · m 的膝盖扭矩，并集成了 3D LiDAR 和深度摄像头。该低成本平台具有多模态传感、板载计算、ROS 支持和可更换电池，为训练和评估 世界模型（World Models） 提供了实用的真实机器人测试平台。

B。指标

指标评估 世界模型（World Models） 捕获动态、推广到未见过的场景以及利用额外资源进行扩展的能力。我们将它们组织为三个抽象级别：§IV-B1 像素预测质量、§IV-B2 状态级理解和 §IV-B3 任务性能，代表从低级信号保真度到高级目标实现的进展。

1) 像素生成质量： 在最基本的层面上，世界模型（World Models） 通过其重建感官输入和生成真实序列的能力进行评估。指标评估图像保真度、时间一致性和感知相似性，提供模型捕获原始环境动态程度的定量测量。

Fréchet Inception Distance (FID) [251]. FID is a metric for assessing the realism and diversity of generated images. It compares real and generated image distributions in the feature space of an ImageNet-pretrained Inception-v3 [252], modeling embeddings as Gaussians with means $\boldsymbol{\mu}_x, \boldsymbol{\mu}_y$ and covariances $\boldsymbol{\Sigma}_x, \boldsymbol{\Sigma}_y$. Defined as

$$\mathrm{FID}(x, y) = \|\boldsymbol{\mu}_x - \boldsymbol{\mu}_y\|_2^2 + \mathrm{Tr}\left(\boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_y - 2(\boldsymbol{\Sigma}_x\boldsymbol{\Sigma}_y)^{1/2}\right), \tag{6}$$

a lower FID denotes a closer alignment between real and generated distributions. By comparing the first and second moments, it penalizes fidelity loss (mean shift) and mode collapse (covariance mismatch), offering a holistic measure of generative performance.

Fréchet Video Distance (FVD) [253]. FVD extends FID to videos, evaluating both per-frame quality and temporal consistency. It replaces the image-based Inception network with an I3D [254] pretrained on Kinetics-400 [255]. Using the same Fréchet distance as Eq. (6) on motion-aware features, FVD yields a holistic video quality score. A lower value indicates a closer alignment of distributions in appearance and dynamics while penalizing temporal artifacts like unnatural motion or flickering.

Structural Similarity Index Measure (SSIM) [256]. SSIM is a perceptual metric for image quality that compares luminance, contrast, and structure between a generated image and its reference. For two patches x and y with means $\boldsymbol{\mu}_x$, $\boldsymbol{\mu}_y$, variances $\boldsymbol{\Sigma}_x^2, \boldsymbol{\Sigma}_y^2$, and covariance $\boldsymbol{\Sigma}_{xy}$, SSIM is defined as

$$\mathrm{SSIM}(x, y) = \frac{(2\boldsymbol{\mu}_x\boldsymbol{\mu}_y + C_1)(2\boldsymbol{\Sigma}_{xy} + C_2)}{(\boldsymbol{\mu}_x^2 + \boldsymbol{\mu}_y^2 + C_1)(\boldsymbol{\Sigma}_x^2 + \boldsymbol{\Sigma}_y^2 + C_2)}. \tag{7}$$

The final score is obtained by averaging SSIM over sliding windows, and values closer to 1 indicate higher similarity.

Peak Signal-to-Noise Ratio (PSNR) [257]. PSNR measures pixel-wise distortion between a reconstruction and its reference. Let the mean-squared error (MSE) over N pixels be

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2, \tag{8}$$

and let MAX denote the maximum possible pixel value(*e.g.*, 255 for RGB or 1 for normalized images). Then

$$\mathrm{PSNR}(x, y) = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}}\right). \tag{9}$$

Higher PSNR values indicate lower distortion and greater fidelity.

Learned Perceptual Image Patch Similarity (LPIPS) [258]. LPIPS is a metric that correlates with human judgments by comparing features extracted from pretrained networks. Let $\hat{f}_x^l$ and $\hat{f}_y^l$ denote the unit-normalized activations at layer l for inputs x and y, and w_l the channel-wise weights. LPIPS is defined as

$$\mathrm{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l}\sum_{h,w}\left\|w_l \odot \left(\hat{f}_{h,w,x}^l - \hat{f}_{h,w,y}^l\right)\right\|_2^2. \tag{10}$$

Lower LPIPS values imply greater similarity, offering enhanced fidelity compared to pixel-based metrics and robustness against distortions.

VBench [259]. VBench is a comprehensive benchmark for video generation that assesses performance across 16 dimensions grouped into two categories: Video Quality (*e.g.*, subject consistency, motion smoothness) and Video-Condition Consistency (*e.g.*, object class, human action). It provides carefully curated prompt suites and large-scale human preference annotations to ensure strong perceptual alignment, thereby enabling fine-grained evaluation of model capabilities and limitations.

2) State-level Understanding: Beyond pixel fidelity, state-level understanding assesses whether models capture objects, layouts, and semantics, and can predict their evolution. Metrics span semantic, BEV, and 3D segmentation, detection, occupancy, geometry, and trajectory accuracy, emphasizing structural understanding beyond appearance.

mean Intersection over Union (mIoU). mIoU evaluates semantic segmentation by averaging the Intersection over Union (IoU) across classes. For class c,

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{11}$$

where TP, FP, and FN denote the true positives, false positives, and false negatives. IoU quantifies overlap with the ground truth while penalizing segmentation errors. The dataset-level score is

$$\mathrm{mIoU} = \frac{1}{|C|}\sum_{c \in C}\mathrm{IoU}_c\,. \tag{12}$$

A higher mIoU reflects more precise semantic scene understanding.

mean Average Precision (mAP). mAP evaluates detection and instance segmentation by averaging per-class Average Precision (AP). For a class c at IoU threshold τ, predictions are ranked by confidence and matched one-to-one with ground truths when $\mathrm{IoU} \geq \tau$ with unmatched predictions counted as FP and unmatched ground truths as FN. Precision and recall are

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{13}$$

Let $P_{c,\tau}(r)$ denote the precision-recall envelope obtained via monotonic interpolation. The AP for class c at threshold τ is

$$\mathrm{AP}_{c,\tau} = \int_0^1 P_{c,\tau}(r)\mathrm{d}r. \tag{14}$$

mAP averages AP across classes and thresholds T:

$$\mathrm{mAP} = \frac{1}{|C|}\sum_{c \in C}\left(\frac{1}{|T|}\sum_{\tau \in T}\mathrm{AP}_{c,\tau}\right). \tag{15}$$

A higher mAP indicates better instance recognition, more accurate localization, and more calibrated confidence estimates.

Displacement Error. Displacement error metrics assess state-level understanding by measuring spatial accuracy for keypoints, object centers, and trajectory waypoints. The L2 trajectory error computes the Euclidean distance between predicted and

原文第 13 页 | 中文翻译初稿

13

Fréchet 起始距离（FID）[251]。 FID 是评估生成图像的真实性和多样性的指标。它比较了 ImageNet 预训练 Inception-v3 [252] 特征空间中的真实图像分布和生成图像分布，将嵌入建模为高斯分布，其均值 μ x, μ y 和协方差 Σ x, Σ y 。定义为

FID(x, y) = ‖ μ x − μ y‖ 2 2 + Tr ⎛ Σ x + Σ y − 2(Σ x Σ y) 1 / 2⎞

,
(6)

较低的 FID 表示真实分布和生成分布之间更接近。通过比较第一和第二时刻，它会惩罚保真度损失（均值漂移）和模式崩溃（协方差不匹配），从而提供生成性能的整体衡量标准。

Fréchet 视频距离（FVD）[253]。 FVD 将 FID 扩展到视频，评估每帧质量和时间一致性。它将基于图像的 Inception 网络替换为在 Kinetics-400 [255] 上预训练的 I3D [254]。使用与方程式相同的 Fréchet 距离。(6) 在运动感知功能上，FVD 产生整体视频质量得分。较低的值表示外观和动态分布更接近，同时会惩罚不自然的运动或闪烁等时间伪影。

结构相似性指数测量（SSIM）[256]。 SSIM 是一种图像质量感知指标，可比较生成图像与其参考图像之间的亮度、对比度和结构。对于两个补丁 x 和 y，均值为 μ x 、 μ y 、方差 Σ2 _x__,_Σ2 y，以及协方差 Σ_xy_，SSIM定义为

(2 μ x μ y + C 1)(2 Σ xy + C 2)
SSIM(x, y) = (7)
(μ 2 x + μ 2 y + C 1)(Σ 2 x + Σ 2 y + C 2) .

最终得分是通过对滑动窗口上的SSIM进行平均而获得的，值越接近1表示相似度越高。

峰值信噪比（PSNR）[257]。 PSNR 测量重建与其参考之间的像素级失真。令 N 个像素的均方误差 (MSE) 为

N
MSE = N 1 ⎛ (xi − yi) 2 , (8)
i =1

并让 MAX 表示最大可能的像素值（例如，RGB 为 255，标准化图像为 1）。然后

PSNR(x, y) = 10 · log10 。 (9)
⎛ MAXMSE2 ⎞

PSNR 值越高表示失真越低且保真度越高。

学习感知图像块相似度（LPIPS）[258]。 LPIPS 是一种通过比较从预训练网络中提取的特征来与人类判断相关联的指标。让 _f_ˆ _x__l_ 和 _f_ˆ _y__l_表示单位归一化激活层_l_for 输入 x 和 y 以及 wl 通道权重。 LPIPS 定义为

1 2
LPIPS(x, y) = ∑ HlWl ∑ ∑∑ wl ⊙ ∑ f ˆ h,w,x l − f ˆ h,w,y l ∑ ∑ ∑∑2 .
l h,w
(10)

较低的 LPIPS 值意味着更大的相似性，与基于像素的指标相比，提供更高的保真度和抗失真的鲁棒性。

VBbench [259]

。 VBench 是视频生成的综合基准，可评估分为两类的 16 个维度的性能：视频质量（例如，主体一致性、运动平滑度）和视频条件一致性（例如，对象类、人类动作）。它提供精心策划的提示套件和大规模的人类偏好注释，以确保强大的感知一致性，从而实现对模型功能和局限性的细粒度评估。

2) 状态级理解： 除了像素保真度之外，状态级理解还评估模型是否捕获对象、布局和语义，并可以预测它们的演变。指标涵盖语义、BEV 和 3D 分割、检测、占用、几何形状和轨迹精度，强调超越外观的结构理解。

平均交并集 (mIoU)。 mIoU 通过平均跨类的交并集 (IoU) 来评估语义分割。对于 c 类，

TP
IoU = (11)
TP + FP + FN ,

其中 TP、FP 和 FN 表示真阳性、假阳性和假阴性。 IoU 量化与真实情况的重叠，同时惩罚分割错误。数据集级别的分数是

1
mIoU = ∑ IoU c 。 (12)
|C|
c∈C

较高的 mIoU 反映了更精确的语义场景理解。

平均精度 (mAP)。 mAP 通过平均每类平均精度 (AP) 来评估检测和实例分割。对于 IoU 阈值 τ 的类_c_，预测按置信度排序，并在 IoU ≥ τ 时与基本事实一对一匹配，不匹配的预测算作 FP，不匹配的基本事实算作 FN。准确率和召回率是

TP TP
精度 = 召回率 = (13)
TP + FP , TP + FN .

令 Pc,τ (r) 表示通过单调插值获得的精确召回包络。 c 类在阈值 τ 处的 AP 为

AP c,τ = Pc,τ (r)d r。 (14)
0 1

mAP 跨类别和阈值的 AP 平均值 T ：

1 1
mAP = AP c,τ 。 (15)
|C| |T |
cεC τ εT

较高的 mAP 表示更好的实例识别、更准确的定位以及更校准的置信度估计。

位移误差。位移误差指标通过测量关键点、对象中心和轨迹路径点的空间精度来评估状态级理解。 L2 轨迹误差计算预测和预测之间的欧几里德距离

ground-truth waypoints. Common variants include Average Displacement Error (ADE), which calculates the average displacement, and Final Displacement Error (FDE), which measures the displacement at the final step. Lower values indicate more accurate localization.

Chamfer Distance (CD) [260]. CD quantifies geometric similarity between a prediction S_1 and ground truth S_2 by summing squared nearest-neighbor distances across the two sets:

$$\mathrm{CD}(S_1, S_2) = \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2^2 + \sum_{y \in S_2} \min_{x \in S_1} \|x - y\|_2^2 . \tag{16}$$

Unlike pixel-level metrics, CD captures surfaces, occupancy, BEV, and 3D structures, and its differentiability enables use as both a training loss and an evaluation metric that complements IoU.

3) Task Performance: Ultimately, the value of a world model lies in supporting effective decision-making, with task-level metrics assessing goal achievement under safety and efficiency constraints in embodied settings.

Success Rate (SR). SR quantifies performance as the fraction of evaluation episodes that satisfy a predefined success condition. In navigation and manipulation, the condition is typically binary, such as reaching a target or placing an object correctly. In autonomous driving, the requirement is stricter, demanding route completion without collisions or major violations. The final SR is reported as the average of binary outcomes across all test episodes.

Sample Efficiency (SE). SE quantifies the samples needed to reach target performance. It is evaluated by fixed-budget benchmarks (*e.g.*, Atari-100k), data-performance curves, or in robotics by the demonstrations required to achieve a given success rate.

Reward. In RL, the reward is a signal r_t at timestep t. The goal is to maximize the discounted return $G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$. Results are reported as cumulative reward or average return, often with normalization for cross-task comparison.

Collision. Safety is evaluated with collision-based metrics. The primary measure, collision rate, is the proportion of evaluation episodes with at least one collision and is common in indoor navigation. In autonomous driving, exposure-normalized variants are used, such as collisions per kilometer or collisions per hour.

V. Performance Comparison

Given the proliferation of world-model variants and heterogeneous metrics, we organize comparisons by task objectives and rely on standardized benchmarks, reporting concise tables that highlight each method's strengths and limitations.

A. Pixel Generation

Generation on nuScenes. Driving video generation is treated as a world-modeling task that synthesizes plausible scene dynamics in fixed-length clips. Typical protocols produce short sequences and evaluate quality with *FID* for appearance fidelity and *FVD* for temporal consistency. For a fair comparison on the nuScenes validation split, recent approaches have achieved remarkable progress, as shown in Tab. IV. DrivePhysica delivers the best visual fidelity, while MiLA achieves the strongest temporal coherence, together establishing new state-of-the-art performance.

TABLE IV
PERFORMANCE COMPARISON OF VIDEO GENERATION ON THE NUSCENES.

Method	Pub.	Resolution	FID↓	FVD↓
MagicDrive3D [90]	arXiv'24	224 × 400	20.7	164.7
Delphi [92]	arXiv'24	512 × 512	15.1	113.5
Drive-WM [94]	CVPR'24	192 × 384	15.8	122.7
GenAD [96]	CVPR'24	256 × 448	15.4	184.0
DriveDreamer [97]	ECCV'24	128 × 192	52.6	452.0
Vista [102]	NeurIPS'24	576 × 1024	6.9	89.4
DrivePhysica [221]	arXiv'24	256 × 448	**4.0**	38.1
DrivingWorld [140]	arXiv'24	512 × 1024	7.4	90.9
DriveDreamer-2 [103]	AAAI'25	256 × 448	11.2	55.7
UniFuture [213]	arXiv'25	320 × 576	11.8	99.9
MiLA [196]	arXiv'25	360 × 640	4.1	**14.9**
GeoDrive [177]	arXiv'25	480 × 720	4.1	61.6
LongDWM [195]	arXiv'25	480 × 720	12.3	102.9
MaskGWM [110]	CVPR'25	288 × 512	8.9	65.4
GEM [108]	CVPR'25	576 × 1024	10.5	158.5
Epona [155]	ICCV'25	512 × 1024	7.5	82.8
STAGE [205]	IROS'25	512 × 768	11.0	242.8
DriVerse [115]	ACMMM'25	480 × 832	18.2	95.2

B. Scene Understanding

4D Occupancy Forecasting on Occ3D-nuScenes. 4D occupancy forecasting is treated as a representative world modeling task. Given 2 s of past 3D occupancy, models predict the subsequent 3 s of scene dynamics. Evaluation follows the Occ3D-nuScenes protocol and reports *mIoU* and per horizon *IoU*. As summarized in Tab. V, we compare methods by input modality, auxiliary supervision, and ego trajectory usage to reveal design choices for spatiotemporal forecasting. Methods using occupancy inputs outperform camera-only variants, and adding auxiliary supervision with a GT ego trajectory further mitigates performance decay at 2-3 s. Among all methods, COME (with GT ego) achieves the best average mIoU and per-horizon IoU.

C. Control Tasks

Evaluation on DMC. Most studies probe the capacity of world models to learn control-relevant dynamics, typically adopting a pixel-based setting with $64 \times 64 \times 3$ observations. The primary metric is *Episode Return*, defined as the cumulative reward over 1 000 steps, with a theoretical maximum of 1 000 given $r_t \in [0, 1]$. For comparability, Tab. VI reports the step budget and summarizes performance by task score and task count. The results indicate improved data efficiency, with recent models reaching strong performance in far fewer training steps. However, inconsistent evaluation protocols and task subsets impede a fair assessment of generalization, and building a

原文第 14 页 | 中文翻译初稿

14

地面实况航点。常见的变体包括计算平均位移的平均位移误差 (ADE) 和测量最后一步位移的最终位移误差 (FDE)。值越低表示定位越准确。

倒角距离 (CD) [260]。 CD 通过对两个集合的最近邻距离平方求和来量化预测 S 1 和真实值 S 2 之间的几何相似性：

CD(S 1 , S 2) = ∑ y min εS 2 ‖x − y‖ 2 2 + ∑ x min εS 1 ‖x − y‖ 2 2 .
xεS 1 yεS 2
(16)

与像素级指标不同，CD 捕获表面、占用率、BEV 和 3D 结构，其可微性可以用作训练损失和补充 IoU 的评估指标。

3) 任务绩效： 最终，世界模型（World Model） 的价值在于支持有效的决策，并通过任务级指标评估具体环境中安全和效率约束下的目标实现情况。

成功率（SR）。 SR 将性能量化为满足预定义成功条件的评估片段的分数。在导航和操纵中，条件通常是二元的，例如到达目标或正确放置物体。在自动驾驶方面，要求更加严格，要求完成路线，不发生碰撞或重大违规。最终 SR 报告为所有测试集的二元结果的平均值。

样本效率（SE）。 SE 量化了达到目标性能所需的样本。它通过固定预算基准（例如，Atari-100k）、数据性能曲线或在机器人技术中通过实现给定成功率所需的演示来评估。

报酬。在强化学习中，奖励是时间步 t 处的信号 rt 。目标是最大化折扣回报 Gt = ∑ ∞k =0_γkrt_+k+1.Resultsarereportedascumulativereward 或平均回报，通常进行跨任务比较的标准化。

碰撞。安全性通过基于碰撞的指标进行评估。主要衡量标准是碰撞率，是至少发生一次碰撞的评估事件的比例，在室内导航中很常见。在自动驾驶中，使用曝光归一化变量，例如每公里的碰撞次数或每小时的碰撞次数。

五、性能对比

鉴于世界模型变体和异构指标的激增，我们按任务目标组织比较，并依赖标准化基准，报告简明的表格，突出每种方法的优点和局限性。

一个。像素生成

nuScenes 上的生成。驾驶视频生成被视为一项世界建模任务，在固定长度的剪辑中合成合理的场景动态。典型协议产生

表四

NUSCENES 上视频生成的性能比较。

方法	发布	分辨率	FID_↓	FVD↓_
MagicDrive3D [90]	arXiv'24	224_×_400	20.7	164.7

德尔福[92]	arXiv'24	512_×_512	15.1	113.5
Drive-WM [94]	CVPR’24	192_×_384	15.8	122.7
GenAD [96]	CVPR'24	256_×_448	15.4	184.0
DriveDreamer [97]	ECCV’24	128_×_192	52.6	452.0
Vista [102]	NeurIPS’24	576_×_1024	6.9	89.4
DrivePhysica [221]	arXiv’24	256_×_448	4.0	38.1
驾驶世界[140]	arXiv'24	512_×_1024	7.4	90.9
DriveDreamer-2 [103]	AAAI'25	256_×_448	11.2	55.7
UniFuture [213]	arXiv'25	320_×_576	11.8	99.9
米拉[196]	arXiv'25	360_×_640	4.1	14.9
GeoDrive [177]	arXiv'25	480_×_720	4.1	61.6
LongDWM [195]	arXiv'25	480_×_720	12.3	102.9
MaskGWM [110]	CVPR’25	288_×_512	8.9	65.4
创业板[108]	CVPR’25	576_×_1024	10.5	158.5
艾波娜[155]	ICCV’25	512_×_1024	7.5	82.8
舞台[205]	IROS'25	512_×_768	11.0	242.8
DriVerse [115]	ACMMM’25	480_×_832	18.2	95.2

短序列并使用 FID 评估外观保真度和 FVD 评估时间一致性的质量。为了对 nuScenes 验证分割进行公平比较，最近的方法取得了显着的进展，如表 1 所示。四． DrivePhysica 提供最佳的视觉保真度，而 MiLA 实现最强的时间连贯性，共同建立新的最先进的性能。

B。场景理解

Occ3D-nuScenes 上的 4D 占用预测。 4D 占用预测被视为具有代表性的世界建模任务。给定过去 2 秒的 3D 占用情况，模型可以预测接下来 3 秒的场景动态。评估遵循 Occ3D-nuScenes 协议并报告 mIoU 和每个水平线 IoU 。正如表中所总结的。 V，我们通过输入模态、辅助监督和自我轨迹使用来比较方法，以揭示时空预测的设计选择。使用占用输入的方法优于仅使用相机的变体，并且使用 GT 自我轨迹添加辅助监督进一步减轻了 2-3 秒的性能衰减。在所有方法中，COME（带有 GT ego）实现了最好的平均 mIoU 和每水平 IoU。

C。控制任务

对DMC的评价。大多数研究探讨 世界模型（World Models） 学习控制相关动力学的能力，通常采用基于像素的设置，具有 64 × 64 × 3 个观测值。主要指标是 Episode Return ，定义为超过 1 000 步的累积奖励，在给定 rt ∈ [0，1] 的情况下，理论最大值为 1 000。为了便于比较，表。 VI 报告步骤预算并按任务分数和任务计数总结绩效。结果表明数据效率得到提高，最近的模型以更少的训练步骤达到了强大的性能。然而，不一致的评估协议和任务子集阻碍了对泛化的公平评估，并构建了

TABLE V
PERFORMANCE COMPARISON OF 4D OCCUPANCY FORECASTING ON THE OCC3D-NUSCENES BENCHMARK[1].

Method	Input	Aux. Sup	Ego traj.	mIoU (%) ↑					IoU (%) ↑				
				Recon.	1s	2s	3s	Avg.	Recon.	1s	2s	3s	Avg.
Copy & Paste[2]	Occ	None	Pred.	66.38	14.91	10.54	8.52	11.33	62.29	24.47	19.77	17.31	20.52
OccWorld-O [99]	Occ	None	Pred.	66.38	25.78	15.14	10.51	17.14	62.29	34.63	25.07	20.18	26.63
OccLLaMA-O [18]	Occ	None	Pred.	75.20	25.05	19.49	15.26	19.93	63.76	34.56	28.53	24.41	29.17
RenderWorld-O [163]	Occ	None	Pred.	-	28.69	18.89	14.83	20.80	-	37.74	28.41	24.08	30.08
DTT-O [104]	Occ	None	Pred.	85.50	37.69	29.77	25.10	30.85	92.07	76.60	74.44	72.71	74.58
DFIT-OccWorld-O [181]	Occ	None	Pred.	-	31.68	21.29	15.18	22.71	-	40.28	31.24	25.29	32.27
COME-O [220]	Occ	None	Pred.	-	30.57	19.91	13.38	21.29	-	36.96	28.26	21.86	29.03
DOME-O [100]	Occ	None	GT	83.08	35.11	25.89	20.29	27.10	77.25	43.99	35.36	29.74	36.36
COME-O [220]	Occ	None	GT	-	42.75	32.97	26.98	34.23	-	50.57	43.47	38.36	44.13
OccWorld-T [99]	Camera	Semantic LiDAR	Pred.	7.21	4.68	3.36	2.63	3.56	10.66	9.32	8.23	7.47	8.34
OccWorld-S [99]	Camera	None	Pred.	0.27	0.28	0.26	0.24	0.26	4.32	5.05	5.01	4.95	5.00
RenderWorld-S [163]	Camera	None	Pred.	-	2.83	2.55	2.37	2.58	-	14.61	13.61	12.98	13.73
COME-S [220]	Camera	None	Pred.	-	25.57	18.35	13.41	19.11	-	45.36	37.06	30.46	37.63
OccWorld-D [99]	Camera	Occ	Pred.	18.63	11.55	8.10	6.22	8.62	22.88	18.90	16.26	14.43	16.53
OccWorld-F [99]	Camera	Occ	Pred.	20.09	8.03	6.91	3.54	6.16	35.61	23.62	18.13	15.22	18.99
OccLLaMA-F [18]	Camera	Occ	Pred.	37.38	10.34	8.66	6.98	8.66	38.92	25.81	23.19	19.97	22.99
DFIT-OccWorld-F [181]	Camera	Occ	Pred.	-	13.38	10.16	7.96	10.50	-	19.18	16.85	15.02	17.02
DTT-F [104]	Camera	Occ	Pred.	43.52	24.87	18.30	15.63	19.60	54.31	38.98	37.45	31.89	36.11
DOME-F [100]	Camera	Occ	GT	75.00	24.12	17.41	13.24	18.25	74.31	35.18	27.90	23.44	28.84
COME-F [220]	Camera	Occ	GT	-	26.56	21.73	18.49	22.26	-	48.08	43.84	40.28	44.07

[1] **Note**: Method variants are denoted by their input source: O for ground-truth; camera-based variants include D (TPVFormer), F (FBOCC), T (TPVFormer with semantic LiDAR), and S (self-supervised TPVFormer).
[2] **Copy & Paste**: A naive baseline that repeats the final input frame for all future predictions.

TABLE VI
PERFORMANCE COMPARISON ON THE DMC BENCHMARK[1].

Method	Step	Episode Return↑				Avg. / Total
		Reacher Easy	Cheetah Run	Finger Spin	Walker Walk	
PlaNet [38]	5M	469	496	495	945	333/20
Dreamer [10]	5M	935	895	499	962	823/20
Dreaming [117]	500k	905	566	762	469	610/12
TransDreamer [28]	2M	-	865	-	933	893/4
DreamerPro [118]	1M	873	897	811	-	857/6
MWM [41]	1M	-	670		-	690/7
HRSSM [25]	500k	910	-	960	-	938/3
DisWM [119]	1M	960	820	-	920	879/5

[1] **Note**: Performance comparison on the DMC. Underlined entries indicate scores approximated from their respective reward curves. Average scores (Avg.) are provided as a coarse indicator, given the varying difficulty across tasks.

TABLE VII
PERFORMANCE COMPARISON FOR MANIPULATION TASKS ON RLBENCH.

	Criteria	Methods				
		VidMan [55]	ManiGaussian [53]	ManiGaussian++ [80]	DreMa [60]	TesserAct [78]
Setting	Episode	125	25	25	250	100
	Pixel	224	128	256	128	512
	Depth		✓	✓	✓	✓
	Language	✓	✓	✓		✓
	Proprioception	✓	✓	✓		
	Characteristic	IDM	GS	Bimanual	GS	DiT
Success Rate (%)	Stack Blocks	48	12	-	12	-
	Close Jar	88	28	-	51	44
	Open Drawer	94	76	-	-	80
	Sweep to Dustpan	93	64	92	-	56
	Slide Block	98	24	-	62	-
	Avg.[1] / Total	67/18	45/10	35/10	25/9	63/10

[1] **Avg.**: Average scores are reported only as a coarse indicator, given varying task difficulty.

broadly transferable model across tasks, modalities, and datasets remains an open challenge.

Evaluation on RLBench. RLBench evaluates manipulation with a 7-DoF simulated Franka arm and is widely used to assess whether world models capture task-relevant dynamics and support conditioned action generation. The primary metric is *Success Rate*, defined as the fraction of episodes that reach the goal within the step limit. As summarized in Tab. VII, implementations differ in episode budgets, resolution, and modalities, which complicates like-for-like comparison. Despite this heterogeneity, several trends are evident. Recent methods increasingly leverage multimodal inputs and adopt stronger backbones such as 3DGS and DiT. VidMan achieves a high average success rate on the broadest task, revealing IDM as a promising architectural direction.

Planning on nuScenes. Open-loop planning is treated as a world modeling task on the nuScenes validation split, where models predict ego motion from a limited history. Methods observe 2 s of past trajectories and forecast the next 3 s as 2D BEV waypoints. Evaluation reports *L2* error and *collision rate* at multiple horizons, and Tab. VIII summarizes results by input modality, auxiliary supervision, and metric settings. Under this shared protocol, a clear tradeoff emerges. UniAD+DriveWorld achieves the lowest *L2* with extensive auxiliary supervision, whereas SSR attains the best collision rate with competitive *L2* without extra supervision. Camera-based methods now surpass models that use privileged occupancy, reflecting the growing maturity of E2E planning.

VI. CHALLENGES AND TRENDS

This section reviews the open challenges and emerging directions for world models in embodied AI. We organize them across three dimensions: §VI-A Data & Evaluation, §VI-B Computational Efficiency, and §VI-C Modeling Strategies.

A. Data & Evaluation

Challenges. From a data perspective, the central challenge lies in the scarcity and heterogeneity of existing corpora. Although embodied AI spans diverse domains such as navigation,

原文第 15 页 | 中文翻译初稿

15

表五

OCC3D-NUSCENES 基准上 4D 占用预测的性能比较1。

方法	输入	AuxSu	Eotraj		mI	oU (%)	↑			Io	U (%)	↑	
	p	。 p	g .	侦察。	1s	2s	3s	平均	侦察。	1s	2s	3s	平均
复制并粘贴2	Occ	无	预测	66.38	14.91	10.54	8.52	11.33	62.29	24.47	19.77	17.31	20.52
OccWorld-O [99]	Occ	无	预测	66.38	25.78	15.14	10.51	17.14	62.29	34.63	25.07	20.18	26.63
OccLLaMA-O [18]	Occ	无	预测	75.20	25.05	19.49	15.26	19.93	63.76	34.56	28.53	24.41	29.17
RenderWorld-O [163]	Occ	无	预测	-	28.69	18.89	14.83	20.80	-	37.74	28.41	24.08	30.08
DTT-O [104]	Occ	无	预测	85.50	37.69	29.77	25.10	30.85	92.07	76.60	74.44	72.71	74.58
DFIT-OccWorld-O [181]	Occ	无	预测	-	31.68	21.29	15.18	22.71	-	40.28	31.24	25.29	32.27
COME-O [220]	Occ	无	预测	-	30.57	19.91	13.38	21.29	-	36.96	28.26	21.86	29.03
DOME-O [100]	Occ	无	GT	83.08	35.11	25.89	20.29	27.10	77.25	43.99	35.36	29.74	36.36
COME-O [220]	Occ	无	GT	-	42.75	32.97	26.98	34.23	-	50.57	43.47	38.36	44.13
OccWorld-T [99]	相机	语义激光雷达	预测	7.21	4.68	3.36	2.63	3.56	10.66	9.32	8.23	7.47	8.34
OccWorld-S [99]	相机	无	预测	0.27	0.28	0.26	0.24	0.26	4.32	5.05	5.01	4.95	5.00
RenderWorld-S [163]	相机	无	预测	-	2.83	2.55	2.37	2.58	-	14.61	13.61	12.98	13.73
COME-S [220]	相机	无	预测	-	25.57	18.35	13.41	19.11	-	45.36	37.06	30.46	37.63
OccWorld-D [99]	相机	Occ	预测	18.63	11.55	8.10	6.22	8.62	22.88	18.90	16.26	14.43	16.53
OccWorld-F [99]	相机	Occ	预测	20.09	8.03	6.91	3.54	6.16	35.61	23.62	18.13	15.22	18.99
OccLLaMA-F [18]	相机	Occ	预测	37.38	10.34	8.66	6.98	8.66	38.92	25.81	23.19	19.97	22.99
DFIT-OccWorld-F [181]	相机	Occ	预测	-	13.38	10.16	7.96	10.50	-	19.18	16.85	15.02	17.02
DTT-F [104]	相机	Occ	预测	43.52	24.87	18.30	15.63	19.60	54.31	38.98	37.45	31.89	36.11
DOME-F [100]	相机	Occ	GT	75.00	24.12	17.41	13.24	18.25	74.31	35.18	27.90	23.44	28.84
COME-F [220]	相机	Occ	GT	-	26.56	21.73	18.49	22.26	-	48.08	43.84	40.28	44.07

1 注意：方法变体由其输入源表示：O 代表真实值；基于摄像头的变体包括 D (TPVFormer)、F (FBOCC)、T（带有语义 LiDAR 的 TPVFormer）和 S（自监督 TPVFormer）。 2 复制和粘贴：一个简单的基线，为所有未来的预测重复最终输入帧。

表 VI DMC 基准的性能比较1。
表七

RLBENCH 上操纵任务的性能比较。

|方法|步骤|轻松到达|剧集
猎豹奔跑|返回_↑_
手指旋转|行者行走|平均。 / 总计|
|---|---|---|---|---|---|---|
|环网[38]|5M|469|496|495|945|333/20|
|梦想家[10]|5M|935|895|499|962|823/20|

|梦想[117]|500k|905|566|762|469|610/12|
|TransDreamer [28]|2M|-|865|-|933|893
/4|
|DreamerPro [118]|1M|873|897|811|-|857/6|
|MWM [41]|1M|-|670||-|690
/7|
|HRSSM [25]|500k|910|-|960|-|938
/3|
|DisWM [119]|1M|960|820|-|920|879
/5|

1 注：DMC 上的性能比较。带下划线的条目表示根据各自的奖励曲线得出的近似分数。考虑到任务之间的难度不同，平均分数（Avg.）作为粗略指标提供。

|条件|VidMan [55|]
ManiGaussian [53|方法
]
ManiGaussian++ [80|]
DreMa [60|]
TesserAct [78]|
|---|---|---|---|---|---|
|集|125|25|25|250|100|
|像素|224|128|256|128|512|
|ting
深度||✓|✓|✓|✓|
|设置
语言|✓|✓|✓||✓|
|本体感觉|✓|✓|✓|||
|特点|IDM|GS|双手|GS|DiT|
|%)
堆栈块|48|12|-|12|-|
|e (
关闭罐子|88|28|-|51|44|
|老鼠
打开抽屉|94|76|-|-|80|
|ss
扫到簸箕|93|64|92|-|56|
|ucce
滑块|98|24|-|62|-|
|S
平均1 /总计|67/18|45/10|35/10|25/9|63/10|

1 平均：考虑到不同的任务难度，平均分数仅作为粗略指标报告。

跨任务、模式和数据集的广泛可转移模型仍然是一个开放的挑战。

RLBench 上的评估。 RLBench 使用 7-DoF 模拟 Franka 臂评估操作，并广泛用于评估 世界模型（World Models） 是否捕获任务相关的动态并支持条件动作生成。主要指标是 Success Rate ，定义为在步骤限制内达到目标的情节分数。正如表中所总结的。 VII，实施方案在剧集预算、分辨率和模式方面存在差异，这使得同类比较变得复杂。尽管存在这种异质性，但有几个趋势是显而易见的。最近的方法越来越多地利用多模式输入并采用更强大的骨干网，例如 3DGS 和 DiT。 VidMan 在最广泛的任务上实现了较高的平均成功率，表明 IDM 是一个有前途的架构方向。

规划 nuScenes。开环规划被视为 nuScenes 验证拆分上的世界建模任务，其中模型根据有限的历史预测自我运动。方法观察过去 2 秒的轨迹并预测接下来的 3 秒作为 2D BEV 航路点。评估报告_L2_错误和_碰撞率_

在多个层面上，以及选项卡。 VIII 按输入方式、辅助监督和指标设置总结结果。在这个共享协议下，出现了明显的权衡。 UniAD+DriveWorld 通过广泛的辅助监督实现了最低的 L2，而 SSR 在没有额外监督的情况下通过竞争性的 L2 获得了最佳碰撞率。基于摄像头的方法现在超越了使用特权占用的模型，反映出端到端规划的日益成熟。

六．挑战和趋势

本节回顾了 具身智能（Embodied AI） 中 世界模型（World Models） 的开放挑战和新兴方向。我们从三个维度组织它们：§VI-A 数据和评估、§VI-B 计算效率和§VI-C 建模策略。

A。数据与评估

挑战。从数据的角度来看，核心挑战在于现有语料库的稀缺性和异质性。尽管 具身智能（Embodied AI） 跨越导航等不同领域，

TABLE VIII
PERFORMANCE COMPARISON FOR OPEN-LOOP PLANNING ON THE NUSCENES VALIDATION SPLIT[1].

Method	Input	Aux. Sup.[2]	L2 (m) ↓				Collision Rate (%) ↓			
			1s	2s	3s	Avg.	1s	2s	3s	Avg.
UniAD [261]	Camera	Map & Box & Motion & Tracklets & Occ	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31
UniAD+DriveWorld [93]	Camera	Map & Box & Motion & Tracklets & Occ	0.34	0.67	1.07	0.69	0.04	0.12	0.41	0.19
GenAD [98]	Camera	Map & Box & Motion	0.36	0.83	1.55	0.91	0.06	0.23	1.00	0.43
FSDrive [107]	Camera	Map & Box & QA	0.40	0.89	1.60	0.96	0.07	0.12	1.02	0.40
OccWorld-T [99]	Camera	Semantic LiDAR	0.54	1.36	2.66	1.52	0.12	0.40	1.59	0.70
Doe-1 [141]	Camera	QA	0.50	1.18	2.11	1.26	0.04	0.37	1.19	0.53
SSR [167]	Camera	None	0.24	0.65	1.36	0.75	0.00	0.10	0.36	0.15
OccWorld-S [99]	Camera	None	0.67	1.69	3.13	1.83	0.19	1.28	4.59	2.02
Epona [155]	Camera	None	0.61	1.17	1.98	1.25	0.01	0.22	0.85	0.36
RenderWorld [163]	Camera	None	0.48	1.30	2.67	1.48	0.14	0.55	2.23	0.97
Drive-OccWorld [164]	Camera	None	0.32	0.75	1.49	0.85	0.05	0.17	0.64	0.29
OccWorld-D [99]	Camera	Occ	0.52	1.27	2.41	1.40	0.12	0.40	2.08	0.87
OccWorld-F [99]	Camera	Occ	0.45	1.33	2.25	1.34	0.08	0.42	1.71	0.73
OccLLaMA-F [18]	Camera	Occ	0.38	1.07	2.15	1.20	0.06	0.39	1.65	0.70
DTT-F [104]	Camera	Occ	0.35	1.01	1.89	1.08	0.08	0.33	0.91	0.44
DFIT-OccWorld-V [181]	Camera	Occ	0.42	1.14	2.19	1.25	0.09	0.19	1.37	0.55
NeMo [168]	Camera	Occ	0.39	0.74	1.39	0.84	0.00	0.09	0.82	0.30
OccWorld-O [99]	Occ	None	0.43	1.08	1.99	1.17	0.07	0.38	1.35	0.60
OccLLaMA-O [18]	Occ	None	0.37	1.02	2.03	1.14	0.04	0.24	1.20	0.49
RenderWorld-O [163]	Occ	None	0.35	0.91	1.84	1.03	0.05	0.40	1.39	0.61
DTT-O [104]	Occ	None	0.32	0.91	1.76	1.00	0.08	0.32	0.51	0.30
DFIT-OccWorld-O [181]	Occ	None	0.38	0.96	1.73	1.02	0.07	0.39	0.90	0.45

[1] **Note**: Results are reported following the UniAD [261] protocol. Method variants are denoted by their input source: O (ground-truth), V (camera-predicted), D (TPVFormer), F (FBOCC), *etc*.
[2] **Aux. Sup.**: Abbreviation for auxiliary supervision, which refers to additional training signals beyond ground-truth trajectories.

manipulation, and autonomous driving, a unified large-scale dataset remains lacking. This fragmentation constrains the capacity of world models and substantially hinders their ability to generalize, especially when model scale grows faster than available task-relevant data. Multi-agent interaction data are particularly underrepresented, limiting progress on socially aware and cooperative world models.

Evaluation practices face similar limitations. Most metrics, such as FID and FVD emphasize pixel fidelity while ignoring physical consistency, dynamics, and causality. As a result, models may excel at appearance-level realism yet exhibit physically implausible or causally inconsistent rollouts. Existing benchmarks (*e.g.*, EWM-Bench [262]) begin to introduce more structured metrics, but remain task-specific and lack cross-domain standards, and there is still a gap between how video simulators (*e.g.*, Sora-style models) are evaluated and how embodied controllers are judged.

Future Directions. Recent initiatives such as OpenDV-2K [96], VideoMix22M [14], and VisionMix163M [85] highlight the growing focus on large-scale pretraining and broader modality coverage, yet resources remain fragmented and domain specific. A key direction is to construct unified, multimodal, cross-domain benchmarks that couple scenes, actions, and rewards, and that explicitly cover multi-agent interaction and safety-critical events. In parallel, evaluation protocols should move beyond perceptual realism toward metrics that assess physical consistency, causal reasoning, and long-horizon error accumulation. Bridging the evaluation gap between video-centric world simulators and embodied action benchmarks is essential if large video models are to serve as reliable substrates for downstream control.

B. Computational Efficiency

Challenges. Embodied AI tasks encounter significant challenges in computational efficiency, particularly in real-time settings. Although Transformers and diffusion-based decoders achieve strong performance, their high inference costs and memory footprint conflict with the latency and energy constraints of onboard robotic platforms. This tension is exacerbated by the mismatch between rapidly growing model sizes and relatively modest compute budgets on edge devices. As a result, many practical systems still rely on compact recurrent models and global latent vectors, accepting limited capacity in exchange for predictable real-time behavior.

Future Directions. To address this challenge, future research should emphasize architectures and training schemes that jointly optimize fidelity and efficiency. Model compression techniques such as quantization, pruning, low-rank and sparse computation are natural candidates to reduce latency and memory while preserving control-relevant performance. Sequence models such as SSMs offer linear-time scaling and show promise for long-horizon reasoning under tight compute budgets. More broadly, it may be beneficial to design world models as adaptive systems that can adjust resolution, rollout horizon, or modality usage to available resources. For example, high-capacity video-diffusion components may be used primarily for offline data generation or simulation, while lightweight cores are reserved for online control and safety monitoring.

C. Modeling Strategy

Challenges. Despite rapid progress, world models still struggle with long-horizon temporal dynamics and efficient spatial representations. A central difficulty is balancing recurrent simulation and global prediction: autoregressive designs are compact and sample-efficient but suffer from error accumulation, whereas global prediction improves multi-step

原文第 16 页 | 中文翻译初稿

16

表八

NUSCENES 验证分割的开环规划性能比较1。

方法	输入	AuxSu2		L2 (	m) ↓			碰撞	比率 (%) ↓	
		。页	1s	2s	3s	平均	1s	2s	3s	平均
UniAD [261]	相机	地图&盒子&运动&轨迹&Occ	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31
UniAD+DriveWorld [93]	相机	地图&盒子&运动&轨迹&Occ	0.34	0.67	1.07	0.69	0.04	0.12	0.41	0.19
GenAD [98]	相机	地图、盒子和运动	0.36	0.83	1.55	0.91	0.06	0.23	1.00	0.43
FSDrive [107]	相机	地图、盒子和 QA	0.40	0.89	1.60	0.96	0.07	0.12	1.02	0.40
OccWorld-T [99]	相机	语义激光雷达	0.54	1.36	2.66	1.52	0.12	0.40	1.59	0.70
Doe-1 [141]	相机	QA	0.50	1.18	2.11	1.26	0.04	0.37	1.19	0.53
SSR [167]	相机	无	0.24	0.65	1.36	0.75	0.00	0.10	0.36	0.15
OccWorld-S [99]	相机	无	0.67	1.69	3.13	1.83	0.19	1.28	4.59	2.02
Epona [155]	相机	无	0.61	1.17	1.98	1.25	0.01	0.22	0.85	0.36
渲染世界 [163]	相机	无	0.48	1.30	2.67	1.48	0.14	0.55	2.23	0.97
Drive-OccWorld [164]	相机	无	0.32	0.75	1.49	0.85	0.05	0.17	0.64	0.29
OccWorld-D [99]	相机	Occ	0.52	1.27	2.41	1.40	0.12	0.40	2.08	0.87
OccWorld-F [99]	相机	Occ	0.45	1.33	2.25	1.34	0.08	0.42	1.71	0.73
OccLLaMA-F [18]	相机	Occ	0.38	1.07	2.15	1.20	0.06	0.39	1.65	0.70
DTT-F [104]	相机	Occ	0.35	1.01	1.89	1.08	0.08	0.33	0.91	0.44
DFIT-OccWorld-V [181]	相机	Occ	0.42	1.14	2.19	1.25	0.09	0.19	1.37	0.55
NeMo [168]	相机	Occ	0.39	0.74	1.39	0.84	0.00	0.09	0.82	0.30
OccWorld-O [99]	Occ	无	0.43	1.08	1.99	1.17	0.07	0.38	1.35	0.60
OccLLaMA-O [18]	Occ	无	0.37	1.02	2.03	1.14	0.04	0.24	1.20	0.49
RenderWorld-O [163]	Occ	无	0.35	0.91	1.84	1.03	0.05	0.40	1.39	0.61
DTT-O [104]	Occ	无	0.32	0.91	1.76	1.00	0.08	0.32	0.51	0.30
DFIT-OccWorld-O [181]	Occ	无	0.38	0.96	1.73	1.02	0.07	0.39	0.90	0.45

1 注：结果按照 UniAD [261] 方案报告。方法变体由其输入源表示：O（地面实况）、V（相机预测）、D（TPVFormer）、F（FBOCC）等。 2 辅助。高级。 ：辅助监督的缩写，指的是超出真实轨迹的额外训练信号。

操纵和自动驾驶，仍然缺乏统一的大规模数据集。这种碎片限制了 世界模型（World Models） 的容量，并极大地阻碍了它们的泛化能力，特别是当模型规模增长速度快于可用的任务相关数据时。多智能体交互数据的代表性尤其不足，限制了社交意识和合作性 世界模型（World Models） 的进展。

评估实践也面临着类似的限制。大多数指标（例如 FID 和 FVD）强调像素保真度，而忽略物理一致性、动态性和因果关系。因此，模型可能在外观层面的真实性方面表现出色，但表现出物理上不可信或因果不一致的 展开轨迹（Rollouts）。现有的基准（例如，EWM-Bench [262]）开始引入更多结构化指标，但仍然针对特定任务且缺乏跨域标准，并且如何评估视频模拟器（例如，Sora 风格模型）与如何判断实体控制

器之间仍然存在差距。

未来的方向

。最近的举措，如 OpenDV2K [96]、VideoMix22M [14] 和 VisionMix163M [85] 突显了人们对大规模预训练和更广泛模态覆盖范围的日益关注，但资源仍然分散且特定于领域。一个关键方向是构建统一的、多模式的、跨领域的基准，将场景、动作和奖励结合起来，并明确涵盖多智能体交互和安全关键事件。与此同时，评估协议应该超越感知现实主义，转向评估物理一致性、因果推理和 长时程（Long-Horizon） 错误累积的指标。如果大型视频模型要作为下游控制的可靠基础，那么弥合以视频为中心的世界模拟器和具体动作基准之间的评估差距至关重要。

B。计算效率

挑战。 具身智能（Embodied AI） 任务在计算效率方面遇到重大挑战，特别是在实时设置中。尽管 Transformer 和基于扩散的解码器实现了强大的性能，但它们的高推理成本和内存占用与机载机器人平台的延迟和能量限制相冲突。快速增长的模型大小与边缘设备上相对适度的计算预算之间的不匹配加剧了这种紧张局势。因此，许多实际系统仍然依赖于紧凑的循环模型和全局潜在向量，接受有限的容量以换取可预测的实时行为。

未来的方向。为了应对这一挑战，未来的研究应该强调共同优化保真度和效率的架构和训练方案。量化、剪枝、低秩和稀疏计算等模型压缩技术是减少延迟和内存，同时保留控制相关性能的自然候选技术。SSM 等序列模型提供线性时间缩放，并在计算预算紧张的情况下显示出长期推理的前景。更广泛地说，将 世界模型（World Models） 设计为自适应系统可能是有益的，可以调整分辨率、展开轨迹（Rollout）范围或可用资源的模态使用。例如，大容量视频扩散组件可主要用于离线数据生成或模拟，而轻量级核心则用于在线控制和安全监控。

C。建模策略

挑战。尽管取得了快速进展，世界模型（World Models） 仍然在 长时程（Long-Horizon） 时间动态和高效空间表示方面苦苦挣扎。一个中心困难是平衡循环仿真和全局预测：自回归设计紧凑且样本效率高，但会受到误差累积的影响，而全局预测则改进了多步骤

coherence at the cost of heavy computation and weaker closed-loop interactivity. On the spatial side, efficiency and inductive bias remain bottlenecks. Latent vectors, token sequences, spatial grids, and decomposed rendering representations each exhibit distinct trade-offs between expressiveness, geometric fidelity, and computational cost. Highly realistic video or 3D generative models often capture correlations in appearance without robust causal structure, leading to visually plausible yet physically incorrect predictions.

Future Directions. Several promising avenues have emerged to address these bottlenecks. SSM-based architectures (*e.g.*, Mamba) align naturally with autoregressive modeling and offer linear-time scalability, while masked prediction frameworks (*e.g.*, JEPA-style objectives) support global, parallel prediction and stronger representation learning. Hybrid designs that combine local autoregression with global or masked updates, possibly supported by explicit memory modules or hierarchical planning, could reduce error accumulation and improve long-horizon stability.

On the spatial axis, future work should seek representations that encode geometry and physics more explicitly while remaining tractable for control. For example, hybrid token-grid or neural field-based models can expose causal structure and uncertainty to downstream policies. Looking ahead, a unifying perspective is to view the world model as shared infrastructure that coordinates perception, prediction, and decision-making for embodied agents. Realizing this perspective will require architectures that tightly integrate temporal and spatial modeling, scale effectively with model and data size, and support principled multi-agent, multi-task, and multi-modal interactions.

VII. Conclusion

This survey has organized world models for embodied AI within a three-axis framework spanning decision coupling, temporal modeling, and spatial representation. Using this lens, we mapped methods in robotics, autonomous driving, and general-purpose video modeling into a unified design space, summarized representative architectures and benchmarks, and analyzed how different combinations of these axes give rise to distinct trade-offs in fidelity, efficiency, and interactivity.

Looking ahead, progress in embodied world models is likely to be driven by three converging trends: scaling general-purpose video and 3D world models, tightening their coupling to control through action- and reward-conditioned training, and improving robustness via causally and physically grounded objectives. Rather than a single unified benchmark, we expect families of cross-domain, multimodal suites that jointly assess prediction quality, decision making, and safety under sim-to-real shift. Although world models are unlikely to close the simulation-reality gap entirely, advances in geometry, physics, and uncertainty modeling should make simulated rollouts more informative for downstream planning and policy learning, positioning world models as shared predictive infrastructure within embodied AI systems.

References

[1] D. Batra, A. X. Chang, S. Chernova, A. J. Davison, J. Deng, V. Koltun, S. Levine, J. Malik, I. Mordatch, R. Mottaghi *et al.*, "Rearrangement: A challenge for embodied AI," *arXiv preprint arXiv:2011.01975*, 2020. 1

[2] T. Gupta, W. Gong, C. Ma, N. Pawlowski, A. Hilmkil, M. Scetbon, M. Rigter, A. Famoti, A. J. Llorens, J. Gao *et al.*, "The essential role of causality in foundation world models for embodied AI," *arXiv preprint arXiv:2402.06665*, 2024. 1

[3] Y. Liu, W. Chen, Y. Bai, X. Liang, G. Li, W. Gao, and L. Lin, "Aligning cyber space with physical world: A comprehensive survey on embodied AI," *IEEE/ASME Transactions on Mechatronics*, pp. 1–28, 2025. 1

[4] J. Ding, Y. Zhang, Y. Shang, Y. Zhang, Z. Zong, J. Feng, Y. Yuan, H. Su, N. Li, N. Sukiennik *et al.*, "Understanding world or predicting future? S comprehensive survey of world models," *ACM Computing Surveys*, 2024. 1

[5] A. Clark, *Being there: Putting brain, body, and world together again*. MIT press, 1998. 1

[6] L. W. Barsalou, "Perceptions of perceptual symbols," *Behavioral and brain sciences*, vol. 22, no. 4, pp. 637–660, 1999. 1

[7] K. Friston, "The free-energy principle: a unified brain theory?" *Nature reviews neuroscience*, vol. 11, no. 2, pp. 127–138, 2010. 1

[8] P. Fung, Y. Bachrach, A. Celikyilmaz, K. Chaudhuri, D. Chen, W. Chung, E. Dupoux, H. Jégou, A. Lazaric, A. Majumdar *et al.*, "Embodied AI agents: Modeling the world," *arXiv preprint arXiv:2506.22355*, 2025. 1

[9] D. Ha and J. Schmidhuber, "Recurrent world models facilitate policy evolution," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2018, pp. 2455–2467. 1, 4

[10] D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi, "Dream to Control: Learning behaviors by latent imagination," in *Int. Conf. Learn. Represent.*, 2020, pp. 1–13. 1, 4, 15

[11] D. Hafner, T. P. Lillicrap, M. Norouzi, and J. Ba, "Mastering atari with discrete world models," in *Int. Conf. Learn. Represent.*, 2021, pp. 1–15. 1, 4

[12] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse control tasks through world models," *Nature*, vol. 640, no. 8059, pp. 647–653, 2025. 1, 3, 4

[13] OpenAI, "Video generation models as world simulators," 2024, accessed: 2025-09-14. [Online]. Available: https://openai.com/index/video-generation-models-as-world-simulators/ 1, 10

[14] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, A. Zholus *et al.*, "V-JEPA 2: Self-supervised video models enable understanding, prediction and planning," *arXiv preprint arXiv:2506.09985*, 2025. 1, 3, 4, 10, 12, 16

[15] A. Venkatraman, M. Hebert, and J. Bagnell, "Improving multi-step prediction of learned time series models," in *AAAI Conf. Artif. Intell.*, vol. 29, no. 1, 2015. 1

[16] K. Asadi, D. Misra, and M. Littman, "Lipschitz continuity in model-based reinforcement learning," in *Int. Conf. Mach. Learn.*, 2018, pp. 264–273. 1

[17] B. Agro, Q. Sykora, S. Casas, T. Gilles, and R. Urtasun, "UnO: Unsupervised occupancy fields for perception and forecasting," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 14 487–14 496. 1, 10

[18] J. Wei, S. Yuan, P. Li, Q. Hu, Z. Gan, and W. Ding, "OccLLaMA: An occupancy-language-action generative world model for autonomous driving," *arXiv preprint arXiv:2409.03272*, 2024. 1, 5, 7, 15, 16

[19] Z. Zhu, X. Wang, W. Zhao, C. Min, N. Deng, M. Dou, Y. Wang, B. Shi, K. Wang, C. Zhang *et al.*, "Is Sora a World Simulator? A comprehensive survey on general world models and beyond," *arXiv preprint arXiv:2405.03520*, 2024. 1

[20] Y. Guan, H. Liao, Z. Li, J. Hu, R. Yuan, G. Zhang, and C. Xu, "World models for autonomous driving: An initial survey," *IEEE Transactions on Intelligent Vehicles*, pp. 1–17, 2024. 1

[21] T. Feng, W. Wang, and Y. Yang, "A survey of world models for autonomous driving," *arXiv preprint arXiv:2501.11260*, 2025. 1

[22] Y. Lu, X. Ren, J. Yang, T. Shen, Z. Wu, J. Gao, Y. Wang, S. Chen, M. Chen, S. Fidler *et al.*, "InfiniCube: Unbounded and controllable dynamic 3D driving scene generation with world-guided video models," in *Int. Conf. Comput. Vis.*, 2025. 3, 5, 9

[23] Y. Chen, Y. Wang, and Z. Zhang, "DrivingGPT: Unifying driving world modeling and planning with multi-modal autoregressive transformers," *arXiv preprint arXiv:2412.18607*, 2024. 3, 6

[24] R. D. Smallwood and E. J. Sondik, "The optimal control of partially observable Markov processes over a finite horizon," *Operations research*, vol. 21, no. 5, pp. 1071–1088, 1973. 2

[25] R. Sun, H. Zang, X. Li, and R. Islam, "Learning latent dynamic robust representations for world models," in *Int. Conf. Mach. Learn.*, 2024, pp. 47 234–47 260. 2, 5, 15

原文第 17 页 | 中文翻译初稿

17 号

一致性是以大量计算和较弱的闭环交互性为代价的。在空间方面，效率和归纳偏差仍然是瓶颈。潜在向量、标记序列、空间网格和分解的渲染表示都在表现力、几何保真度和计算成本之间表现出不同的权衡。高度逼真的视频或 3D 生成模型通常会捕获外观上的相关性，而没有强大的因果结构，从而导致视觉上合理但物理上不正确的预测。

未来的方向。已经出现了一些有希望的途径来解决这些瓶颈。基于 SSM 的架构（例如，Mamba）自然地与自回归建模保持一致，并提供线性时间可扩展性，而屏蔽预测框架（例如，JEPA 风格的目标）支持全局、并行预测和更强的表示学习。将局部自回归与全局或屏蔽更新相结合的混合设计，可能受到显式内存模块或分层规划的支持，可以减少错误累积并提高长期稳定性。

在空间轴上，未来的工作应该寻求更明确地编码几何和物理的表示，同时保持易于控制。例如，混合令牌网格或基于神经场的模型可以暴露下游策略的因果结构和不确定性。展望未来，一个统一的观点是将 世界模型（World Model）视为协调具身智能体的感知、预测和决策的共享基础设施。实现这一观点需要架构紧密集成时间和空间建模，根据模型和数据大小有效扩展，并支持有原则的多智能体、多任务和多模式交互。

七．结论

这项调查在涵盖决策耦合、时间建模和空间表示的三轴框架内组织了 世界模型（World Models）和 具身智能（Embodied AI）。使用这个镜头，我们将机器人、自动驾驶和 通用（General-Purpose）视频建模中的方法映射到统一的设计空间中，总结了代表性架构和基准，并分析了这些轴的不同组合如何在保真度、效率和交互性方面产生不同的权衡。

展望未来，世界模型（World Models）的进展可能会受到三个融合趋势的推动：缩放 通用（General-Purpose）视频和 3D 世界模型（World Models），通过行动和奖励条件训练加强它们与控制的耦合，以及通过因果和物理目标提高鲁棒性。我们期望跨域、多模式套件系列能够共同评估模拟-真实转变下的预测质量、决策和安全性，而不是单一的统一基准。尽管 世界模型（World Models）不太可能完全缩小模拟与现实的差距，但几何、物理和不确定性建模的进步应该使模拟的 展开轨迹（Rollouts）为下游规划和策略学习提供更多信息，将 世界模型（World Models）定位为 具身智能（Embodied AI）系统内的共享预测基础设施。

参考文献

[1] D. Batra、A. X. Chang、S. Chernova、A. J. Davison、J. Deng、V. Koltun、S. Levine、J. Malik、I. Mordatch、R. Mottaghi 等人，“重排：

具身智能（Embodied AI）的挑战，”arXiv 预印本 arXiv:2011.01975，2020 年。

1

- [2] T. Gupta、W. Kong、C. Ma、N. Pawlowski、A. Hilmkil、M. Scetbon、M. Rigter、A. Famoti、A. J. Llorens、J. Gau 等人，“因果关系在 具身智能（Embodied AI）基础 世界模型（World Models）中的重要作用，”arXiv 预印本arXiv:2402.06665, 2024. 1

- [3] Y. Liu、W. Chen、Y. Bai、X. Liang、G. Li、W. Gau 和 L. Lin，“使网络空间与物理世界保持一致：对 具身智能（Embodied AI）的综合调查”，IEEE/ASME Transactions on Mechatronics，第 1-28 页，

2025 年。 1

- [4] J. Ding, Y. 张, Y. Shang, Y. 张, Z. Zong, J. Feng, Y. Yuan, H. Su, N. Li, N. Sukiennik et al. , “理解世界还是预测未来？ 世界模型（World Models） 的综合调查，”ACM 计算调查，2024. 1

- [5] A. Clark，身临其境：将大脑、身体和世界再次放在一起。麻省理工学院出版社，1998. 1

- [6] L. W. Barsalou，“感知符号的感知”，行为与脑科学，卷。 22、没有。 4，第 637-660 页，1999 年。 1

- [7] K. Friston，“自由能原理：统一的大脑理论？” 自然评论神经科学，卷。 11、没有。 2，第 127-138 页，2010 年。1

- [8] P. Fung、Y. Bachrach、A. Celikyilmaz、K. Chaudhuri、D. Chen、W. Chung、E. Dupoux、H. Jégou、A. Lazaric、A. Majumdar 等人，“具身智能（Embodied AI） 智能体：模拟世界”，arXiv 预印本 arXiv:2506.22355 , 2025. 1

- [9] D. Ha 和 J. Schmidhuber，“经常性 世界模型（World Models） 促进策略演变”，载于 Annu。会议。神经。通知。过程。系统，2018 年，第 2455-2467 页。1, 4

- [10] D. Hafner、T. Lillicrap、J. Ba 和 M. Norouzi，“控制梦想：通过潜在想象力学习行为”，《Int》。会议。学习。代表._，2020 年，第 1-13 页。 1、4、15

- [11] D. Hafner、T. P. Lillicrap、M. Norouzi 和 J. Ba，“使用离散 世界模型（World Models） 掌握 atari”，载于_Int。会议。学习。代表._，2021 年，第 1-15 页。 1, 4

- [12] D. Hafner、J. Pasukonis、J. Ba 和 T. Lillicrap，“通过 世界模型（World Models） 掌握多样化的控制任务”，Nature，卷。 640，没有。 8059，第 647-653 页，2025 年。1,3,4

- [13] OpenAI，“视频生成模型作为世界模拟器”，2024 年，访问时间：2025-09-14。 [在线]。可用：https://openai.com/index/video- Generation-models-as-world-simulators/ 1, 10

- [14] M. Assran、A. Bardes、D. Fan、Q. Garrido、R. Howes、M. Muckley、A. Rizvi、C. Roberts、K. Sinha、A. Zholus et al. ，“V-JEPA 2：自监督视频模型实现理解、预测和规划”，arXiv 预印本 arXiv:2506.09985， 2025 年 1、3、4、10、12、16

- [15] A. Venkatraman、M. Hebert 和 J. Bagnell，“改进学习时间序列模型的多步预测”，载于_AAAI Conf。阿蒂夫。情报._，卷。 29、没有。 2015年1月1日

- [16] K. Asadi、D. Misra 和 M. Littman，“基于模型的 强化学习（Reinforcement Learning） 中的 Lipschitz 连续性”，载于_Int。会议。马赫。学习._，2018 年，第 264-273 页。 1

- [17] B. Agro、Q. Sykora、S. Casas、T. Gilles 和 R. Urtasun，“UnO：用于感知和预测的无监督占用场”，载于 _IEEE Conf。计算。维斯。模式识别，2024 年，第 14 487-14 496 页。1, 10

- [18] J. Wei、S. Yuan、P. Li、Q. Hu、Z. Gan 和 W. Ding，“OccLLaMA：用于自动驾驶的占用语言动作生成 世界模型（World Model）”，arXiv 预印本 arXiv:2409.03272，2024. 1, 5, 7, 15, 16

- [19] 朱子、王X、赵W、C

。 Min, N. Deng, M. Dou, Y. Wang, B. Shi, K. Wang, C. Zhang et al. , “Sora 是一个世界模拟器吗？对一般 世界模型（World Models） 及其他内容的综合调查”, arXiv 预印本 arXiv:2405.03520 , 2024. 1

- [20] Y.guan、H.Liao、Z.Li、J.Hu、R.Yuan、G.Zhang 和 C.Xu，“自动驾驶的 世界模型（World Models）：初步调查”，IEEE 智能车辆交易，第 1-17 页，2024 年。 1

- [21] T. Feng、W. Wang 和 Y. Yang，“自动驾驶 世界模型（World Models） 的调查”，arXiv 预印本 arXiv:2501.11260，2025. 1

- [22] Y. Lu, X. Ren, J. Yang, T. Shen, Z. Wu, J. Gau, Y. Wang, S. Chen, M. Chen, S. Fidler et al.，“InfiniCube：使用世界引导视频模型生成无界且可控的动态 3D 驾驶场景”，《Int.》。会议。计算。视觉_，2025. 3, 5, 9

- [23] Y. Chen、Y. Wang 和 Z. Zhang，“DrivingGPT：使用多模态自回归变压器统一驾驶世界建模和规划”，arXiv 预印本 arXiv:2412.18607，2024. 3, 6

- [24] R. D. Smallwood 和 E. J. Sondik，“有限范围内部分可观测马尔可夫过程的最优控制”，运筹学，卷。21、没有。 5，第 1071-1088 页，1973 年。2

- [25] R. Sun、H. Zang、X. Li 和 R. Islam，“学习 世界模型（World Models） 的潜在动态鲁棒表示”，载于_Int。会议。马赫。学习._，2024 年，第 47 234-47 260. 2, 5, 15 页

[26] G. Zhai, X. Zhang, and N. Navab, "Recurrent world model with tokenized latent states," in *Int. Conf. Learn. Represent. Worksh.*, 2025, pp. 1–7. 2

[27] E. Aljalbout, M. Krinner, A. Romero, and D. Scaramuzza, "Accelerating model-based reinforcement learning with state-space world models," in *Int. Conf. Learn. Represent. Worksh.*, 2025. 2

[28] C. Chen, Y.-F. Wu, J. Yoon, and S. Ahn, "TransDreamer: Reinforcement learning with transformer world models," *arXiv preprint arXiv:2202.09481*, 2022. 2, 4, 6, 15

[29] J. Robine, M. Höftmann, T. Uelwer, and S. Harmeling, "Transformer-based world models are happy with 100k interactions," in *Int. Conf. Learn. Represent.*, 2023, pp. 1–13. 2, 4, 6

[30] W. Zhang, G. Wang, J. Sun, Y. Yuan, and G. Huang, "STORM: Efficient stochastic transformer based world models for reinforcement learning," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2023, pp. 27 147–27 166. 2

[31] C. Zhu, R. Yu, S. Feng, B. Burchfiel, P. Shah, and A. Gupta, "Unified World Models: Coupling video and action diffusion for pretraining on large robotic datasets," in *Robotics: Science and Systems*, 2025. 2

[32] H. Qi, H. Yin, A. Zhu, Y. Du, and H. Yang, "Strengthening generative robot policies through predictive world modeling," *arXiv preprint arXiv:2502.00622*, 2025. 2

[33] J. Guo, X. Ma, Y. Wang, M. Yang, H. Liu, and Q. Li, "FlowDreamer: A RGB-D world model with flow-based motion representations for robot manipulation," *arXiv preprint arXiv:2505.10075*, 2025. 2

[34] S. Huang, L. Chen, P. Zhou, S. Chen, Z. Jiang, Y. Hu, Y. Liao, P. Gao, H. Li, M. Yao *et al.*, "EnerVerse: Envisioning embodied future space for robotics manipulation," *arXiv preprint arXiv:2501.01895*, 2025. 2, 4, 7

[35] H. A. Alhaija, J. Alvarez, M. Bala, T. Cai, T. Cao, L. Cha, J. Chen, M. Chen, F. Ferroni, S. Fidler *et al.*, "Cosmos-Transfer1: Conditional world generation with adaptive multimodal control," *arXiv preprint arXiv:2503.14492*, 2025. 2

[36] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian Splatting for real-time radiance field rendering," *ACM Transactions on Graphics*, vol. 42, no. 4, pp. 1–14, 2023. 3

[37] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as Neural Radiance Fields for view synthesis," in *Eur. Conf. Comput. Vis.*, 2020, pp. 405–421. 3

[38] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson, "Learning latent dynamics for planning from pixels," in *Int. Conf. Mach. Learn.*, 2019, pp. 2555–2565. 4, 15

[39] K. Paster, S. A. McIlraith, and J. Ba, "Planning from pixels using inverse dynamics models," in *Int. Conf. Learn. Represent.*, 2021, pp. 1–12. 4, 6

[40] M. Pan, X. Zhu, Y. Wang, and X. Yang, "Iso-Dream: Isolating and leveraging noncontrollable visual dynamics in world models," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2022, pp. 23 178–23 191. 4, 6

[41] Y. Seo, D. Hafner, H. Liu, F. Liu, S. James, K. Lee, and P. Abbeel, "Masked world models for visual control," in *Conf. Robot. Learn.*, 2023, pp. 1332–1344. 4, 6, 15

[42] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar *et al.*, "Inner Monologue: Embodied reasoning through planning with language models," in *Conf. Robot. Learn.*, 2023, pp. 1769–1782. 4, 7

[43] P. Wu, A. Escontrela, D. Hafner, P. Abbeel, and K. Goldberg, "DayDreamer: World models for physical robot learning," in *Conf. Robot. Learn.*, 2023, pp. 2226–2240. 4, 5

[44] V. Micheli, E. Alonso, and F. Fleuret, "Transformers are sample-efficient world models," in *Int. Conf. Learn. Represent.*, 2023, pp. 1–14. 4, 6

[45] X. Wang, Z. Zhu, G. Huang, B. Wang, X. Chen, and J. Lu, "WorldDreamer: Towards general world models for video generation via predicting masked tokens," *arXiv preprint arXiv:2401.09985*, 2024. 4, 10

[46] T. Yoneda, J. Fang, P. Li, H. Zhang, T. Jiang, S. Lin, B. Picker, D. Yunis, H. Mei, and M. R. Walter, "Statler: State-maintaining language models for embodied reasoning," in *IEEE Int. Conf. Robot. Autom.*, 2024, pp. 15 083–15 091. 4, 7

[47] J. Xiang, G. Liu, Y. Gu, Q. Gao, Y. Ning, Y. Zha, Z. Feng, T. Tao, S. Hao, Y. Shi *et al.*, "Pandora: Towards general world model with natural language actions and video states," *arXiv preprint arXiv:2406.09455*, 2024. 4, 9

[48] X. Gu, Y.-J. Wang, X. Zhu, C. Shi, Y. Guo, Y. Liu, and J. Chen, "Advancing Humanoid Locomotion: Mastering challenging terrains with denoising world model learning," in *Robotics: Science and Systems*, 2024. 4, 6

[49] S. Zhou, Y. Du, J. Chen, Y. Li, D.-Y. Yeung, and C. Gan, "RoboDreamer: Learning compositional world models for robot imagination," in *Int. Conf. Mach. Learn.*, 2024, pp. 61 885–61 896. 4, 8

[50] J. Bruce, M. D. Dennis, A. Edwards, J. Parker-Holder, Y. Shi, E. Hughes, M. Lai, A. Mavalankar, R. Steigerwald, C. Apps *et al.*, "Genie: Generative interactive environments," in *Int. Conf. Mach. Learn.*, 2024, pp. 4603–4623. 4, 8

[51] A. Bardes, Q. Garrido, J. Ponce, X. Chen, M. Rabbat, Y. LeCun, M. Assran, and N. Ballas, "Revisiting feature prediction for learning visual representations from video," *Trans. Mach. Learn. Res.*, vol. 2024, pp. 1–14, 2024. 4, 10

[52] L. Zhang, M. Kan, S. Shan, and X. Chen, "PreLAR: World model pre-training with learnable action representation," in *Eur. Conf. Comput. Vis.*, 2024, pp. 185–201. 4, 5

[53] G. Lu, S. Zhang, Z. Wang, C. Liu, J. Lu, and Y. Tang, "ManiGaussian: Dynamic gaussian splatting for multi-task robotic manipulation," in *Eur. Conf. Comput. Vis.*, 2024, pp. 349–366. 4, 7, 15

[54] M. Zawalski, W. Chen, K. Pertsch, O. Mees, C. Finn, and S. Levine, "Robotic control via embodied Chain-of-Thought reasoning," in *Conf. Robot. Learn.*, 2025, pp. 3157–3181. 4, 7

[55] Y. Wen, J. Lin, Y. Zhu, J. Han, H. Xu, S. Zhao, and X. Liang, "VidMan: Exploiting implicit dynamics from video diffusion model for effective robot manipulation," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2024, pp. 41 051–41 075. 4, 8, 15

[56] J. Wu, S. Yin, N. Feng, X. He, D. Li, J. Hao, and M. Long, "iVideoGPT: Interactive videoGPTs are scalable world models," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2024, pp. 68 082–68 119. 4, 8

[57] Q. He, W. Liang, C. Hao, G. Sun, and J. Tian, "GLAM: Global-local variation awareness in mamba-based world model," in *AAAI Conf. Artif. Intell.*, 2025, pp. 17 105–17 113. 4, 6

[58] B. Lin, Y. Nie, Z. Wei, J. Chen, S. Ma, J. Han, H. Xu, X. Chang, and X. Liang, "NavCot: Boosting LLM-based Vision-and-Language navigation via learning disentangled reasoning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 7, pp. 5945–5945, 2025. 4, 7

[59] J. Guo, Y. Ye, T. He, H. Wu, Y. Jiang, T. Pearce, and J. Bian, "MineWorld: A real-time and open-source interactive world model on minecraft," *arXiv preprint arXiv:2504.08388*, 2025. 4, 6

[60] L. Barcellona, A. Zadaianchuk, D. Allegro, S. Papa, S. Ghidoni, and E. Gavves, "Dream to Manipulate: Compositional world models empowering robot imitation learning with imagination," in *Int. Conf. Learn. Represent.*, 2025. 4, 7, 15

[61] F. Petri, L. Asprino, and A. Gangemi, "Learning local causal world models with state space models and attention," *arXiv preprint arXiv:2505.02074*, 2025. 4, 9

[62] J. Wu, S. Yin, N. Feng, and M. Long, "RLVR-World: Training world models with reinforcement learning," *arXiv preprint arXiv:2505.13934*, 2025. 4, 9

[63] N. Savov, N. Kazemi, D. Zhang, D. P. Paudel, X. Wang, and L. Van Gool, "StateSpaceDiffuser: Bringing long context to diffusion world models," *arXiv preprint arXiv:2505.22246*, 2025. 4, 9

[64] J. Chen, H. Zhu, X. He, Y. Wang, J. Zhou, W. Chang, Y. Zhou, Z. Li, Z. Fu, J. Pang *et al.*, "DeepVerse: 4D autoregressive video generation as a world model," *arXiv preprint arXiv:2506.01103*, 2025. 4, 9

[65] X. Yang, B. Li, S. Xu, N. Wang, C. Ye, Z. Chen, M. Qin, Y. Ding, X. Jin, H. Zhao *et al.*, "ORV: 4D occupancy-centric robot video generation," *arXiv preprint arXiv:2506.03079*, 2025. 4, 10

[66] A. Bar, G. Zhou, D. Tran, T. Darrell, and Y. LeCun, "Navigation world models," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 15 791–15 801. 4, 7

[67] J. Cen, C. Yu, H. Yuan, Y. Jiang, S. Huang, J. Guo, X. Li, Y. Song, H. Luo, F. Wang *et al.*, "WorldVLA: Towards autoregressive action world model," *arXiv preprint arXiv:2506.21539*, 2025. 4, 6

[68] H. Chen, B. Wang, J. Guo, T. Zhang, Y. Hou, X. Huang, C. Tie, and L. Shao, "World4Omni: A zero-shot framework from image generation world model to robotic manipulation," *arXiv preprint arXiv:2506.23919*, 2025. 4, 8

[69] Z. Wang, K. Wang, L. Zhao, P. Stone, and J. Bian, "Dyn-O: Building structured world models with object-centric representations," *arXiv preprint arXiv:2507.03298*, 2025. 4, 6

[70] G. Zhou, H. Pan, Y. LeCun, and L. Pinto, "DINO-WM: World models on pre-trained visual features enable zero-shot planning," in *Int. Conf. Mach. Learn.*, 2025. 4, 7

[71] X. Chi, C.-K. Fan, H. Zhang, X. Qi, R. Zhang, A. Chen, C.-M. Chan, W. Xue, Q. Liu, S. Zhang *et al.*, "Empowering world models with reflection for embodied video prediction," in *Int. Conf. Mach. Learn.*, 2025. 4, 8

原文第 18 页 | 中文翻译初稿

18

- [26] G. Zhai、X. Zhang 和 N. Navab，“具有标记化潜在状态的循环 世界模型（World Model）”，载于 Int。会议。学习。代表。 Worksh.，2025 年，第 1-7 页。 2

- [27] E. Aljalbout、M. Krinner、A. Romero 和 D. Scaramuzza，“利用状态空间 世界模型（World Models） 加速基于模型的 强化学习（Reinforcement Learning）”，载于_Int。会议。学习。代表。工作._, 2025. 2

- [28] C. Chen，Y.-F。 Wu、J. Yoon 和 S. Ahn，“TransDreamer：带有变压器 世界模型（World Models） 的 强化学习（Reinforcement Learning）”，arXiv 预印本 arXiv:2202.09481，2022. 2, 4, 6, 15

- [29] J. Robine、M. Höftmann、T. Uelwer 和 S. Harmeling，“基于 Transformer 的 世界模型（World Models） 对 10 万次交互感到满意”，载于 Int。会议。学习。代表.，2023 年，第 1-13 页。 2, 4, 6

- [30] 张伟、王刚、孙杰、袁宇、G

。 Huang，“STORM：基于 世界模型（World Models） 的高效随机变压器，适用于 强化学习（Reinforcement Learning）”，载于 Annu。会议。神经。通知。过程。系统.，2023 年，第 27 147-27 166 页。2

- [31] C. Zhu, R. Yu, S. Feng, B. Burchfiel, P. Shah, and A. Gupta, “Unified 世界模型（World Models）: Coupling video and action diffusion for pretraining on large robotic datasets,” in Robotics: Science and Systems , 2025. 2

- [32] H. Qi、H. Yin、A. Zhu、Y. Du 和 H. Yang，“通过预测世界建模加强生成机器人策略”，arXiv 预印本 arXiv:2502.00622，2025. 2

- [33] J. Guo, X. Ma, Y. Wang, M. Yang, H. Liu, and Q. Li, “FlowDreamer: A RGB-D 世界模型（World Model） with flow-based motion representations for robot manipulation,” arXiv preprint arXiv:2505.10075 , 2025. 2

- [34] 黄S.黄，L.陈，P.周，S

。陈正江、胡勇、廖勇、高鹏、李慧明

。 Yao et al. ，“EnerVerse：展望机器人操作的未来空间”，arXiv 预印本 arXiv:2501.01895，2025. 2, 4, 7

- [35] H. A. Alhaija, J. Alvarez, M. Bala, T. Cai, T. Cao, L. Cha, J. Chen, M. Chen, F. Ferroni, S. Fidler et al.，“Cosmos-Transfer1：自适应多模态控制的条件世界生成”，arXiv 预印本 arXiv:2503.14492，2025. 2

- [36] B. Kerbl、G. Kopanas、T. Leimkühler 和 G. Drettakis，“用于实时辐射场渲染的 3D 高斯泼溅”，ACM Transactions on Graphics，卷。 42、没有。 4，第 1-14 页，2023 年。3

- [37] B. Mildenhall、P. P. Srinivasan、M. Tancik、J. T. Barron、R. Ramamoorthi 和 R. Ng，“NeRF：将场景表示为神经辐射场以进行视图合成”，载于 Eur。会议。计算。 Vis.，2020 年，第 405-421 页。 3

- [38] D. Hafner、T. Lillicrap、I. Fischer、R. Villegas、D. Ha、H. Lee 和 J. Davidson，“从像素学习规划的潜在动态”，《Int》。会议。马赫。学习._，2019 年，第 2555-2565 页。 4、15

- [39] K. Paster、S. A. McIlraith 和 J. Ba，“使用逆动态模型从像素进行规划”，载于_Int。会议。学习。代表._，2021 年，第 1-12 页。 4, 6

- [40] M. Pan、X. Zhu、Y. Wang 和 X. Yang，“Iso-Dream：隔离和利用 世界模型（World Models） 中的不可控视觉动态”，载于 Annu。会议。神经。通知。过程。系统.，2022 年，第 23 178-23 191. 4, 6 页

- [41] Y. Seo、D. Hafner、H. Liu、F. Liu、S. James、K. Lee 和 P. Abbeel，“用于视觉控制的蒙版 世界模型（World Models）”，参见 Conf。机器人。学习.，2023 年，第 1332-1344 页。 4、6、15

- [42] W. 黄、F. 夏、T. 肖、H. 陈、J. 梁、P. Florence、A. 曾、J

。汤普森，I. 莫达奇，Y

。 Chebotar_et al._，“内心独白：通过语言模型规划进行具体推理”，收录于 Conf。机器人。 Learn.，2023 年，第 1769-1782 页。 4, 7

- [43] P. Wu、A. Escontrela、D. Hafner、P. Abbeel 和 K. Goldberg，“DayDreamer：用于物理机器人学习的 世界模型（World Models）”，参见 Conf。机器人。学习.，2023 年，第 2226-2240 页。 4, 5

- [44] V. Micheli、E. Alonso 和 F. Fleuret，“Transformers 具有样本效率 世界模型（World Models）”，载于 Int。会议。学习。代表.，2023 年，第 1-14 页。 4, 6

- [45] X. Wang、Z. Zhu、G. Huang、B. Wang、X. Chen 和 J. Lu，“WorldDreamer：通过预测屏蔽标记实现视频生成的通用 世界模型（World Models）”，arXiv 预印本 arXiv:2401.09985，2024. 4, 10

- [46] T. Yoneda、J. Fang、P. Li、H. Zhang、T. Jiang、S. Lin、B. Picker、D. Yunis、H. Mei 和 M. R. Walter，“Statler：用于体现推理的状态维护语言模型”，载于 IEEE Int。会议。机器人。 Autom.，2024 年，第 15 083-15 091. 4, 7 页

- [47] J.向，G.刘，Y.顾，Q.高，Y.宁，Y.查，Z.冯，T.陶，S.浩，Y.石等人，“潘多拉：通过自然语言动作和视频状态走向通用世界模型（World Model）”，arXiv预印本arXiv：2406.09455，2024年。 4, 9

- [48] X.顾，Y.-J。 Wang、X. Zhu、C. Shi、Y.Guo、Y.Liu 和 J.Chen，“推进人形运动：通过去噪 世界模型（World Model） 学习掌握具有挑战性的地形”，《机器人：科学与系统》，2024 年 4 月 6 日

- [49] 周圣，杜宇，陈建，李宇，D.-Y。 Yeung 和 C. Gan，“RoboDreamer：学习组合 世界模型（World Models） 以实现机器人想象力”，载于_Int。会议。马赫。学习._，2024 年，第 61 885-61 896 页。4, 8

- [50] J. Bruce、M. D. Dennis、A. Edwards、J. Parker-Holder、Y. Shi、E. Hughes、M. Lai、A. Mavalankar、R. Steigerwald、C. Apps et al.，“精灵：生成交互环境”，载于_Int。会议。马赫。学习._，2024 年，第 4603-4623 页。 4, 8

- [51] A. Bardes、Q. Garrido、J. Ponce、X. Chen、M. Rabbat、Y. LeCun、M. Assran 和 N. Ballas，“重新审视从视频中学习视觉表示的特征预测”，Trans。马赫。学习。 Res.，卷。 2024 年，第 1-14 页，2024. 4, 10

- [52] L.Zhang、M.Kan、S.Shan 和 X.Chen，“PreLAR：世界模型（World Model） 具有可学习动作表示的预训练”，载于 Eur。会议。计算。 Vis.，2024 年，第 185-201 页。 4, 5

- [53] G. Lu、S. 张、Z. Wang、C. Liu、J. Lu 和 Y. Tang，“ManiGaussian：用于多任务机器人操作的动态高斯泼溅”，载于 Eur。会议。计算。 Vis.，2024 年，第 349-366 页。 4、7、15

- [54] M. Zawalski、W. Chen、K. Pertsch、O. Mees、C. Finn 和 S. Levine，“通过体现的思想链推理进行机器人控制”，参见 Conf。机器人。学习.，2025 年，第 3157-3181 页。 4, 7

- [55] Y. Wen、J. Lin、Y. Zhu、J. Han、H. Xu、S. Zhao 和 X. Liang，“VidMan：利用视频扩散模型的隐式动力学进行有效的机器人操作”，载于 Annu。会议。神经。通知。过程。系统.，2024 年，第 41 051-41 075 页。4,8,15

- [56] J. Wu、S. Yin、N. Feng、X. He、D. Li、J.hao 和 M. Long，“iVideoGPT：交互式视频GPT 是可扩展的 世界模型（World Models）”，载于 Annu。会议。神经。通知。过程。系统.，2024 年，第 68 082-68 119. 4, 8 页

- [57] Q.He、W.Liang、C.Hao、G.Sun 和 J.Tian，“GLAM：基于曼巴的 世界模型（World Model） 中的全局局部变化感知”，载于_AAAI Conf。阿蒂夫。 Intel._，2025 年，第 17 105-17 113 页

。 4, 6

- [58] B. Lin、Y. Nie、Z. Wei、J. Chen、S. Ma、J. Han、H. Xu、X. Chang 和 X. Liang，“NavCot：通过学习解缠推理促进基于 LLM 的视觉和语言导航”，IEEE Trans。模式肛门。马赫。情报.，卷。 47、没有。 7，第 5945-5945 页，2025 年。4, 7

- [59] J.Guo、Y.Ye、T.He、H.Wu、Y.Jiang、T.Pearce 和 J.Bian，“MineWorld：Minecraft 上的实时开源交互式 世界模型（World Model）”，arXiv 预印本 arXiv:2504.08388，2025. 4, 6

- [60] L. Barcellona、A. Zadaianchuk、D. Allegro、S. Papa、S. Ghidoni 和 E. Gavves，“操纵梦想：组合 世界模型（World Models） 赋予机器人 模仿学习（Imitation Learning） 想象力”，载于_Int。会议。学习。代表._，2025. 4, 7, 15

- [61] F. Petri、L. Asprino 和 A. Gangemi，“利用状态空间模型和注意力学习局部因果 世界模型（World Models）”，arXiv 预印本 arXiv:2505.02074，2025. 4, 9

- [62] J. Wu、S. Yin、N. Feng 和 M. Long，“RLVR-World：使用 强化学习（Reinforcement Learning） 训练 世界模型（World Models）”，arXiv 预印本 arXiv:2505.13934，2025. 4, 9

- [63] N. Savov、N. Kazemi、D. Zhu、D. P. Paudel、X. Wang 和 L. Van Gool，“StateSpaceDiffuser：为扩散带来长上下文 世界模型（World Models）”，arXiv 预印本 arXiv:2505.22246，2025. 4, 9

- [64] J. Chen, H. Zhu, X. He, Y. Wang, J. Zhou, W. Chang, Y. Zhou, Z. Li, Z. Fu, J. Pang et al.，“DeepVerse：4D 自回归视频生成作为 世界模型（World Model）”，arXiv 预印本 arXiv:2506.01103，2025。 4, 9

- [65] X. Yang, B. Li, S. Xu, N. Wang, C. Ye, Z. Chen, M.qin, Y. Ding, X. Jin, H. Zhao et al.，“ORV: 4D obsancy-centric Robot video Generation,” arXiv preprint arXiv:2506.03079，2025. 4, 10

- [66] A. Bar、G. Zhou、D. Tran、T. Darrell 和 Y. LeCun，“Navigation 世界模型（World Models）”，载于 _IEEE Conf。计算。维斯。模式识别，2025，第 15 791-15 801. 4, 7

- [67] J. Cen, C. Yu, H. Yuan, Y. Jiang, S. Huang, J.Guo, X. Li, Y. Song, H. Luo, F. Wang et al.，“WorldVLA: Towards autoregressive action 世界模型（World Model）,” arXiv preprint arXiv:2506.21539，2025. 4, 6

- [68] H. Chen、B. Wang、J.Guo、T.Zhang、Y.Hou、X.Huang、C.Tie 和 L.Shao，“World4Omni：从图像生成 世界模型（World Model） 到机器人操作的零样本框架”，arXiv 预印本 arXiv:2506.23919，2025. 4, 8

- [69] Z. Wang、K. Wang、L. Zhu、P. Stone 和 J. Bian，“Dyn-O：使用以对象为中心的表示构建结构化世界模型（World Models）”，arXiv 预印本 arXiv:2507.03298，2025. 4, 6

- [70] G. Zhou、H. Pan、Y. LeCun 和 L. Pinto，“DINO-WM：预训练视觉特征上的 世界模型（World Models） 可实现零样本规划”，载于 Int。会议。马赫。学习. , 2025. 4, 7

- [71] X. Chi，C.-K。范，张，X.齐，R.张，A.陈，C.-M。 Chan、W.Xue、Q.Liu、S.Zhang_et al.，“利用体现视频预测的反射为 世界模型（World Models） 提供支持”，《Int》。会议。马赫。学习. , 2025. 4, 8

[72] S. Gao, S. Zhou, Y. Du, J. Zhang, and C. Gan, "AdaWorld: Learning adaptable world models with latent actions," in *Int. Conf. Mach. Learn.*, 2025. 4, 8

[73] Y. Yang, J. Liu, Z. Zhang, S. Zhou, R. Tan, J. Yang, Y. Du, and C. Gan, "MindJourney: Test-time scaling with world models for spatial reasoning," *arXiv preprint arXiv:2507.12508*, 2025. 4, 9

[74] Y. Chai, L. Deng, R. Shao, J. Zhang, L. Xing, H. Zhang, and Y. Liu, "GAF: Gaussian action field as a dvnamic world model for robotic mlanipulation," *arXiv preprint arXiv:2506.14135*, 2025. 4, 7

[75] X. Mao, S. Lin, Z. Li, C. Li, W. Peng, T. He, J. Pang, M. Chi, Y. Qiao, and K. Zhang, "Yume: An interactive world generation model," *arXiv preprint arXiv:2507.17744*, 2025. 4, 9

[76] X. Chen, H. Wei, P. Zhang, C. Zhang, K. Wang, Y. Guo, R. Yang, Y. Wang, X. Xiao, L. Zhao *et al.*, "villa-X: Enhancing latent action modeling in Vision-Language-Action models," *arXiv preprint arXiv:2507.23682*, 2025. 4, 8

[77] A. Team, H. Zhu, Y. Wang, J. Zhou, W. Chang, Y. Zhou, Z. Li, J. Chen, C. Shen, J. Pang *et al.*, "Aether: Geometric-aware unified world modeling," in *Int. Conf. Comput. Vis.*, 2025. 4, 10

[78] H. Zhen, Q. Sun, H. Zhang, J. Li, S. Zhou, Y. Du, and C. Gan, "TesserAct: Learning 4D embodied world models," in *Int. Conf. Comput. Vis.*, 2025. 4, 8, 15

[79] E. Zhou, Y. Qin, Z. Yin, Y. Huang, R. Zhang, L. Sheng, Y. Qiao, and J. Shao, "MineDreamer: Learning to follow instructions via Chain-of-Imagination for simulated-world control," in *Int. Conf. Intell. Robot. Syst.*, 2025. 4, 7

[80] T. Yu, G. Lu, Z. Yang, H. Deng, S. S. Chen, J. Lu, W. Ding, G. Hu, Y. Tang, and Z. Wang, "ManiGaussian++: General robotic bimanual manipulation with hierarchical gaussian world model," in *Int. Conf. Intell. Robot. Syst.*, 2025. 4, 7, 15

[81] L. Maes, Q. L. Lidec, D. Scieur, Y. LeCun, and R. Balestriero, "LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from pixels," *arXiv preprint arXiv:2603.19312*, 2026. 4, 5

[82] S. Ye, Y. Ge, K. Zheng, S. Gao, S. Yu, G. Kurian, S. Indupuru, Y. L. Tan, C. Zhu, J. Xiang *et al.*, "World action models are zero-shot policies," *arXiv preprint arXiv:2602.15922*, 2026. 4, 6

[83] W. Huang, Y.-W. Chao, A. Mousavian, M.-Y. Liu, D. Fox, K. Mo, and L. Fei-Fei, "PointWorld: Scaling 3D world models for In-The-Wild robotic manipulation," *arXiv preprint arXiv:2601.03782*, 2026. 4, 7

[84] M. J. Kim, Y. Gao, T.-Y. Lin, Y.-C. Lin, Y. Ge, G. Lam, P. Liang, S. Song, M.-Y. Liu, C. Finn *et al.*, "Cosmos Policy: Fine-Tuning video models for visuomotor control and planning," *arXiv preprint arXiv:2601.16163*, 2026. 4, 8

[85] L. Mur-Labadia, M. Muckley, A. Bar, M. Assran, K. Sinha, M. Rabbat, Y. LeCun, N. Ballas, and A. Bardes, "V-JEPA 2.1: Unlocking dense features in video self-supervised learning," *arXiv preprint arXiv:2603.14482*, 2026. 4, 10, 12, 16

[86] N. Agarwal, A. Ali, J. Allen, M. Antolini, A. Aubame, A. Azzolini, J. Bai, M. Bala, Y. Balaji, J. Bapst *et al.*, "Cosmos 3: Omnimodal world models for Physical AI," *arXiv preprint arXiv:2606.02800*, 2026. 4, 10

[87] A. Hu, G. Corrado, N. Griffiths, Z. Murez, C. Gurau, H. Yeo, A. Kendall, R. Cipolla, and J. Shotton, "Model-based imitation learning for urban driving," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2022, pp. 20 703–20 716. 5

[88] L. Zhang, Y. Xiong, Z. Yang, S. Casas, R. Hu, and R. Urtasun, "Copilot4D: Learning unsupervised world models for autonomous driving via discrete diffusion," in *Int. Conf. Learn. Represent.*, 2024. 5, 9

[89] Z. Gao, Y. Mu, C. Chen, J. Duan, P. Luo, Y. Lu, and S. E. Li, "Enhance sample efficiency and robustness of End-to-End urban autonomous driving via semantic masked world model," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 10, pp. 13 067–13 079, 2024. 5

[90] R. Gao, K. Chen, Z. Li, L. Hong, Z. Li, and Q. Xu, "MagicDrive3D: Controllable 3D generation for any-view rendering in street scenes," *arXiv preprint arXiv:2405.14475*, 2024. 5, 10, 14

[91] L. Wang, W. Zheng, Y. Ren, H. Jiang, Z. Cui, H. Yu, and J. Lu, "OccSora: 4D occupancy generation models as world simulators for autonomous driving," *arXiv preprint arXiv:2405.20337*, 2024. 5, 10

[92] E. Ma, L. Zhou, T. Tang, Z. Zhang, D. Han, J. Jiang, K. Zhan, P. Jia, X. Lang, H. Sun *et al.*, "Unleashing generalization of End-to-End autonomous driving with controllable long video generation," *arXiv preprint arXiv:2406.01349*, 2024. 5, 9, 14

[93] C. Min, D. Zhao, L. Xiao, J. Zhao, X. Xu, Z. Zhu, L. Jin, J. Li, Y. Guo, J. Xing *et al.*, "DriveWorld: 4D pre-trained scene understanding via world models for autonomous driving," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 15 522–15 533. 5, 7, 16

[94] Y. Wang, J. He, L. Fan, H. Li, Y. Chen, and Z. Zhang, "Driving into the Future: Multiview visual forecasting and planning with world model for autonomous driving," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 14 749–14 759. 5, 8, 14

[95] Z. Yang, L. Chen, Y. Sun, and H. Li, "Visual point cloud forecasting enables scalable autonomous driving," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 14 673–14 684. 5, 9

[96] J. Yang, S. Gao, Y. Qiu, L. Chen, T. Li, B. Dai, K. Chitta, P. Wu, J. Zeng, P. Luo *et al.*, "Generalized predictive model for autonomous driving," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 14 662–14 672. 5, 8, 12, 14, 16

[97] X. Wang, Z. Zhu, G. Huang, X. Chen, J. Zhu, and J. Lu, "DriveDreamer: Towards real-world-drive world models for autonomous driving," in *Eur. Conf. Comput. Vis.*, 2024, pp. 55–72. 5, 7, 14

[98] W. Zheng, R. Song, X. Guo, C. Zhang, and L. Chen, "GenAD: Generative End-to-End autonomous driving," in *Eur. Conf. Comput. Vis.*, 2024, pp. 87–104. 5, 7, 16

[99] W. Zheng, W. Chen, Y. Huang, B. Zhang, Y. Duan, and J. Lu, "OccWorld: Learning a 3D occupancy world model for autonomous driving," in *Eur. Conf. Comput. Vis.*, 2024, pp. 55–72. 5, 7, 15, 16

[100] S. Gu, W. Yin, B. Jin, X. Guo, J. Wang, H. Li, Q. Zhang, and X. Long, "DOME: Taming diffusion model into high-fidelity controllable occupancy world model," *arXiv preprint arXiv:2410.10429*, 2024. 5, 9, 15

[101] R. Tian, B. Li, X. Weng, Y. Chen, E. Schmerling, Y. Wang, B. Ivanovic, and M. Pavone, "Tokenize the world into object-level knowledge to address long-tail events in autonomous driving," in *Conf. Robot. Learn.*, 2025, pp. 3656–3673. 5, 8

[102] S. Gao, J. Yang, L. Chen, K. Chitta, Y. Qiu, A. Geiger, J. Zhang, and H. Li, "Vista: A generalizable driving world model with high fidelity and versatile controllability," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2024, pp. 91 560–91 596. 5, 9, 14

[103] G. Zhao, X. Wang, Z. Zhu, X. Chen, G. Huang, X. Bao, and X. Wang, "DriveDreamer-2: LLM-enhanced world models for diverse driving video generation," in *AAAI Conf. Artif. Intell.*, 2025, pp. 10 412–10 420. 5, 10, 14

[104] H. Xu, P. Peng, G. Tan, Y. Chang, Y. Zhao, and Y. Tian, "Delta-Triplane transformers as occupancy world models," *arXiv preprint arXiv:2503.07338*, 2025. 5, 8, 15, 16

[105] H. Bian, L. Kong, H. Xie, L. Pan, Y. Qiao, and Z. Liu, "DynamicCity: Large-scale 4D occupancy generation from dynamic scenes," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–14. 5, 10

[106] V. Zyrianov, H. Che, Z. Liu, and S. Wang, "LidarDM: Generative LiDAR simulation in a generated world," in *IEEE Int. Conf. Robot. Autom.*, 2025. 5, 9

[107] S. Zeng, X. Chang, M. Xie, X. Liu, Y. Bai, Z. Pan, M. Xu, and X. Wei, "FutureSightDrive: Thinking visually with spatio-temporal CoT for autonomous driving," *arXiv preprint arXiv:2505.17685*, 2025. 5, 7, 16

[108] M. Hassan, S. Stapf, A. Rahimi, P. Rezende, Y. Haghighi, D. Brüggemann, I. Katircioglu, L. Zhang, X. Chen, S. Saha *et al.*, "GEM: A generalizable ego-vision multimodal world model for fine-grained ego-motion, object dynamics, and scene composition control," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 22 404–22 415. 5, 9, 14

[109] S. Zuo, W. Zheng, Y. Huang, J. Zhou, and J. Lu, "GaussianWorld: Gaussian world model for streaming 3D occupancy prediction," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 6772–6781. 5, 9

[110] J. Ni, Y. Guo, Y. Liu, R. Chen, L. Lu, and Z. Wu, "MaskGWM: A generalizable driving world model with video mask reconstruction," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 22 381–22 391. 5, 10, 14

[111] G. Zhao, C. Ni, X. Wang, Z. Zhu, X. Zhang, Y. Wang, G. Huang, X. Chen, B. Wang, Y. Zhang *et al.*, "DriveDreamer4D: World models are effective data machines for 4D driving scene representation," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 12 015–12 026. 5, 10

[112] C. Ni, G. Zhao, X. Wang, Z. Zhu, W. Qin, G. Huang, C. Liu, Y. Chen, Y. Wang, X. Zhang *et al.*, "ReconDreamer: Crafting world models for driving scene reconstruction via online restoration," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 1559–1569. 5, 10

[113] Y. Li, Y. Wang, Y. Liu, J. He, L. Fan, and Z. Zhang, "End-to-End driving with online trajectory evaluation via BEV world model," in *Int. Conf. Comput. Vis.*, 2025. 5, 7

[114] X. Zhou, D. Liang, S. Tu, X. Chen, Y. Ding, D. Zhang, F. Tan, H. Zhao, and X. Bai, "HERMES: A unified self-driving world model for simultaneous 3D scene understanding and generation," in *Int. Conf. Comput. Vis.*, 2025. 5, 10

原文第 19 页 | 中文翻译初稿

19

- [72] S.Gao、S.Zhou、Y.Du、J.Zhang 和 C.Gan，“AdaWorld：通过潜在动作学习适应性 世界模型（World Models）”，载于 Int。会议。马赫。学习. , 2025. 4, 8

- [73] Y. Yang、J. Liu、Z. Zhang、S. Zhou、R. Tan、J. Yang、Y. Du 和 C. Gan，“MindJourney：使用世界模型（World Models） 进行空间推理的测试时间缩放”，arXiv 预印本 arXiv:2507.12508，2025. 4, 9

- [74] Y. Chai、L. Deng、R. Shao、J. Zhang、L. Xing、H. Zhang 和 Y. Liu，“GAF：高斯作用场作为机器人操作的动态 世界模型（World Model）”，arXiv 预印本 arXiv:2506.14135，2025. 4, 7

- [75] X. Mao、S. Lin、Z. Li、C. Li、W. Peng、T. He、J. Pang、M. Chi、Y. Qiao 和 K. Zhang，“Yume：一种交互式世界生成模型”，arXiv 预印本 arXiv:2507.17744，2025. 4, 9

- [76] X. Chen, H. Wei, P.Zhang, C.Zhang, K.Wang, Y.Guo, R.Yang, Y.Wang, X.Xiao, L.Zhao_et al._，“villa-X：增强 视觉语言动作（Vision-Language-Action） 模型中的潜在动作建模”，arXiv 预印本 arXiv:2507.23682，2025. 4, 8

- [77] A. Team、H. Zhu、Y. Wang、J. Zhou、W. Chang、Y. Zhou、Z. Li、J. Chen、C. Shen、J. Pang 等人，“以太：几何感知的统一世界建模”，载于_Int。会议。计算。视觉_ , 2025. 4, 10

- [78] H.Zhen、Q.Sun、H.Zhang、J.Li、S.Zhou、Y.Du 和 C.Gan，“TesserAct：学习 4D 体现 世界模型（World Models）”，载于_Int。会议。计算。视觉_ , 2025. 4, 8, 15

- [79] E. Zhou、Y.qin、Z. Yin、Y. Huang、R. Zhang、L. Shen、Y. Qiao 和 J. Shao，“MineDreamer：学习通过想象链遵循指令进行模拟世界控制”，载于_Int。会议。英特尔。机器人。系统_ , 2025. 4, 7

- [80] T. Yu、G. Lu、Z. Yang、H. Deng、S. S. Chen、J. Lu、W. Ding、G. Hu、Y. Tang 和 Z. Wang，“ManiGaussian++：采用分层高斯 世界模型（World Model） 的通用机器人双手操作”，载于_Int。会议。英特尔。机器人。系统_ , 2025. 4, 7, 15

- [81] L. Maes、Q. L. Lidec、D. Scieur、Y. LeCun 和 R. Balestriero，“LeWorldModel：来自像素的稳定端到端联合嵌入预测架构”，arXiv 预印本 arXiv:2603.19312，2026. 4, 5

- [82] S. Ye, Y. Ge, K. Cheng, S. Gau, S. Yu, G. Kurian, S. Indupuru, Y. L. Tan, C. Zhu, J. Xian et al., “世界行动模型是零样本策略”, arXiv 预印本 arXiv:2602.15922 , 2026. 4, 6

- [83] 黄伟，Y.-W。 Chao, A. Mousavian, M.-Y. Liu、D. Fox、K. Mo 和 L. Fei-Fei，“PointWorld：用于野外机器人操作的缩放 3D 世界模型（World Models）”，arXiv 预印本 arXiv:2601.03782，2026. 4, 7

- [84] M. J. Kim，Y. Gau，T.-Y。林，Y.-C。林宇戈、林刚、梁鹏、宋美宇。 Liu, C. Finn et al.，“Cosmos Policy：微调视频模型以实现视觉运动控制和规划”，arXiv 预印本 arXiv:2601.16163，2026. 4, 8

- [85] L. Mur-Labadia、M. Muckley、A. Bar、M. Assran、K. Sinha、M. Rabbat、Y. LeCun、N. Ballas 和 A. Bardes，“V-JEPA 2.1：解锁视频自监督学习中的密集特征”，arXiv 预印本 arXiv:2603.14482，2026. 4， 10、12、16

- [86] N. Agarwal、A. Ali、J. Allen、M. Antolini、A. Aubame、A. Azzolini、J. Bai、M. Bala、Y. Balaji、J. Bapst 等人，“Cosmos 3：物理 AI 的全模态 世界模型（World Models）”，arXiv 预印本

arXiv:2606.02800，2026年4月10日

- [87] A. Hu、G. Corrado、N. Griffiths、Z. Murez、C. Gurau、H. Yeo、A. Kendall、R. Cipolla 和 J

。 Shotton，“用于城市驾驶的基于模型的 模仿学习（Imitation Learning）”，载于 Annu。会议。神经。通知。过程。系统.，2022 年，第 20 703-20 716. 5 页

- [88] L.Zhang、Y.Xiong、Z.Yang、S.Casas、R.Hu 和 R.Urtasun，“Copilot4D：通过离散扩散学习用于自动驾驶的无监督 世界模型（World Models）”，载于_Int。会议。学习。代表._，2024. 5, 9

- [89] Z. 高、Y. Mu、C. Chen、J. Duan、P. Luo、Y. Lu 和 S. E. Li，“通过语义屏蔽 世界模型（World Model） 提高端到端城市自动驾驶的样本效率和鲁棒性”，IEEE Trans。英特尔。运输。系统.，卷。 25、没有。 10，第 13 067-13 079 页，2024 年。 5

- [90] R. 高、K. Chen、Z. Li、L. Hong、Z. Li 和 Q. Xu，“MagicDrive3D：街道场景中任意视图渲染的可控 3D 生成”，arXiv 预印本 arXiv:2405.14475，2024. 5, 10, 14

- [91] L. Wang、W. Cheng、Y. Ren、H. Jiang、Z. Cui、H. Yu 和 J. Lu，“OccSora：4D 占用生成模型作为自动驾驶的世界模拟器”，arXiv 预印本 arXiv:2405.20337，2024. 5, 10

- [92] E. Ma, L. Zhou, T. Tang, Z. Zhang, D. Han, J. Jiang, K. Zhan, P. Jia, X. Lang, H. Sun et al.，“通过可控长视频生成释放端到端自动驾驶的泛化”，arXiv 预印本 arXiv:2406.01349，2024. 5, 9, 14

- [93] C. Min, D. Zhu, L. Xiao, J. Zhu, X. Xu, Z. Zhu, L. Jin, J. Li, Y.Guo, J. Xing et al.，“DriveWorld：通过 世界模型（World Models） 实现自动驾驶的 4D 预训练场景理解”，载于_IEEE Conf.计算。维斯。模式识别，2024 年，第 15 522-15 533 页。5,7,16

- [94] Y. Wang、J. He、L. Fan、H. Li、Y. Chen 和 Z. Zhang，“驶向未来：使用 世界模型（World Model）进行自动驾驶的多视图视觉预测和规划”，载于 _IEEE Conf.计算。维斯。模式识别，2024 年，第 14 749-14 759 页。5,8,14

- [95] Z. Yang、L. Chen、Y. Sun 和 H. Li，“视觉点云预测实现可扩展的自动驾驶”，载于 _IEEE Conf.计算。维斯。模式识别，2024，第 14 673-14 684. 5, 9

- [96] J. Yang、S. Gau、Y. Qiu、L. Chen、T. Li、B. Dai、K. Chitta、P. Wu、J. Zeng、P. Luo et al.，“自动驾驶的广义预测模型”，载于_IEEE Conf.计算。维斯。模式识别，2024 年，第 14 662-14 672 页。5,8,12,14,16

- [97] X. Wang、Z. Zhu、G. Huang、X. Chen、J. Zhu 和 J. Lu，“DriveDreamer：走向自动驾驶的现实世界驱动 世界模型（World Models）”，载于_Eur。会议。计算。 Vis._，2024 年，第 55-72 页。 5、7、14

- [98] W. 郑、R. 宋、X. 郭、C. 张和 L. 陈，“GenAD：生成式端到端自动驾驶”，载于 Eur。会议。计算。Vis.，2024 年，第 87-104 页。 5、7、16

- [99] W. Cheng、W. Chen、Y. Huang、B. Zhang、Y. Duan 和 J. Lu，“OccWorld：学习用于自动驾驶的 3D 占用 世界模型（World Model）”，载于 Eur。会议。计算。 Vis.，2024 年，第 55-72 页。 5, 7, 15, 16

- [100] S. Gu, W. Yin, B. Jin, X.Guo, J. Wang, H. Li, Q. Zhang, and X. Long, “DOME: 将扩散模型驯化为高保真可控占用 世界模型（World Model），” arXiv 预印本 arXiv:2410.10429 , 2024. 5, 9, 15

- [101] R. Tian、B. Li、X. Weng、Y. Chen、E. Schmerling、Y. Wang、B. Ivanovic 和 M. Pavone，“将世界标记为对象级知识以解决自动驾驶中的长尾事件”，参见_Conf。机器人。学习._，2025 年，第 3656-3673 页。 5, 8

- [102] 高圣，J

。 Yang、L. Chen、K. Chitta、Y. Qiu、A. Geiger、J. Zhang 和 H. Li，“Vista：具有高保真度和多功能可控性的通用驾驶 世界模型（World Model）”，载于 Annu。会议。神经。通知。过程。系统.，2024 年，第 91 560-91 596. 5, 9, 14

- [103] G. Zhu、X. Wang、Z. Zhu、X. Chen、G. Huang、X. Bao 和 X. Wang，“DriveDreamer-2：用于生成多样化驾驶视频的 LLM 增强型 世界模型（World Models）”，载于 AAAI Conf.阿蒂夫。 Intel.，2025 年，第 10 412-10 420. 5, 10, 14 页

- [104] H. Xu、P. Peng、G. Tan、Y. Chang、Y. Zhao 和 Y. Tian，“DeltaTriplane 变压器作为占用 世界模型（World Models）”，arXiv 预印本 arXiv:2503.07338，2025. 5, 8, 15, 16

- [105] H. Bian、L. Kong、H. Xie、L. Pan、Y. Qiao 和 Z. Liu，“DynamicCity：动态场景的大规模 4D 占用生成”，载于_Int。会议。学习。代表._，2025 年，第 1-14 页。 5, 10

- [106] V. Zyrianov、H. Che、Z. Liu 和 S. Wang，“LidarDM：生成世界中的生成 LiDAR 模拟”，载于 IEEE Int。会议。机器人。自动化，2025. 5, 9

- [107] S. Zeng、X. Chang、M. Xie、X. Liu、Y. Bai、Z. Pan、M. Xu 和 X. Wei，“FutureSightDrive：利用时空 CoT 进行视觉思考，实现自动驾驶”，arXiv 预印本 arXiv:2505.17685，2025. 5, 7, 16

- [108] M. Hassan, S. Stapf, A. Rahimi, P. Rezende, Y. Haghighi, D. Brüggemann, I. Katircioglu, L. Zhang, X. Chen, S. Saha et al.，“GEM：一种可概括的自我视觉多模态 世界模型（World Model），用于细粒度自我运动、物体动力学和场景构图控制，”在 _IEEE 会议中。计算。维斯。模式识别，2025 年，第 22 404-22 415 页。5,9,14

- [109] S. Zuo、W. Cheng、Y. Huang、J. Zhou 和 J. Lu，“GaussianWorld：用于流式 3D 占用预测的高斯 世界模型（World Model）”，载于 _IEEE Conf。计算。维斯。模式识别，2025 年，第 6772-6781 页。 5, 9

- [110] J. Ni、Y.Guo、Y.Liu、R.Chen、L.Lu 和 Z.Wu，“MaskGWM：具有视频掩模重建功能的通用驱动世界模型（World Model）”，载于 _IEEE Conf。计算。维斯。模式识别，2025，第 22 381-22 391. 5, 10, 14

- [111] G.Zhao、C.Ni、X.Wang、Z.Zhu、X.Zhang、Y.Wang、G.Huang、X.Chen、B.Wang、Y.Zhang_et al._，“DriveDreamer4D：世界模型（World Models） 是用于 4D 驾驶场景表示的有效数据机”，见 _IEEE Conf.计算。维斯。模式识别，2025，第 12 015-12 026. 5, 10

- [112] C. Ni, G. Zhu, X. Wang, Z. Zhu, W.qin, G. Huang, C. Liu, Y. Chen, Y. Wang, X.Zhang_et al._，“ReconDreamer：通过在线恢复驱动场景重建的 世界模型（World Models）”，载于_IEEE Conf.计算。维斯。模式识别，2025 年，第 1559-1569 页。 5, 10

- [113] Y. Li、Y. Wang、Y. Liu、J. He、L. Fan 和 Z. Zhang，“通过 BEV 世界模型（World Model） 进行在线轨迹评估的端到端驾驶”，载于_Int。会议。计算。视觉_，2025. 5, 7

- [114] X. Zhou、D. Liang、S. Tu、X. Chen、Y. Ding、D. Zhu、F. Tan、H. Zhu 和 X. Bai，“HERMES：用于同步 3D 场景理解和生成的统一自动驾驶 世界模型（World Model）”，载于_Int。会议。计算。视觉_，2025. 5, 10

[115] X. Li, C. Wu, Z. Yang, Z. Xu, D. Liang, Y. Zhang, J. Wan, and J. Wang, "DriVerse: Navigation world model for driving simulation via multimodal trajectory prompting and motion alignment," in *ACM Int. Conf. Multimedia*, 2025. 5, 9, 14

[116] E. Karypidis, I. Kakogeorgiou, S. Gidaris, and N. Komodakis, "DINO-Foresight: Looking into the future with DINO," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2025, pp. 163 779–163 811. 5, 10

[117] M. Okada and T. Taniguchi, "Dreaming: Model-based reinforcement learning by latent imagination without reconstruction," in *IEEE Int. Conf. Robot. Autom.*, 2021, pp. 4209–4215. 4, 15

[118] F. Deng, I. Jang, and S. Ahn, "DreamerPro: Reconstruction-free model-based reinforcement learning with prototypical representations," in *Int. Conf. Mach. Learn.*, 2022, pp. 4956–4975. 4, 15

[119] Q. Wang, Z. Zhang, B. Xie, X. Jin, Y. Wang, S. Wang, L. Zheng, X. Yang, and W. Zeng, "Disentangled world models: Learning to transfer semantic knowledge from distracting videos for reinforcement learning," in *Int. Conf. Comput. Vis.*, 2025. 5, 15

[120] Y. Wang, M. Verghese, and J. Schneider, "Latent policy steering with embodiment-agnostic pretrained world models," *arXiv preprint arXiv:2507.13340*, 2025. 5

[121] C. Sancaktar, C. Gumbsch, A. Zadaianchuk, P. Kolev, and G. Martius, "SENSEI: Semantic exploration guided by foundation models to learn versatile world models," in *Int. Conf. Mach. Learn.*, 2025. 5

[122] V. D. Nguyen, Z. Yang, C. L. Buckley, and A. Ororbia, "SR-AIF: Solving sparse-reward robotic tasks from pixels with active inference and world models," in *IEEE Int. Conf. Robot. Autom.*, 2025, pp. 6510–6518. 5

[123] J. Lanier, K. Kim, A. Karamzade, Y. Liu, A. Sinha, K. He, D. Corsi, and R. Fox, "Adapting world models with latent-state dynamics residuals," *arXiv preprint arXiv:2504.02252*, 2025. 5

[124] H. Wang, X. Ye, F. Tao, C. Pan, A. Mallik, B. Yaman, L. Ren, and J. Zhang, "AdaWM: Adaptive world model based planning for autonomous driving," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–13. 5

[125] H. Lai, J. Cao, J. Xu, H. Wu, Y. Lin, T. Kong, Y. Yu, and W. Zhang, "World model-based perception for visual legged locomotion," in *IEEE Int. Conf. Robot. Autom.*, 2025. 5

[126] Y. Wang, R. Yu, S. Wan, L. Gan, and D.-C. Zhan, "FOUNDER: Grounding foundation models in world models for open-ended embodied decision making," in *Int. Conf. Mach. Learn.*, 2025. 5

[127] I. Nematollahi, B. DeMoss, A. L. Chandra, N. Hawes, W. Burgard, and I. Posner, "LUMOS: Language-conditioned imitation learning with world models," in *IEEE Int. Conf. Robot. Autom.*, 2025. 5

[128] A. Popov, A. Degirmenci, D. Wehr, S. Hegde, R. Oldja, A. Kamenev, B. Douillard, D. Nistér, U. Muller, R. Bhargava *et al.*, "Mitigating covariate shift in imitation learning for autonomous vehicles using latent space generative world models," in *IEEE Int. Conf. Robot. Autom.*, 2025. 5

[129] Y. Qu, Z. Huang, Z. Sheng, J. Chen, S. Chen, and S. Labi, "VL-SAFE: Vision-Language guided safety-aware reinforcement learning with world models for autonomous driving," *arXiv preprint arXiv:2505.16377*, 2025. 5

[130] H. Wang, D. Gao, and J. Zhang, "Ego-centric learning of communicative world models for autonomous driving," *arXiv preprint arXiv:2506.08149*, 2025. 5

[131] R. G. Goswami, P. Krishnamurthy, Y. LeCun, and F. Khorrami, "OSVI-WM: One-shot visual imitation for unseen tasks using world-model-guided trajectory generation," *arXiv preprint arXiv:2505.20425*, 2025. 6

[132] C. Li, A. Krause, and M. Hutter, "Robotic world model: A neural network simulator for robust policy optimization in robotics," *arXiv preprint arXiv:2501.10100*, 2025. 6

[133] W. Liu, H. Zhao, C. Li, J. Biswas, B. Okal, P. Goyal, Y. Chang, and S. Pouya, "X-MOBILITY: End-to-end generalizable navigation via world modeling," in *IEEE Int. Conf. Robot. Autom.*, 2025. 6

[134] W. Sun, L. Chen, Y. Su, B. Cao, Y. Liu, and Z. Xie, "Learning humanoid locomotion with world model reconstruction," *arXiv preprint arXiv:2502.16230*, 2025. 6

[135] P. Agrawal, A. V. Nair, P. Abbeel, J. Malik, and S. Levine, "Learning to Poke by Poking: Experiential learning of intuitive physics," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2016. 6

[136] X. Yao, J. Gao, and C. Xu, "NavMorph: A self-evolving world model for vision-and-language navigation in continuous environments," in *Int. Conf. Comput. Vis.*, 2025. 6

[137] M. Burchi and R. Timofte, "Learning transformer-based world models with contrastive predictive coding," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–14. 6

[138] T. Feng, X. Wang, Z. Zhou, R. Wang, Y. Zhan, G. Li, Q. Li, and W. Zhu, "EvoAgent: Agent autonomous evolution with continual world model for long-horizon tasks," *arXiv preprint arXiv:2502.05907*, 2025. 6

[139] Z. Chen, J. Huo, Y. Chen, and Y. Gao, "RoboHorizon: An LLM-assisted multi-view world model for long-horizon robotic manipulation," *arXiv preprint arXiv:2501.06605*, 2025. 6

[140] X. Hu, W. Yin, M. Jia, J. Deng, X. Guo, Q. Zhang, X. Long, and P. Tan, "DrivingWorld: Constructing world model for autonomous driving via video gpt," *arXiv preprint arXiv:2412.19505*, 2024. 6, 14

[141] W. Zheng, Z. Xia, Y. Huang, S. Zuo, J. Zhou, and J. Lu, "Doe-1: Closed-loop autonomous driving with large world model," *arXiv preprint arXiv:2412.09627*, 2024. 6, 16

[142] L. Xiao, J.-J. Liu, S. Yang, X. Li, X. Ye, W. Yang, and J. Wang, "Learning multiple probabilistic decisions from latent world model in autonomous driving," in *IEEE Int. Conf. Robot. Autom.*, 2025. 6

[143] A. B. Vasudevan, N. Peri, J. Schneider, and D. Ramanan, "Planning with adaptive world models for autonomous driving," in *IEEE Int. Conf. Robot. Autom.*, 2025, pp. 14 938–14 945. 6

[144] A. Dedieu, J. Ortiz, X. Lou, C. Wendelken, J. S. Guntupalli, W. Lehrach, M. Lazaro-Gredilla, and K. P. Murphy, "Improving transformer world models for data-efficient RL," in *Int. Conf. Mach. Learn.*, 2025. 6

[145] J. Lyu, Z. Li, X. Shi, C. Xu, Y. Wang, and H. Wang, "DyWA: Dynamics-adaptive world action model for generalizable non-prehensile manipulation," in *Int. Conf. Comput. Vis.*, 2025. 6

[146] L. Chen, Y. Wang, S. Tang, Q. Ma, T. He, W. Ouyang, X. Zhou, H. Bao, and S. Peng, "EgoAgent: A joint predictive agent model in egocentric worlds," in *Int. Conf. Comput. Vis.*, 2025. 6

[147] A. Scannell, M. Nakhaeinezhadfard, K. Kujanpää, Y. Zhao, K. S. Luck, A. Solin, and J. Pajarinen, "Discrete codebook world models for continuous control," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–16. 6

[148] S. Yin, J. Wu, S. Huang, X. Su, X. He, J. HAO, and M. Long, "Trajectory world models for heterogeneous environments," in *Int. Conf. Mach. Learn.*, 2025. 6

[149] S. Hamdan and F. Güney, "CarFormer: Self-driving with learned object-centric representations," in *Eur. Conf. Comput. Vis.*, 2024, pp. 177–193. 6

[150] Y. Jeong, J. Chun, S. Cha, and T. Kim, "Object-centric world model for language-guided manipulation," in *Int. Conf. Learn. Represent. Worksh.*, 2025, pp. 1–13. 6

[151] V. Micheli, E. Alonso, and F. Fleuret, "Efficient world models with context-aware tokenization," in *Int. Conf. Mach. Learn.*, 2024, pp. 35 623–35 638. 6

[152] S. Wang, Z. Fei, Q. Cheng, S. Zhang, P. Cai, J. Fu, and X. Qiu, "World Modeling Makes a Better Planner: Dual preference optimization for embodied task planning," in *Annu. Meet. Assoc. Comput. Linguistics*, 2025. 6

[153] K. Zhang, P. Ren, B. Lin, J. Lin, S. Ma, H. Xu, and X. Liang, "PIVOT-R: Primitive-driven waypoint-aware world model for robotic manipulation," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2024, pp. 54 105–54 136. 6

[154] Y. Chen, J. Wei, C. Xu, B. Li, M. Tomizuka, A. Bajcsy, and R. Tian, "Reimagination with Test-time Observation Interventions: Distractor-robust world model predictions for visual model predictive control," in *Robotics: Science and Systems Worksh.*, 2025. 6

[155] K. Zhang, Z. Tang, X. Hu, X. Pan, X. Guo, Y. Liu, J. Huang, L. Yuan, Q. Zhang, X.-X. Long *et al.*, "Epona: Autoregressive diffusion world model for autonomous driving," in *Int. Conf. Comput. Vis.*, 2025. 7, 14, 16

[156] M. Goff, G. Hogan, G. Hotz, A. du Parc Locmaria, K. Raczy, H. Schäfer, A. Shihadeh, W. Zhang, and Y. Yousfi, "Learning to drive from a world model," in *IEEE Conf. Comput. Vis. Pattern Recog. Worksh.*, 2025, pp. 1964–1973. 7

[157] S. Tan, J. Lambert, H. Jeon, S. Kulshrestha, Y. Bai, J. Luo, D. Anguelov, M. Tan, and C. M. Jiang, "SceneDiffuser++: City-scale traffic simulation via a generative world model," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 1570–1580. 7

[158] X. Yu, B. Peng, R. Xu, M. Galley, H. Cheng, S. Nath, J. Gao, and Z. Yu, "Dyna-Think: Synergizing reasoning, acting, and world model simulation in AI agents," *arXiv preprint arXiv:2506.00320*, 2025. 7

[159] Z. Zhao, W. Zhang, H. Huang, K. Liu, J. Gao, G. Wang, and K. Chen, "RIG: Synergizing reasoning and imagination in End-to-End generalist policy," *arXiv preprint arXiv:2503.24388*, 2025. 7

[160] J. Gkountouras, M. Lindemann, P. Lippe, E. Gavves, and I. Titov, "Language agents meet causality–bridging llms and causal world models," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–15. 7

原文第 20 页 | 中文翻译初稿

- [115] X. Li、C. Wu、Z. Yang、Z. Xu、D. Liang、Y. Zhang、J. Wan 和 J. Wang，“DriVerse：通过多模态轨迹提示和运动对齐进行驾驶模拟的导航 世界模型（World Model）”，载于_ACM Int。会议。多媒体_，2025. 5, 9, 14

- [116] E. Karypidis、I. Kakogeorgiou、S. Gidaris 和 N. Komodakis，“DINOForesight：与 DINO 一起展望未来”，载于 Annu。会议。神经。通知。过程。系统.，2025 年，第 163 779-163 811. 5, 10 页

- [117] M. Okada 和 T. Taniguchi，“梦想：基于模型的 强化学习（Reinforcement Learning），无需重构，通过潜在想象”，载于 IEEE Int。会议。机器人。 Autom.，2021 年，第 4209-4215 页。 4、15

- [118] F. Deng、I. Jang 和 S. Ahn，“DreamerPro：具有原型表示的基于无重建模型的 强化学习（Reinforcement Learning）”，载于_Int。会议。马赫。学习._，2022 年，第 4956-4975 页。 4、15

- [119] Q. Wang、Z. Zhang、B. Xie、X. Jin、Y. Wang、S. Wang、L. Cheng、X. Yang 和 W. Zeng，“解开 世界模型（World Models）：学习从 强化学习（Reinforcement Learning）分散注意力的视频中转移语义知识”，载于_Int。会议。计算。视觉_, 2025. 5, 15

- [120] Y. Wang、M. Verghese 和 J. Schneider，“采用与实施例无关的预训练 世界模型（World Models）进行潜在策略引导”，arXiv 预印本 arXiv:2507.13340，2025. 5

- [121] C. Sancaktar、C. Gumbsch、A. Zadaianchuk、P. Kolev 和 G. Martius，“SENSEI：基础模型（Foundation Models）引导的语义探索以学习多功能 世界模型（World Models）”，载于_Int。会议。马赫。学习._, 2025. 5

- [122] V. D. Nguyen、Z. Yang、C. L. Buckley 和 A. Ororbia，“SR-AIF：通过主动推理和 世界模型（World Models）的像素解决稀疏奖励机器人任务”，载于 IEEE Int。会议。机器人。 Autom.，2025 年，第 6510-6518 页。5

- [123] J. Lanier、K. Kim、A. Karamzade、Y. Liu、A. Sinha、K. He、D. Corsi 和 R. Fox，“利用潜在状态动力学残差调整 世界模型（World Models）”，arXiv 预印本 arXiv:2504.02252，2025. 5

- [124] H. Wang、X. Ye、F. Tao、C. Pan、A. Mallik、B. Yaman、L. Ren 和 J. Zhang，“AdaWM：基于自适应 世界模型（World Model）的自动驾驶规划”，载于_Int。会议。学习。代表._，2025 年，第 1-13 页。 5

- [125] H. Lai、J. Cao、J. Xu、H. Wu、Y. Lin、T. Kong、Y. Yu 和 W. Zhang，“基于 世界模型（World Model）的视觉腿部运动感知”，载于 _IEEE Int。会议。机器人。自动化, 2025. 5

- [126] Y. Wang、R. Yu、S. Wan、L. Gan 和 D.-C。 Zhan，“创始人：落地关联（Grounding）基础模型（Foundation Models）in 世界模型（World Models），用于开放式具体决策，”in Int。会议。马赫。学习. , 2025. 5

- [127] I. Nematollahi、B. DeMoss、A. L. Chandra、N. Hawes、W. Burgard 和 I. Posner，“LUMOS：语言条件 模仿学习（Imitation Learning）和 世界模型（World Models）”，载于_IEEE Int。会议。机器人。自动化, 2025. 5

- [128] A. Popov、A. Degirmenci、D. Wehr、S. Hegde、R. Oldja、A. Kamenev、B. Douillard、D. Nistér、U. Muller、R. Bhargava 等人，“使用潜在空间生成技术减轻自动驾驶汽车 模仿学习（Imitation Learning）中的协变量偏移世界模型（World Models），”在 _IEEE Int。会议。机器人。自动化, 2025. 5

- [129] Y. Qu、Z. Huang、Z. Shen、J. Chen、S. Chen 和 S. Labi，“VL-SAFE：用于自动驾驶的视觉语言引导安全感知 强化学习（Reinforcement Learning） 和 世界模型（World Models）”，arXiv 预印本 arXiv:2505.16377，2025. 5

- [130] H.王，D.高，J

。张，“用于自动驾驶的交流 世界模型（World Models） 的以自我为中心的学习”，arXiv 预印本 arXiv:2506.08149，2025. 5

- [131] R. G. Goswami、P. Krishnamurthy、Y. LeCun 和 F. Khorrami，“OSVIWM：使用世界模型引导轨迹生成对看不见的任务进行一次性视觉模仿”，arXiv 预印本 arXiv:2505.20425，2025. 6

- [132] C. Li、A. Krause 和 M. Hutter，“机器人 世界模型（World Model）：用于机器人技术中稳健策略优化的神经网络模拟器”，arXiv 预印本 arXiv:2501.10100，2025. 6

- [133] W. Liu、H. Zhu、C. Li、J. Biswas、B. Okal、P. Goyal、Y. Chang 和 S. Pouya，“X-MOBILITY：通过世界建模实现端到端通用导航”，载于 _IEEE Int。会议。机器人。自动化, 2025. 6

- [134] W. Sun、L. Chen、Y. Su、B. Cao、Y. Liu 和 Z. Xie，“通过 世界模型（World Model） 重建学习人形运动”，arXiv 预印本 arXiv:2502.16230，2025. 6

- [135] P. Agrawal、A. V. Nair、P. Abbeel、J. Malik 和 S. Levine，“通过戳戳来学习戳戳：直观物理学的体验式学习”，《Annu》。会议。神经。通知。过程。系统._，2016. 6

- [136] X. Yao、J. Gau 和 C. Xu，“NavMorph：用于连续环境中视觉和语言导航的自我进化 世界模型（World Model）”，载于 Int。会议。计算。视觉., 2025. 6

- [137] M. Burchi 和 R. Timofte，“使用对比预测编码学习基于 Transformer 的 世界模型（World Models）”，载于 Int。会议。学习。代表.，2025 年，第 1-14 页。 6

- [138] T. Feng、X. Wang、Z. Zhou、R. Wang、Y. Zhan、G. Li、Q. Li 和 W. Zhu，“EvoAgent：针对 长时程（Long-Horizon） 任务的持续 世界模型（World Model） 的智能体自主进化”，arXiv 预印本 arXiv:2502.05907，2025. 6

- [139] Z. Chen、J. Huo、Y. Chen 和 Y. Gau，“RoboHorizon：用于 长时程（Long-Horizon） 机器人操作的法学硕士辅助多视图 世界模型（World Model）”，arXiv 预印本 arXiv：2501.06605，2025 年 6 月

- [140] X. Hu、W. Yin、M. Jia、J. Deng、X.Guo、Q.Zhang、X.Long 和 P.Tan，“DrivingWorld：通过视频 gpt 构建 世界模型（World Model） 用于自动驾驶”，arXiv 预印本 arXiv:2412.19505，2024. 6, 14

- [141] W. Cheng、Z. Xia、Y. Huang、S. Zuo、J. Zhou 和 J. Lu，“Doe-1：大型 世界模型（World Model）闭环自动驾驶”，arXiv 预印本 arXiv:2412.09627，2024. 6, 16

- [142] L.肖，J.-J。 Liu、S. Yang、X. Li、X. Ye、W. Yang 和 J. Wang，“从自动驾驶中的潜在 世界模型（World Model） 中学习多个概率决策”，载于 _IEEE Int。会议。机器人。自动化, 2025. 6

- [143] A. B. Vasudevan、N. Peri、J. Schneider 和 D. Ramanan，“利用自适应 世界模型（World Models）进行自动驾驶规划”，载于 IEEE Int。会议。机器人。 Autom.，2025 年，第 14 938-14 945 页。6

- [144] A. Dedieu、J. Ortiz、X. Lou、C. Wendelken、J. S. Guntupalli、W. Lehrach、M. Lazaro-Gredilla 和 K. P. Murphy，“改进变压器 世界模型（World Models） 以实现数据高效的 RL”，载于_Int。会议。马赫。学习._，2025. 6

- [145] J. Lyu、Z. Li、X. Shi、C. Xu、Y. Wang 和 H. Wang，“DyWA：用于广义非抓取操作的动力学自适应世界动作模型”，载于_Int。会议。计算。视觉._, 2025. 6

- [146] L. Chen、Y. Wang、S. Tang、Q. Ma、T. He、W. Ouyang、X. Zhou、H. Bao 和 S. Peng，“EgoAgent：自我中心世界中的联合预测智能体模型”，载于_Int。会议。计算。视觉._, 2025. 6

- [147] A. Scannell，M. Nakhaeinezhadfard，K

。 Kujanpää、Y.Zhao、K.S.Luck、A.Solin 和 J.Pajarinen，“用于连续控制的离散码本 世界模型（World Models）”，载于 Int。会议。学习。代表.，2025 年，第 1-16 页。 6

- [148] S. Yin、J. Wu、S. Huang、X. Su、X. He、J. HAO 和 M. Long，“异构环境的轨迹 世界模型（World Models）”，载于_Int。会议。马赫。学习._，2025. 6

- [149] S. Hamdan 和 F. Güney，“CarFormer：利用学习的以对象为中心的表示进行自动驾驶”，载于 Eur。会议。计算。 Vis.，2024 年，第 177-193 页。 6

- [150] Y. Jeong、J. Chun、S. Cha 和 T. Kim，“用于语言引导操作的以对象为中心的 世界模型（World Model）”，载于_Int。会议。学习。代表。 Worksh._，2025 年，第 1-13 页。 6

- [151] V. Micheli、E. Alonso 和 F. Fleuret，“具有上下文感知标记化的高效 世界模型（World Models）”，载于 Int。会议。马赫。学习.，2024 年，第 35 623-35 638 页。6

- [152] S. Wang、Z. Fei、Q. Cheng、S. Zhang、P. Cai、J. Fu 和 X. Qiu，“世界建模打造更好的规划器：具体任务规划的双重偏好优化”，载于 Annu。见面。副教授。计算。语言学，2025. 6

- [153] K.Zhang、P.Ren、B.Lin、J.Lin、S.Ma、H.Xu 和 X.Liang，“PIVOT-R：用于机器人操作的原始驱动的路径点感知 世界模型（World Model）”，载于 _Annu。会议。神经。通知。过程。系统，2024 年，第 54 105-54 136. 6 页

- [154] Y. Chen、J. Wei、C. Xu、B. Li、M. Tomizuka、A. Bajcsy 和 R. Tian，“通过测试时间观察干预重新想象：视觉模型预测控制的 Distractorrobust 世界模型（World Model） 预测”，《机器人：科学与系统工作》，2025 年。6

- [155] 张坤、唐正、胡晓、潘晓、郭晓、刘勇、黄俊、袁丽、张强、X.-X。 Long et al. ，“Epona：用于自动驾驶的自回归扩散 世界模型（World Model）”，in Int。会议。计算。视觉，2025. 7, 14, 16

- [156] M. Goff、G. Hogan、G. Hotz、A. du Parc Locmaria、K. Raczy、H. Schäfer、A. Shihadeh、W. Zhu 和 Y. Yousfi，“从 世界模型（World Model） 学习驾驶”，载于 IEEE Conf。计算。维斯。模式识别。 Worksh.，2025 年，第 1964-1973 页。 7

- [157] S. Tan、J. Lambert、H. Jeon、S. Kulshrestha、Y. Bai、J. Luo、D. Anguelov、M. Tan 和 C. M. Jiang，“SceneDiffuser++：通过生成式 世界模型（World Model） 进行城市规模交通模拟”，载于 _IEEE Conf.计算。维斯。模式识别，2025 年，第 1570-1580 页。 7

- [158] X. Yu、B. Peng、R. Xu、M. Galley、H. Cheng、S. Nath、J. Gau 和 Z. Yu，“Dyna-Think：在人工智能智能体中协同推理、行动和 世界模型（World Model） 模拟”，arXiv 预印本 arXiv:2506.00320，

2025. 7

- [159] Z.Zhao、W.Zhang、H.Huang、K.Liu、J.Gao、G.Wang 和 K.Chen，“RIG：在端到端通才策略中协同推理和想象力”，arXiv 预印本 arXiv:2503.24388，2025. 7

- [160] J. Gkountouras、M. Lindemann、P. Lippe、E. Gavves 和 I. Titov，“语言智能体满足因果关系--桥接 llms 和因果关系 世界模型（World Models）”，载于_Int。会议。学习。代表._，2025 年，第 1-15 页。 7

[161] T. Yin, Z. Mei, T. Sun, L. Zha, E. Zhou, J. Bao, M. Yamane, O. Sho, and A. Majumdar, "WoMAP: World models for embodied open-vocabulary object localization," in *Robotics: Science and Systems Worksh.*, 2025. 7

[162] Z. Yang, X. Jia, Q. Li, X. Yang, M. Yao, and J. Yan, "Raw2Drive: Reinforcement learning with aligned world models for End-to-End autonomous driving (in CARLA v2)," *arXiv preprint arXiv:2505.16394*, 2025. 7

[163] Z. Yan, W. Dong, Y. Shao, Y. Lu, L. Haiyang, J. Liu, H. Wang, Z. Wang, Y. Wang, F. Remondino *et al.*, "RenderWorld: World model with self-supervised 3D label," in *IEEE Int. Conf. Robot. Autom.*, 2025. 7, 15, 16

[164] Y. Yang, J. Mei, Y. Ma, S. Du, W. Chen, Y. Qian, Y. Feng, and Y. Liu, "Driving in the Occupancy World: Vision-Centric 4D occupancy forecasting and planning via world models for autonomous driving," in *AAAI Conf. Artif. Intell.*, 2025, pp. 9327–9335. 7, 16

[165] X. Li, P. Li, Y. Zheng, W. Sun, Y. Wang, and Y. Chen, "Semi-supervised vision-centric 3D occupancy world model for autonomous driving," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–13. 7

[166] Y. Li, L. Fan, J. He, Y. Wang, Y. Chen, Z. Zhang, and T. Tan, "Enhancing End-to-End autonomous driving with latent world model," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–14. 7

[167] P. Li and D. Cui, "Navigation-guided sparse scene representation for End-to-End autonomous driving," in *Int. Conf. Learn. Represent.*, 2024, pp. 1–13. 7, 16

[168] Z. Huang, J. Zhang, and E. Ohn-Bar, "Neural volumetric world models for autonomous driving," in *Eur. Conf. Comput. Vis.*, 2024, pp. 195–213. 7, 16

[169] Y. Yang, H. Lin, Y. Luo, S. Fu, C. Zheng, X. Yan, S. Mei, K. Tang, S. Cui, and Z. Li, "FASTopoWM: Fast-slow lane segment topology reasoning with latent world models," *arXiv preprint arXiv:2507.23325*, 2025. 7

[170] D. Nie, X. Guo, Y. Duan, R. Zhang, and L. Chen, "WMNav: Integrating vision-language models into world models for object goal navigation," in *Int. Conf. Intell. Robot. Syst.*, 2025. 7

[171] Z. Zhang, Q. Zhang, W. Cui, S. Shi, Y. Guo, G. Han, W. Zhao, J. Sun, J. Cao, J. Wang *et al.*, "Occupancy world model for robots," *arXiv preprint arXiv:2505.05512*, 2025. 7

[172] S. Huang, Q. Chen, X. Zhang, J. Sun, and M. Schwager, "ParticleFormer: A 3D point cloud world model for multi-object, multi-material robotic manipulation," in *Conf. Robot. Learn.*, 2025. 7

[173] J. Abou-Chakra, K. Rana, F. Dayoub, and N. Sünderhauf, "Physically Embodied Gaussian Splatting: A realtime correctable world model for robotics," in *Conf. Robot. Learn.*, 2025, pp. 513–530. 7

[174] T. Jiang, Y. Guan, L. Ma, J. Xu, J. Meng, W. Chen, Z. Zeng, L. Li, D. Wu, and R. Chen, "DexSim2Real^2: Building explicit world model for precise articulated object dexterous manipulation," *IEEE Trans. Robot.*, vol. 41, pp. 4360–4379, 2025. 8

[175] W. Li, H. Zhao, Z. Yu, Y. Du, Q. Zou, R. Hu, and K. Xu, "PIN-WM: Learning physics-informed world models for non-prehensile manipulation," in *Robotics: Science and Systems*, 2025. 8

[176] C. Ning, K. Fang, and W.-C. Ma, "Prompting with the Future: Open-world model predictive control with interactive digital twins," in *Robotics: Science and Systems*, 2025. 8

[177] A. Chen, W. Zheng, Y. Wang, X. Zhang, K. Zhan, P. Jia, K. Keutzer, and S. Zhang, "GeoDrive: 3D geometry-informed driving world model with precise action control," *arXiv preprint arXiv:2505.22421*, 2025. 8, 14

[178] R. Zheng, J. Wang, S. Reed, J. Bjorck, Y. Fang, F. Hu, J. Jang, K. Kundalia, Z. Lin, L. Magne *et al.*, "FLARE: Robot learning with implicit world modeling," in *Robotics: Science and Systems*, 2025. 8

[179] Y. Huang, J. Zhang, S. Zou, X. Liu, R. Hu, and K. Xu, "LaDi-WM: A latent diffusion-based world model for predictive manipulation," in *Conf. Robot. Learn.*, 2025. 8

[180] B. Wang, X. Meng, X. Wang, Z. Zhu, A. Ye, Y. Wang, Z. Yang, C. Ni, G. Huang, and X. Wang, "EmbodieDreamer: Advancing real2sim2real transfer for policy training via embodied world modeling," *arXiv preprint arXiv:2507.05198*, 2025. 8

[181] H. Zhang, Y. Xue, X. Yan, J. Zhang, W. Qiu, D. Bai, B. Liu, S. Cui, and Z. Li, "An efficient occupancy world model via decoupled dynamic flow and image-assisted training," *arXiv preprint arXiv:2412.13772*, 2024. 8, 15, 16

[182] Y. Li, X. Wei, X. Chi, Y. Li, Z. Zhao, H. Wang, N. Ma, M. Lu, and S. Zhang, "ManipDreamer: Boosting robotic manipulation world model with action tree and visual guidance," *arXiv preprint arXiv:2504.16464*, 2025. 8

[183] H. Zhi, P. Chen, S. Zhou, Y. Dong, Q. Wu, L. Han, and M. Tan, "3DFlowAction: Learning cross-embodiment manipulation from 3D flow world model," *arXiv preprint arXiv:2506.06199*, 2025. 8

[184] A. Garg and K. Madhava Krishna, "Imagine-2-Drive: High-fidelity world modeling in carla for autonomous vehicles," in *Int. Conf. Intell. Robot. Syst.*, 2025. 8

[185] Y. Zheng, P. Yang, Z. Xing, Q. Zhang, Y. Zheng, Y. Gao, P. Li, T. Zhang, Z. Xia, P. Jia *et al.*, "World4Drive: End-to-End autonomous driving via intention-aware physical latent world model," in *Int. Conf. Comput. Vis.*, 2025. 8

[186] H. Zhang, Z. Wang, Q. Lyu, Z. Zhang, S. Chen, T. Shu, B. Dariush, K. Lee, Y. Du, and C. Gan, "COMBO: Compositional world models for embodied multi-agent cooperation," in *Int. Conf. Learn. Represent.*, 2025, pp. 1–16. 8

[187] Y. Shang, X. Zhang, Y. Tang, L. Jin, C. Gao, W. Wu, and Y. Li, "RoboScape: Physics-informed embodied world model," *arXiv preprint arXiv:2506.23135*, 2025. 8

[188] R. Bonatti, S. Vemprala, S. Ma, F. Frujeri, S. Chen, and A. Kapoor, "PACT: Perception-action causal transformer for autoregressive robotics pre-training," in *Int. Conf. Intell. Robot. Syst.*, 2023, pp. 3621–3627. 8

[189] F. Baldassarre, M. Szafraniec, B. Terver, V. Khalidov, F. Massa, Y. LeCun, P. Labatut, M. Seitzer, and P. Bojanowski, "Back to the Features: DINO as a foundation for video world models," *arXiv preprint arXiv:2507.19468*, 2025. 8

[190] Y. Huang, W. Zheng, Y. Gao, X. Tao, P. Wan, D. Zhang, J. Zhou, and J. Lu, "Owl-1: Omni world model for consistent long video generation," *arXiv preprint arXiv:2412.09600*, 2024. 8

[191] S. Huang, J. Wu, Q. Zhou, S. Miao, and M. Long, "Vid2World: Crafting video diffusion models to interactive world models," *arXiv preprint arXiv:2505.14357*, 2025. 8

[192] H. Wu, D. Wu, T. He, J. Guo, Y. Ye, Y. Duan, and J. Bian, "Geometry Forcing: Marrying video diffusion and 3D representation for consistent world modeling," *arXiv preprint arXiv:2507.07982*, 2025. 9

[193] T. Chen, X. Hu, Z. Ding, and C. Jin, "Learning world models for interactive video generation," *arXiv preprint arXiv:2505.21996*, 2025. 9

[194] X. Guo, C. Ding, H. Dou, X. Zhang, W. Tang, and W. Wu, "Infinity-Drive: Breaking time limits in driving world models," *arXiv preprint arXiv:2412.01522*, 2024. 9

[195] X. Wang, Z. Wu, and P. Peng, "Longdwm: Cross-granularity distillation for building a long-term driving world model," *arXiv preprint arXiv:2506.01546*, 2025. 9, 14

[196] H. Wang, D. Liu, H. Xie, H. Liu, E. Ma, K. Yu, L. Wang, and B. Wang, "MiLA: Multi-view intensive-fidelity long-term video generation world model for autonomous driving," *arXiv preprint arXiv:2503.15875*, 2025. 9, 14

[197] A. Mousakhan, S. Mittal, S. Galesso, K. Farid, and T. Brox, "Orbis: Overcoming challenges of long-horizon prediction in driving world models," *arXiv preprint arXiv:2507.13162*, 2025. 9

[198] J. Quevedo, S. Ansh Kumar, Y. Sun, V. Suryavanshi, P. Liang, and S. Yang, "WorldGym: World model as an environment for policy evaluation," *arXiv preprint arXiv:2506.00613*, 2025. 9

[199] Y. Li, Y. Zhu, J. Wen, C. Shen, and Y. Xu, "WorldEval: World model as real-world robot policies evaluator," *arXiv preprint arXiv:2505.19017*, 2025. 9

[200] Y. Guan, H. Liao, C. Wang, X. Liu, J. Zhang, and Z. Li, "World model-based End-to-End scene generation for accident anticipation in autonomous driving," *Communications Engineering*, vol. 4, no. 1, p. 144, 2025. 9

[201] R. Po, Y. Nitzan, R. Zhang, B. Chen, T. Dao, E. Shechtman, G. Wetzstein, and X. Huang, "Long-Context State-Space video world models," in *Int. Conf. Comput. Vis.*, 2025. 9

[202] V. L. Guen and N. Thome, "Disentangling physical dynamics from unknown factors for unsupervised video prediction," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 11 474–11 484. 9

[203] X. Liu and H. Tang, "FOLIAGE: Towards physical intelligence world models via unbounded surface evolution," *arXiv preprint arXiv:2506.03173*, 2025. 9

[204] S. Zhou, Y. Du, Y. Yang, L. Han, P. Chen, D.-Y. Yeung, and C. Gan, "Learning 3D persistent embodied world models," *arXiv preprint arXiv:2505.05495*, 2025. 9

[205] J. Wang, Y. Yao, X. Feng, H. Wu, Y. Wang, Q. Huang, Y. Ma, and X. Zhu, "STAGE: A stream-centric generative world model for long-horizon driving-scene simulation," in *Int. Conf. Intell. Robot. Syst.*, 2025. 9, 14

原文第 21 页 | 中文翻译初稿

- [161] T. Yin, Z. Mei, T. Sun, L. Zha, E. Zhou, J. Bao, M. Yamane, O. Sho, and A. Majumdar, “WoMAP: 世界模型（World Models） for embodied open-vocabulary object localization,” in Robotics: Science and Systems Worksh., 2025. 7

- [162] Z. Yang、X. Jia、Q. Li、X. Yang、M. Yao 和 J. Yan，“Raw2Drive：强化学习（Reinforcement Learning） 与对齐的 世界模型（World Models） 用于端到端自动驾驶（在 CARLA v2 中）”，arXiv 预印本 arXiv:2505.16394，2025. 7

- [163] Z. Yan, W. Dong, Y. Shao, Y. Lu, L. Haiyang, J. Liu, H. Wang, Z. Wang, Y. Wang, F. Remondino et al. , “RenderWorld: 世界模型（World Model） with selfsupervised 3D label,” in IEEE Int.会议。机器人。汽车 , 2025. 7, 15, 16

- [164] Y. Yang、J. Mei、Y. Ma、S. Du、W. Chen、Y. Qian、Y. Feng 和 Y. Liu，“在占用世界中驾驶：通过 世界模型（World Models） 实现自动驾驶的以视觉为中心的 4D 占用预测和规划”，载于 AAAI Conf. 阿蒂夫。 Intel.，2025 年，第 9327-9335 页。 7, 16

- [165] X. Li、P. Li、Y. Cheng、W. Sun、Y. Wang 和 Y. Chen，“用于自动驾驶的半监督视觉中心 3D 占用 世界模型（World Model）”，载于_Int。会议。学习。代表._，2025 年，第 1-13 页。 7

- [166] Y. Li、L. Fan、J. He、Y. Wang、Y. Chen、Z.Zhang 和 T. Tan，“利用潜在的 世界模型（World Model） 增强端到端自动驾驶”，载于_Int。会议。学习。代表._，2025 年，第 1-14 页。 7

- [167] P. Li 和 D. Cui，“端到端自动驾驶的导航引导稀疏场景表示”，载于_Int。会议。学习。代表._，2024 年，第 1-13 页。 7, 16

- [168] Z. Huang、J. Zhang 和 E. Ohn-Bar，“用于自动驾驶的神经体积 世界模型（World Models）”，载于 Eur。会议。计算。 Vis.，2024 年，第 195-213 页。 7, 16

- [169] Y. Yang, H. Lin, Y. Luo, S. Fu, C. Cheng, X. Yan, S. Mei, K. Tang, S. Cui, and Z. Li, “FASTopoWM: Fast-slow Lane Segment Topology Reasoning with Latent 世界模型（World Models）,” arXiv preprint arXiv:2507.23325 , 2025. 7

- [170] D. Nie、X.Guo、Y.Duan、R.Zhang 和 L.Chen，“WMNav：将视觉语言模型集成到 世界模型（World Models） 中以实现目标目标导航”，载于_Int。会议。英特尔。机器人。系统._, 2025. 7

- [171] Z.Zhang, Q.Zhang, W.Cui, S.Shi, Y.Guo, G.Han, W.Zhao, J.Sun, J.Cao, J.Wang et al., “机器人的占用 世界模型（World Model）”, arXiv 预印本 arXiv:2505.05512 , 2025. 7

- [172] S. Huang、Q. Chen、X. Zhu、J. Sun 和 M. Schwager，“ParticleFormer：用于多对象、多材料机器人操作的 3D 点云 世界模型（World Model）”，参见 Conf。机器人。学习. , 2025. 7

- [173] J. Abou-Chakra、K. Rana、F. Dayoub 和 N. Sünderhauf，“物理体现的高斯泼溅：用于机器人技术的实时可校正 世界模型（World Model）”，参见 Conf。机器人。学习.，2025 年，第 513-530 页。 7

- [174] T. Jiang、Y.guan、L.Ma、J.Xu、J.Meng、W.Chen、Z.Zeng、L.Li、D.Wu 和 R.Chen，“DexSim2Real2：构建用于精确铰接对象灵巧操作的显式 世界模型（World Model）”，IEEE Trans。机器人.，卷。 41，第 4360-4379 页，2025 年。 8

- [175] W. Li、H. Zhu、Z. Yu、Y. Du、Q. Zou、R. Hu 和 K. Xu，“PINWM：学习物理信息 世界模型（World Models）用于非抓取操作”，《机器人：科学与系统》，2025 年。 8

- [176] C. Ning，K. Fang，W.-C. Ma，“提示未来：使用交互式数字孪生的开放世界模型预测控制”，《机器人：科学与系统》，2025 年

。 8

- [177] A. Chen、W. Cheng、Y. Wang、X. Zhang、K. Zhan、P. Jia、K. Keutzer 和 S. Zhang，“GeoDrive：具有精确动作控制的 3D 几何信息驾驶 世界模型（World Model）”，arXiv 预印本 arXiv:2505.22421，2025. 8, 14

- [178] R. Cheng、J. Wang、S. Reed、J. Bjorck、Y. Fang、F. Hu、J. Jang、K. Kundalia、Z. Lin、L. Magne 等人，“FLARE：使用隐式世界建模进行机器人学习”，《机器人学：科学与系统》，2025 年。 8

- [179] Y. Huang、J. Zhang、S. Zou、X. Liu、R. Hu 和 K. Xu，“LaDi-WM：用于预测操作的基于潜在扩散的 世界模型（World Model）”，参见 Conf。机器人。学习. , 2025. 8

- [180] B. Wang、X.Meng、X.Wang、Z.Zhu、A.Ye、Y.Wang、Z.Yang、C.Ni、G.Huang 和 X.Wang，“EmbodieDreamer：通过体现世界建模推进用于策略训练的 real2仿真到现实（Sim2Real） 迁移”，arXiv 预印本 arXiv:2507.05198，2025 年。 8

- [181] H.Zhang、Y.Xue、X.Yan、J.Zhang、W.Qiu、D.Bai、B.Liu、S.Cui 和 Z.Li，“通过解耦动态流和图像辅助训练实现高效占用 世界模型（World Model）”，arXiv 预印本 arXiv:2412.13772，2024 年 8 月 15 日， 16

- [182] Y. Li、X. Wei、X. Chi、Y. Li、Z. Zhao、H. Wang、N. Ma、M. Lu 和 S. Zhang，“ManipDreamer：通过动作树和视觉引导增强机器人操作 世界模型（World Model）”，arXiv 预印本 arXiv:2504.16464，2025. 8

- [183] H.zhi、P.Chen、S.Zhou、Y.Dong、Q.Wu、L.Han 和 M.Tan，“3DFlowAction：从 3D 流中学习跨实施例操作 世界模型（World Model）”，arXiv 预印本 arXiv：2506.06199，2025. 8

- [184] A. Garg 和 K. Madhava Krishna，“Imagine-2-Drive：用于自动驾驶车辆的 Carla 中的高保真世界建模”，载于 Int。会议。英特尔。机器人。系统., 2025. 8

- [185] Y.Zheng，P.Yang，Z.Xing，Q.Zhang，Y.Zheng，Y.Gao，P.Li，T.Zhang，Z.Xia，P.Jia_et al._，“World4Drive：通过意图感知物理潜在 世界模型（World Model） 进行端到端自动驾驶”，in_Int.会议。计算。视觉._ , 2025. 8

- [186] H.Zhang、Z.Wang、Q.Lyu、Z.Zhang、S.Chen、T.Shu、B.Dariush、K.Lee、Y.Du 和 C.Gan，“COMBO：用于体现多智能体合作的组合 世界模型（World Models）”，载于_Int。会议。学习。代表._，2025 年，第 1-16 页。 8

- [187] Y. Shang、X. Zhang、Y. Tang、L. Jin、C. Gau、W. Wu 和 Y. Li，“RoboScape：物理信息体现世界模型（World Model）”，arXiv 预印本 arXiv:2506.23135，2025. 8

- [188] R. Bonatti、S. Vemprala、S. Ma、F. Frujeri、S. Chen 和 A. Kapoor，“PACT：自回归机器人预训练的感知-动作因果变换器”，载于_Int。会议。英特尔。机器人。 Syst._，2023 年，第 3621-3627 页。 8

- [189] F. Baldassarre、M. Szafraniec、B. Terver、V. Khalidov、F. Massa、Y. LeCun、P. Labatut、M. Seitzer 和 P. Bojanowski，“回到特征：DINO 作为视频 世界模型（World Models） 的基础”，arXiv 预印本 arXiv:2507.19468， 2025年8月

- [190] Y. Huang、W. Cheng、Y. Taka、X.Tao、P. Wan、D.Zhang、J. Zhou 和 J. Lu，“Owl-1：Omni 世界模型（World Model） 用于一致的长视频生成”，arXiv 预印本 arXiv:2412.09600，2024. 8

- [191] S. Huang、J. Wu、Q. Zhou、S. Miao 和 M. Long，“Vid2World：制作交互式视频扩散模型 世界模型（World Models）”，arXiv 预印本 arXiv:2505.14357，2025. 8

- [192] H. 吴、D. 吴、T. 赫、J. 郭、Y. 叶、Y. 段和 J

。 Bian，“几何强制：结合视频扩散和 3D 表示以实现一致的世界建模”，arXiv 预印本 arXiv:2507.07982，2025. 9

- [193] T. Chen、X. Hu、Z. Ding 和 C. Jin，“学习 世界模型（World Models） 用于交互式视频生成”，arXiv 预印本 arXiv:2505.21996，2025. 9

- [194] X.Guo、C.Ding、H.Dou、X.Zhang、W.Tang 和 W.Wu，“InfinityDrive：打破驾驶时间限制 世界模型（World Models）”，arXiv 预印本 arXiv:2412.01522，2024. 9

- [195] X. Wang、Z. Wu 和 P. Peng，“Longdwm：构建长期驱动的跨粒度蒸馏 世界模型（World Model）”，arXiv 预印本 arXiv:2506.01546，2025. 9, 14

- [196] H. Wang、D. Liu、H. Xie、H. Liu、E. Ma、K. Yu、L. Wang 和 B. Wang，“MiLA：用于自动驾驶的多视图高保真长期视频生成 世界模型（World Model）”，arXiv 预印本 arXiv:2503.15875，2025. 9, 14

- [197] A. Mousakhan、S. Mittal、S. Galesso、K. Farid 和 T. Brox，“Orbis：克服 长时程（Long-Horizon） 预测在驱动 世界模型（World Models） 方面的挑战”，arXiv 预印本 arXiv:2507.13162，2025. 9

- [198] J. Quevedo、S. Ansh Kumar、Y. Sun、V. Suryavanshi、P. Liang 和 S. Yang，“WorldGym：世界模型（World Model） 作为策略评估的环境”，arXiv 预印本 arXiv:2506.00613，2025. 9

- [199] Y. Li、Y. Zhu、J. Wen、C. Shen 和 Y. Xu，“WorldEval：世界模型（World Model） 作为现实世界的机器人策略评估器”，arXiv 预印本 arXiv:2505.19017，2025. 9

- [200] Y.guan，H.Liao，C.Wang，X.Liu，J.Zhang，Z.Li，“基于 世界模型（World Model） 的自动驾驶事故预测端到端场景生成”，《通信工程》，第 1 卷。 4、没有。 1，p。 144, 2025. 9

- [201] R. Po、Y. Nitzan、R. Zhu、B. Chen、T. Dao、E. Shechtman、G. Wetzstein 和 X. Huang，“长上下文状态空间视频 世界模型（World Models）”，载于_Int。会议。计算。视觉._，2025. 9

- [202] V. L. Guen 和 N. Thome，“将物理动力学与未知因素分离以进行无监督视频预测”，载于 _IEEE Conf。计算。维斯。模式识别，2020，第 11 474-11 484. 9

- [203] X. Liu 和 H. Tang，“FOLIAGE：通过无界表面进化实现物理智能 世界模型（World Models）”，arXiv 预印本 arXiv:2506.03173，2025. 9

- [204] 周三，杜宇，杨宇，韩丽，陈鹏，D.-Y。 Yeung 和 C. Gan，“学习 3D 持久性体现 世界模型（World Models）”，arXiv 预印本 arXiv:2505.05495，2025. 9

- [205] J. Wang、Y. Yao、X. Feng、H. Wu、Y. Wang、Q. Huang、Y. Ma 和 X. Zhu，“STAGE：用于长视野驾驶场景模拟的以流为中心的生成式 世界模型（World Model）”，载于_Int。会议。英特尔。机器人。

系统_ , 2025. 9, 14

[206] T. Wu, S. Yang, R. Po, Y. Xu, Z. Liu, D. Lin, and G. Wetzstein, "Video world models with long-term spatial memory," *arXiv preprint arXiv:2506.05284*, 2025. 9

[207] Y. LeCun, "A path towards autonomous machine intelligence," 2022, version 0.9.2, 2022-06-27. Position paper. [Online]. Available: https://openreview.net/pdf?id=BZ5a1r-kVsf 10

[208] H. Zhu, Z. Dong, K. Topollai, and A. Choromanska, "AD-L-JEPA: Self-supervised spatial world models with joint embedding predictive architecture for autonomous driving with LiDAR data," *arXiv preprint arXiv:2501.04969*, 2025. 10

[209] Y. Zhang, X. Guo, H. Xu, and M. Long, "Consistent world models via foresight diffusion," *arXiv preprint arXiv:2505.16474*, 2025. 10

[210] B. Zhao, R. Tang, M. Jia, Z. Wang, F. Man, X. Zhang, Y. Shang, W. Zhang, C. Gao, W. Wu *et al.*, "AirScape: An aerial generative world model with motion controllability," *arXiv preprint arXiv:2507.08885*, 2025. 10

[211] L. Li, Z. Fan, W. Cong, X. Liu, Y. Yin, M. Foutter, P. Pan, C. You, Y. Wang, Z. Wang *et al.*, "Martian World Models: Controllable video synthesis with physically accurate 3D reconstructions," *arXiv preprint arXiv:2507.07978*, 2025. 10

[212] Y. Yue, Y. Wang, H. Jiang, P. Liu, S. Song, and G. Huang, "EchoWorld: Learning motion-aware world models for echocardiography probe guidance," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025, pp. 25 993–26 003. 10

[213] D. Liang, D. Zhang, X. Zhou, S. Tu, T. Feng, X. Li, Y. Zhang, M. Du, X. Tan, and X. Bai, "Seeing the Future, Perceiving the Future: A unified driving world model for future generation and perception," *arXiv preprint arXiv:2503.13587*, 2025. 10, 14

[214] Y. Zhang, S. Gong, K. Xiong, X. Ye, X. Tan, F. Wang, J. Huang, H. Wu, and H. Wang, "BEVWorld: A multimodal world model for autonomous driving via unified BEV latent space," *arXiv e-prints*, pp. arXiv–2407, 2024. 10

[215] T. Khurana, P. Hu, A. Dave, J. Ziglar, D. Held, and D. Ramanan, "Differentiable raycasting for self-supervised occupancy forecasting," in *Eur. Conf. Comput. Vis.*, 2022, pp. 353–369. 10

[216] T. Khurana, P. Hu, D. Held, and D. Ramanan, "Point cloud forecasting as a proxy for 4D occupancy forecasting," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023, pp. 1116–1124. 10

[217] B. Mersch, X. Chen, J. Behley, and C. Stachniss, "Self-supervised point cloud prediction using 3D spatio-temporal convolutional networks," in *Conf. Robot. Learn.*, 2022, pp. 1444–1454. 10

[218] J. Ma, X. Chen, J. Huang, J. Xu, Z. Luo, J. Xu, W. Gu, R. Ai, and H. Wang, "Cam4DOcc: Benchmark for camera-only 4D occupancy forecasting in autonomous driving applications," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 21 486–21 495. 10

[219] T. Liu, S. Zhao, and N. Rhinehart, "Towards foundational LiDAR world models with efficient latent flow matching," *arXiv preprint arXiv:2506.23434*, 2025. 10

[220] Y. Shi, K. Jiang, Q. Meng, K. Wang, J. Wang, W. Sun, T. Wen, M. Yang, and D. Yang, "COME: Adding scene-centric forecasting control to occupancy world model," *arXiv preprint arXiv:2506.13260*, 2025. 10, 15

[221] Z. Yang, X. Guo, C. Ding, C. Wang, and W. Wu, "Physical informed driving world model," *arXiv preprint arXiv:2412.08410*, 2024. 10, 14

[222] Z. Liu, S. Li, E. Cousineau, S. Feng, B. Burchfiel, and S. Song, "Geometry-aware 4D video generation for robot manipulation," *arXiv preprint arXiv:2507.01099*, 2025. 10

[223] B. Jin, W. Li, B. Yang, Z. Zhu, J. Jiang, H.-a. Gao, H. Sun, K. Zhan, H. Hu, X. Zhang *et al.*, "PosePilot: Steering camera pose for generative world models with self-supervised depth," in *Int. Conf. Intell. Robot. Syst.*, 2025. 10

[224] J. Zhu, Z. Jia, T. Gao, J. Deng, S. Li, F. Liu, P. Jia, X. Lang, and X. Sun, "Other Vehicle Trajectories Are Also Needed: A driving world model unifies ego-other vehicle trajectories in video latent space," *arXiv preprint arXiv:2503.09215*, 2025. 10

[225] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Int. Conf. Intell. Robot. Syst.*, 2012, pp. 5026–5033. 11, 12

[226] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An open urban driving simulator," in *Conf. Robot. Learn.*, 2017. 11, 12

[227] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik *et al.*, "Habitat: A platform for embodied AI research," in *Int. Conf. Comput. Vis.*, 2019, pp. 9339–9347. 11, 12

[228] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac Gym: High performance GPU based physics simulation for robot learning," in *Annu. Conf. Neur. Inform. Process. Syst. Worksh.*, 2021. 11, 12

[229] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar *et al.*, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robot. Autom. Lett.*, vol. 8, no. 6, pp. 3740–3747, 2023. 11, 12

[230] M. G. Bellemare, Y. Naddaf, J. Veness, and M. Bowling, "The arcade learning environment: An evaluation platform for general agents," *Journal of artificial intelligence research*, vol. 47, pp. 253–279, 2013. 11, 12

[231] Y. Tassa, Y. Doron, A. Muldal, T. Erez, Y. Li, D. d. L. Casas, D. Budden, A. Abdolmaleki, J. Merel, A. Lefrancq *et al.*, "Deepmind control suite," *arXiv preprint arXiv:1801.00690*, 2018. 11, 12

[232] T. Yu, D. Quillen, Z. He, R. Julian, K. Hausman, C. Finn, and S. Levine, "Meta-World: A benchmark and evaluation for multi-task and meta reinforcement learning," in *Conf. Robot. Learn.*, 2020, pp. 1094–1100. 11, 12

[233] S. James, Z. Ma, D. R. Arrojo, and A. J. Davison, "RLBench: The robot learning benchmark & learning environment," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 3019–3026, 2020. 11, 12

[234] H. Caesar, J. Kabzan, K. S. Tan, W. K. Fong, E. Wolff, A. Lang, L. Fletcher, O. Beijbom, and S. Omari, "NuPlan: A closed-loop ML-based planning benchmark for autonomous vehicles," *arXiv preprint arXiv:2106.11810*, 2021. 11, 12

[235] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "LIBERO: Benchmarking knowledge transfer for lifelong robot learning," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2023, pp. 44 776–44 791. 11, 12

[236] R. Goyal, S. Ebrahimi Kahou, V. Michalski, J. Materzynska, S. Westphal, H. Kim, V. Haenel, I. Fruend, P. Yianilos, M. Mueller-Freitag *et al.*, "The "Something Something" video database for learning and evaluating visual common sense," in *Int. Conf. Comput. Vis.*, 2017, pp. 5842–5850. 11, 12

[237] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuScenes: A multimodal dataset for autonomous driving," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 11 621–11 631. 11, 12

[238] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine *et al.*, "Scalability in Perception for Autonomous Driving: Waymo Open Dataset," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2020, pp. 2446–2454. 11, 12

[239] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. M. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang, M. Savva, Y. Zhao, and D. Batra, "Habitat-Matterport 3D Dataset (HM3D): 1000 large-scale 3D environments for embodied AI," in *Annu. Conf. Neur. Inform. Process. Syst. Worksh.*, 2021. 11, 12

[240] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu *et al.*, "RT-1: Robotics transformer for real-world control at scale," *arXiv preprint arXiv:2212.06817*, 2022. 11, 12

[241] X. Tian, T. Jiang, L. Yun, Y. Mao, H. Yang, Y. Wang, Y. Wang, and H. Zhao, "Occ3D: A large-scale 3D occupancy prediction benchmark for autonomous driving," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2023, pp. 64 318–64 330. 11, 12

[242] A. O'Neill, A. Rehman, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, A. Jain *et al.*, "Open X-Embodiment: Robotic learning datasets and RT-X models : Open X-embodiment collaboration," in *IEEE Int. Conf. Robot. Autom.*, 2024, pp. 6892–6903. 11, 12

[243] S. Haddadin, S. Parusel, L. Johannsmeier, S. Golz, S. Gabl, F. Walch, M. Sabaghian, C. Jähne, L. Hausperger, and S. Haddadin, "The Franka Emika Robot: A reference platform for robotics research and education," *IEEE Robotics & Automation Magazine*, vol. 29, no. 2, pp. 46–64, 2022. 12

[244] Unitree Robotics, "Unitree Go1," 2021, accessed: 2025-09-25. [Online]. Available: https://www.unitree.com/cn/go1 12

[245] ——, "Unitree G1," 2024, accessed: 2025-09-25. [Online]. Available: https://www.unitree.com/cn/g1 12

[246] Ł. Kaiser, M. Babaeizadeh, P. Miłos, B. Osiński, R. H. Campbell, K. Czechowski, D. Erhan, C. Finn, P. Kozakowski, S. Levine *et al.*, "Model based reinforcement learning for atari," in *Int. Conf. Learn. Represent.*, 2020, pp. 1–14. 11

[247] R. Zellers, J. Lu, X. Lu, Y. Yu, Y. Zhao, M. Salehi, A. Kusupati, J. Hessel, A. Farhadi, and Y. Choi, "MERLOT Reserve: Neural script knowledge through vision and language and sound," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2022, pp. 16 375–16 387. 12

[248] A. Miech, D. Zhukov, J.-B. Alayrac, M. Tapaswi, I. Laptev, and J. Sivic, "HowTo100M: Learning a text-video embedding by watching hundred

原文第 22 页 | 中文翻译初稿

22

- [206] T. Wu、S. Yang、R. Po、Y. Xu、Z. Liu、D. Lin 和 G. Wetzstein，

- “具有长期空间记忆的视频 世界模型（World Models）”，arXiv 预印本 arXiv:2506.05284，2025 年 9 月

- [207] Y. LeCun，“通往自主机器智能的道路”，2022 年，版本 0.9.2，2022-06-27。立场文件。 [在线的]。可用：https://openreview.net/pdf?id=BZ5a1r-kVsf 10

- [208] H. Zhu、Z. Dong、K. Topollai 和 A. Choromanska，“AD-L-JEPA：自监督空间 世界模型（World Models），具有用于 LiDAR 数据自动驾驶的联合嵌入预测架构”，arXiv 预印本 arXiv:2501.04969，2025 年 10 月

- [209] Y.Zhang、X.Guo、H.Xu 和 M.Long，“通过前瞻扩散实现一致的 世界模型（World Models）”，arXiv 预印本 arXiv:2505.16474，2025.10

- [210] B.Zhao、R.Tang、M.Jia、Z.Wang、F.Man、X.Zhang、Y.Shang、W.Zhang、C.Gao、W.Wu_et al._，“AirScape：具有运动可控性的空中生成 世界模型（World Model）”，arXiv 预印本 arXiv:2507.08885，2025.10

- [211] L. Li、Z. Fan、W. Cong、X. Liu、Y. Yin、M. Foutter、P. Pan、C. You、Y. Wang、Z. Wang et al.，“Martian 世界模型（World Models）：具有物理精确 3D 重建的可控视频合成”，arXiv 预印本 arXiv：2507.07978，2025 年。 10

- [212] Y. Yue、Y. Wang、H. Jiang、P. Liu、S. Song 和 G. Huang，“EchoWorld：学习用于超声心动图探头引导的运动感知 世界模型（World Models）”，载于 _IEEE Conf。计算。维斯。模式识别，2025 年，第 25 993-26 003. 10 页

- [213] 梁大，张大，周小，涂斯，冯，李，张，杜明，谭小，白小，“看见未来，感知未来：未来一代与感知的统一驱动世界模型（World Model）”，arXiv预印本arXiv:2503.13587，2025.10， 14

- [214] Y.Zhang、S.Gong、K.Xiong、X.Ye、X.Tan、F.Wang、J.Huang、H.Wu 和 H.Wang，“BEVWorld：通过统一 BEV 潜在空间进行自动驾驶的多模式 世界模型（World Model）”，arXiv e-prints，第 arXiv-2407，2024 年。 10

- [215] T. Khurana、P. Hu、A. Dave、J. Ziglar、D. Held 和 D. Ramanan，“用于自监督占用预测的可微分光线投射”，载于 Eur。会议。计算。 Vis.，2022 年，第 353-369 页。 10

- [216] T. Khurana、P. Hu、D. Held 和 D. Ramanan，“点云预测作为 4D 占用预测的智能体”，载于 _IEEE Conf。计算。维斯。模式识别，2023 年，第 1116-1124 页。 10

- [217] B. Mersch、X. Chen、J. Behley 和 C. Stachniss，“使用 3D 时空卷积网络进行自监督点云预测”，参见 Conf。机器人。学习.，2022 年，第 1444-1454 页。 10

- [218] J. Ma、X. Chen、J. Huang、J. Xu、Z. Luo、J. Xu、W. Gu、R. Ai 和 H. Wang，“Cam4DOcc：自动驾驶应用中仅摄像头 4D 占用预测的基准”，载于 _IEEE Conf.计算。维斯。模式识别，2024 年，第 21 486-21 495 页。10

- [219] T. Liu、S. Zhu 和 N. Rhinehart，“迈向具有高效潜在 流匹配（Flow Matching） 的基础 LiDAR 世界模型（World Models）”，arXiv 预印本 arXiv:2506.23434，2025. 10

- [220] Y. Shi、K. Jiang、Q.Meng、K.Wang、J.Wang、W.Sun、T.Wen、M.Yang 和 D.Yang，“COME：为占用情况添加以场景为中心的预测控制 世界模型（World Model）”，arXiv 预印本 arXiv:2506.13260，2025.10, 15

- [221] 杨子、郭旭、丁春、王春、W

。吴，“物理知情驾驶 世界模型（World Model）”，arXiv 预印本 arXiv:2412.08410，2024 年 10 月 14 日

- [222] Z. Liu、S. Li、E. Cousineau、S. Feng、B. Burchfiel 和 S. Song，“用于机器人操作的几何感知 4D 视频生成”，arXiv 预印本 arXiv:2507.01099，2025. 10

- [223] 金本，李文，杨本，朱子，江建，H.-a。高、H. Sun、K. Zhan、H. Hu、X. Zhang et al.，“PosePilot：利用自监督深度控制生成 世界模型（World Models） 的相机姿势”，《Int》。会议。英特尔。机器人。系统._，2025. 10

- [224] J. Zhu、Z. Jia、T. Gau、J. Deng、S. Li、F. Liu、P. Jia、X. Lang 和 X. Sun，“还需要其他车辆轨迹：驾驶 世界模型（World Model） 在视频潜在空间中统一自我-其他车辆轨迹”，arXiv 预印本 arXiv:2503.09215， 2025年10月

- [225] E. Todorov、T. Erez 和 Y. Tassa，“MuJoCo：基于模型控制的物理引擎”，载于_Int。会议。英特尔。机器人。系统._，2012年，第5026-5033页。11, 12

- [226] A. Dosovitskiy、G. Ros、F. Codevilla、A. Lopez 和 V. Koltun，“CARLA：开放式城市驾驶模拟器”，载于 Conf。机器人。学习.，2017. 11, 12

- [227] M. Savva、A. Kadian、O. Maksymets、Y. Zhao、E. Wijmans、B. Jain、J. Straub、J. Liu、V. Koltun、J. Malik 等人，“栖息地：具身智能（Embodied AI） 研究平台”，载于_Int。会议。计算。 Vis._，2019 年，第 9339-9347 页。 11, 12

- [228] V. Makoviychuk、L. Wawrzyniak、Y. Guo、M. Lu、K. Storey、M. Macklin、D. Hoeller、N. Rudin、A. Allshire、A. Handa 和 G. State，“Isaac Gym：

- 用于机器人学习的基于高性能 GPU 的物理模拟，”in Annu。会议。神经。通知。过程。系统。工作.，2021. 11, 12

- [229] M. Mittal、C. Yu、Q. Yu、J. Liu、N. Rudin、D. Hoeller、J. L. Yuan、R. Singh、Y.Guo、H. Mazhar 等人，“Orbit：交互式机器人学习环境的统一模拟框架”，IEEE Robot。自动。 Lett.，卷。 8、不。 6，第 3740-3747 页，2023 年。11、12

- [230] M. G. Bellemare、Y. Naddaf、J. Veness 和 M. Bowling，“街机学习环境：通用智能体的评估平台”，人工智能研究杂志，卷。 47，第 253-279 页，2013 年。11、12

- [231] Y.塔萨、Y.多伦、A.穆达尔、T.埃雷兹、Y.李、D.d。 L. Casas、D. Budden、A. Abdolmaleki、J. Merel、A. Lefrancq et al.，“Deepmind 控制套件”，arXiv 预印本 arXiv:1801.00690，2018. 11, 12

- [232] T. Yu、D. Quillen、Z. He、R. Julian、K. Hausman、C. Finn 和 S. Levine，“元世界：多任务和元强化学习（Reinforcement Learning） 的基准和评估”，参见 Conf。机器人。学习.，2020 年，第 1094-1100 页。 11, 12

- [233] S. James、Z. Ma、D. R. Arrojo 和 A. J. Davison，“RLBench：机器人学习基准和学习环境”，IEEE Robot。自动。 Lett.，卷。 5、没有。 2，第 3019-3026 页，2020 年。11、12

- [234] H. Caesar、J. Kabzan、K. S. Tan、W. K. Fong、E. Wolff、A. Lang、L. Fletcher、O. Beijbom 和 S. Omari，“NuPlan：基于机器学习的闭环自动驾驶汽车规划基准”，arXiv 预印本 arXiv:2106.11810，2021. 11, 12

- [235] B. Liu、Y. Zhu、C. Gau、Y. Feng、Q. Liu、Y. Zhu 和 P. Stone，“LIBERO：终身机器人学习的基准知识转移”，载于 Annu。会议。神经。通知。过程。系统.，2023 年，第 44 776-44 791. 11, 12 页

- [236] R. Goyal、S. Ebrahimi Kahou、V. Michalski、J. Materzynska、S

。 Westphal、H. Kim、V. Haenel、I. Fruend、P. Yianilos、M. Mueller-Freitag 等人，“用于学习和评估视觉常识的“Something Something”视频数据库”，《Int》。会议。计算。 Vis._，2017 年，第 5842-5850 页。 11, 12

- [237] H. Caesar、V. Bankiti、A. H. Lang、S. Vora、V. E. Liong、Q. Xu、A. Krishnan、Y. Pan、G. Baldan 和 O. Beijbom，“nuScenes：自动驾驶多模态数据集”，载于 _IEEE Conf.计算。维斯。模式识别，2020，第 11 621-11 631. 11, 12

- [238] P. Sun、H. Kretzschmar、X. Dotiwalla、A. Chouard、V. Patnaik、P. Tsui、J.Guo、Y. Zhou、Y. Chai、B. Caine 等人，“自动驾驶感知的可扩展性：Waymo 开放数据集”，载于 _IEEE Conf.计算。维斯。模式识别，2020 年，第 2446-2454 页。 11, 12

- [239] S.K. Ramakrishnan、A. Gokaslan、E. Wijmans、O. Maksymets、A. Clegg、J. M. Turner、E. Undersander、W. Galuba、A. Westbury、A. X. Chang、M. Savva、Y. Zhu 和 D. Batra，“Habitat-Matterport 3D 数据集 (HM3D)：1000 个大型 3D 具身智能（Embodied AI）的环境，”在 Annu 中。会议。神经。通知。过程。系统。工作.，2021. 11, 12

- [240] A. Brohan、N. Brown、J. Carbajal、Y. Chebotar、J. Dabis、C. Finn、K. Gopalakrishnan、K. Hausman、A. Herzog、J. Hsu 等人，“RT-1：用于大规模现实世界控制的机器人变压器”，arXiv 预印本 arXiv:2212.06817， 2022年11月12日

- [241] X. Tian、T. Jiang、L. Yun、Y. Mao、H. Yang、Y. Wang、Y. Wang 和 H. Zhu，“Occ3D：自动驾驶的大规模 3D 占用预测基准”，载于 Annu。会议。神经。通知。过程。系统.，2023 年，第 64 318-64 330. 11, 12 页

- [242] A. O’Neill、A. Rehman、A. Maddukuri、A. Gupta、A. Padalkar、A. Lee、A. Pooley、A. Gupta、A. Mandlekar、A. Jain 等人，“开放 X-体现：机器人学习数据集和 RT-X 模型：开放 X-体现协作”，载于 IEEE Int.会议。机器人。 Autom.，2024 年，第 6892-6903 页。 11, 12

- [243] S. Haddadin、S. Parusel、L. Johannsmeier、S. Golz、S. Gabl、F. Walch、M. Sabaghian、C. Jähne、L. Hausperger 和 S. Haddadin，“Franka Emika 机器人：机器人研究和教育的参考平台”，IEEE 机器人与自动化杂志，卷。 29、没有。 2，第 46-64 页，2022 年。 12

- [244] Unitree Robotics，“Unitree Go1”，2021 年，访问时间：2025 年 9 月 25 日。 [在线的]。获取：https://www.unitree.com/cn/go1 12

- [245] --，《Unitree G1》，2024 年，访问时间：2025 年 9 月 25 日。 [在线的]。获取：https://www.unitree.com/cn/g1 12

- [246] Ł. Kaiser、M. Babaeizadeh、P. Miłos、B. Osi´nski、R. H. Campbell、K. Czechowski、D. Erhan、C. Finn、P. Kozakowski、S. Levine et al.，“基于 Atari 的 强化学习（Reinforcement Learning） 模型”，参见 Int。会议。学习。代表.，2020 年，第 1-14 页。 11

- [247] R. Zellers、J. Lu、X. Lu、Y. Yu、Y. Zhu、M. Salehi、A. Kusupati、J. Hessel、A. Farhadi 和 Y. Choi，“MERLOT Reserve：通过视觉、语言和声音获得神经脚本知识”，载于 _IEEE Conf。计算。维斯。模式识别，2022 年，第 16 375-16 387 页。12

- [248] A. Miech, D. Zhukov, J.-B. Alayrac、M. Tapaswi、I. Laptev 和 J. Sivic，“HowTo100M：通过观看数百个文本视频嵌入来学习

million narrated video clips," in *Int. Conf. Comput. Vis.*, 2019, pp. 2630–2640. 12

[249] J. Carreira, E. Noland, C. Hillier, and A. Zisserman, "A short note on the kinetics-700 human action dataset," *arXiv preprint arXiv:1907.06987*, 2019. 12

[250] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2009, pp. 248–255. 12

[251] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local nash equilibrium," in *Annu. Conf. Neur. Inform. Process. Syst.*, 2017. 13

[252] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2016, pp. 2818–2826. 13

[253] T. Unterthiner, S. Van Steenkiste, K. Kurach, R. Marinier, M. Michalski, and S. Gelly, "Towards Accurate Generative Models of Video: A new metric & challenges," *arXiv preprint arXiv:1812.01717*, 2018. 13

[254] J. Carreira and A. Zisserman, "Quo Vadis, Action Recognition? A new model and the kinetics dataset," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2017, pp. 6299–6308. 13

[255] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, F. Viola, T. Green, T. Back, P. Natsev *et al.*, "The Kinetics human action video dataset," *arXiv preprint arXiv:1705.06950*, 2017. 13

[256] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, 2004. 13

[257] A. Hore and D. Ziou, "Image quality metrics: PSNR vs. SSIM," in *Int. Conf. Pattern Recog.*, 2010, pp. 2366–2369. 13

[258] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2018, pp. 586–595. 13

[259] Z. Huang, Y. He, J. Yu, F. Zhang, C. Si, Y. Jiang, Y. Zhang, T. Wu, Q. Jin, N. Chanpaisit *et al.*, "VBench: Comprehensive benchmark suite for video generative models," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2024, pp. 21 807–21 818. 13

[260] H. Fan, H. Su, and L. J. Guibas, "A point set generation network for 3D object reconstruction from a single image," in *IEEE Conf. Comput. Vis. Pattern Recog.*, 2017, pp. 605–613. 14

[261] Y. Hu, J. Yang, L. Chen, K. Li, C. Sima, X. Zhu, S. Chai, S. Du, T. Lin, W. Wang *et al.*, "Planning-Oriented autonomous driving," in *CVPR*, 2023, pp. 17 853–17 862. 16

[262] H. Yue, S. Huang, Y. Liao, S. Chen, P. Zhou, L. Chen, M. Yao, and G. Ren, "EWMBench: Evaluating scene, motion, and semantic quality in embodied world models," *arXiv preprint arXiv:2505.09694*, 2025. 16

原文第 23 页 | 中文翻译初稿

百万个解说视频剪辑”，载于 Int. Conf. Comput. Vis.，2019 年，第 2630-2640 页。12

- [249] J. Carreira、E. Noland、C. Hillier 和 A. Zisserman，“关于 Kinetics-700 人类行为数据集的简短说明”，arXiv 预印本 arXiv：1907.06987，2019. 12

- [250] J. Deng、W. Dong、R. Socher、L.-J。 Li、K. Li 和 L. Fei-Fei，“ImageNet：大规模分层图像数据库”，载于 _IEEE Conf。计算。维斯。模式识别，2009 年，第 248-255 页。 12

- [251] M. Heusel、H. Ramsauer、T. Unterthiner、B. Nessler 和 S. Hochreiter，“通过两个时间尺度更新规则训练的 GAN 收敛到局部纳什均衡”，《Annu》。会议。神经。通知。过程。系统._，2017. 13

- [252] C. Szegedy、V. Vanhoucke、S. Ioffe、J. Shlens 和 Z. Wojna，“重新思考计算机视觉的初始架构”，载于 _IEEE Conf。计算。维斯。模式识别，2016 年，第 2818-2826 页。 13

- [253] T. Unterthiner、S. Van Steenkiste、K. Kurach、R. Marinier、M. Michalski 和 S. Gelly，“迈向准确的视频生成模型：新的指标和挑战”，arXiv 预印本 arXiv：1812.01717，2018. 13

- [254] J. Carreira 和 A. Zisserman，“Quo Vadis，动作识别？一个新模型和动力学数据集”，载于 IEEE Conf. Comput. Vis. Pattern Recog.，2017 年，第 6299-6308 页。 13

- [255] W. Kay、J. Carreira、K. Simonyan、B. Zhang、C. Hillier、S. Vijayanarasimhan、F. Viola、T. Green、T. Back、P. Natsev et al.，“The Kinetics 人类动作视频数据集”，arXiv 预印本 arXiv:1705.06950，2017. 13

- [256] Z. Wang、A. C. Bovik、H. R. Sheikh 和 E. P. Simoncelli，“图像质量评估：从错误可见性到结构相似性”，IEEE Trans。图像处理.，卷。 13、没有。 4，第 600-612 页，2004 年。 13

- [257] A. Hore 和 D. Ziou，“图像质量指标：PSNR 与 SSIM”，载于 _Int。会议。模式识别，2010 年，第 2366-2369 页。 13

- [258] R.Zhang、P.Isola、A.A.Efros、E.Shechtman 和 O.Wang，“深度特征作为感知度量的不合理有效性”，载于 _IEEE Conf。计算。维斯。模式识别，2018，第 586-595 页。 13

- [259] Z. Huang、Y. He、J. Yu、F. Zhang、C. Si、Y. Jiang、Y. Zhang、T. Wu、Q. Jin、N. Chanpaisit 等人，“VBench：视频生成模型综合基准套件”，载于 _IEEE Conf.计算。维斯。模式识别，2024 年，第 21 807-21 818 页。13

- [260] H. Fan、H. Su 和 L. J. Guibas，“用于从单个图像重建 3D 对象的点集生成网络”，载于 _IEEE Conf。计算。维斯。模式识别，2017 年，第 605-613 页。 14

- [261] Y. Hu, J. Yang, L. Chen, K. Li, C. Sima, X. Zhu, S. Chai, S. Du, T. Lin, W. Wang et al.，“面向规划的自动驾驶”，载于 CVPR，2023 年，第 17 853-17 862 页。 16

- [262] H. Yue、S. Huang、Y. Liao、S. Chen、P. Zhou、L. Chen、M. Yao 和 G. Ren，“EWMBench：评估 世界模型（World Models） 中的场景、运动和语义质量”，arXiv 预印本 arXiv:2505.09694，2025. 16